



NVIDIA OptiX

API Reference Manual

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1 Module Index

1.1 Modules

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2 Class Index

2.1 Class List

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3 Module Documentation

3.1 Device API

Functions

- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2)`

- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5, unsigned int &p6)`
- static `__forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5, unsigned int &p6, unsigned int &p7)`
- static `__forceinline__`
`__device__ void optixSetPayload_0 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_1 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_2 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_3 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_4 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_5 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_6 (unsigned int p)`
- static `__forceinline__`
`__device__ void optixSetPayload_7 (unsigned int p)`
- static `__forceinline__`
`__device__ unsigned int optixGetPayload_0 ()`
- static `__forceinline__`
`__device__ unsigned int optixGetPayload_1 ()`

- static __forceinline__
__device__ unsigned int [optixGetPayload_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_7](#) ()
- static __forceinline__
__device__ unsigned int [optixUndefinedValue](#) ()
- static __forceinline__
__device__ float3 [optixGetWorldRayOrigin](#) ()
- static __forceinline__
__device__ float3 [optixGetWorldRayDirection](#) ()
- static __forceinline__
__device__ float3 [optixGetObjectRayOrigin](#) ()
- static __forceinline__
__device__ float3 [optixGetObjectRayDirection](#) ()
- static __forceinline__
__device__ float [optixGetRayTmin](#) ()
- static __forceinline__
__device__ float [optixGetRayTmax](#) ()
- static __forceinline__
__device__ float [optixGetRayTime](#) ()
- static __forceinline__
__device__ unsigned int [optixGetRayFlags](#) ()
- static __forceinline__
__device__ unsigned int [optixGetRayVisibilityMask](#) ()
- static __forceinline__
__device__ void [optixGetTriangleVertexData](#) ([OptixTraversableHandle](#) gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float3 data[3])
- static __forceinline__
__device__ void [optixGetLinearCurveVertexData](#) ([OptixTraversableHandle](#) gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[2])
- static __forceinline__
__device__ void [optixGetQuadraticBSplineVertexData](#) ([OptixTraversableHandle](#) gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[3])
- static __forceinline__
__device__ void [optixGetCubicBSplineVertexData](#) ([OptixTraversableHandle](#) gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[4])
- static __forceinline__
__device__
[OptixTraversableHandle](#) [optixGetGASTraversableHandle](#) ()

- static __forceinline__
__device__ float optixGetGASMotionTimeBegin (OptixTraversableHandle gas)
- static __forceinline__
__device__ float optixGetGASMotionTimeEnd (OptixTraversableHandle gas)
- static __forceinline__
__device__ unsigned int optixGetGASMotionStepCount (OptixTraversableHandle gas)
- static __forceinline__
__device__ void optixGetWorldToObjectTransformMatrix (float m[12])
- static __forceinline__
__device__ void optixGetObjectToWorldTransformMatrix (float m[12])
- static __forceinline__
__device__ float3 optixTransformPointFromWorldToObjectSpace (float3 point)
- static __forceinline__
__device__ float3 optixTransformVectorFromWorldToObjectSpace (float3 vec)
- static __forceinline__
__device__ float3 optixTransformNormalFromWorldToObjectSpace (float3 normal)
- static __forceinline__
__device__ float3 optixTransformPointFromObjectToWorldSpace (float3 point)
- static __forceinline__
__device__ float3 optixTransformVectorFromObjectToWorldSpace (float3 vec)
- static __forceinline__
__device__ float3 optixTransformNormalFromObjectToWorldSpace (float3 normal)
- static __forceinline__
__device__ unsigned int optixGetTransformListSize ()
- static __forceinline__
__device__
OptixTraversableHandle optixGetTransformListHandle (unsigned int index)
- static __forceinline__
__device__ OptixTransformType optixGetTransformTypeFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const
OptixStaticTransform * optixGetStaticTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const
OptixSRTMotionTransform * optixGetSRTMotionTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const
OptixMatrixMotionTransform * optixGetMatrixMotionTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ unsigned int optixGetInstanceIdFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const float4 * optixGetInstanceTransformFromHandle (OptixTraversableHandle handle)

- static __forceinline__
__device__ const float4 * [optixGetInstanceInverseTransformFromHandle](#)
([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1, unsigned int a2)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1, unsigned int a2, unsigned int a3)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5, unsigned int
a6)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0,
unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5, unsigned int
a6, unsigned int a7)
- static __forceinline__
__device__ unsigned int [optixGetAttribute_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_7](#) ()
- static __forceinline__
__device__ void [optixTerminateRay](#) ()

- static __forceinline__
__device__ void [optixIgnoreIntersection](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPrimitiveIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetSbtGASIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetInstanceId](#) ()
- static __forceinline__
__device__ unsigned int [optixGetInstanceIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetHitKind](#) ()
- static __forceinline__
__device__ [OptixPrimitiveType](#) [optixGetPrimitiveType](#) (unsigned int hitKind)
- static __forceinline__
__device__ bool [optixIsFrontFaceHit](#) (unsigned int hitKind)
- static __forceinline__
__device__ bool [optixIsBackFaceHit](#) (unsigned int hitKind)
- static __forceinline__
__device__ [OptixPrimitiveType](#) [optixGetPrimitiveType](#) ()
- static __forceinline__
__device__ bool [optixIsFrontFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsBackFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleFrontFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleBackFaceHit](#) ()
- static __forceinline__
__device__ float2 [optixGetTriangleBarycentrics](#) ()
- static __forceinline__
__device__ float [optixGetCurveParameter](#) ()
- static __forceinline__
__device__ uint3 [optixGetLaunchIndex](#) ()
- static __forceinline__
__device__ uint3 [optixGetLaunchDimensions](#) ()
- static __forceinline__
__device__ [CUdeviceptr](#) [optixGetSbtDataPointer](#) ()
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1)

- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3, unsigned int exceptionDetail4)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3, unsigned int exceptionDetail4, unsigned int exceptionDetail5)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3, unsigned int exceptionDetail4, unsigned int exceptionDetail5, unsigned int exceptionDetail6)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3, unsigned int exceptionDetail4, unsigned int exceptionDetail5, unsigned int exceptionDetail6, unsigned int exceptionDetail7)
- static __forceinline__
__device__ int [optixGetExceptionCode](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_7](#) ()
- static __forceinline__
__device__
[OptixTraversableHandle](#) [optixGetExceptionInvalidTraversable](#) ()
- static __forceinline__
__device__ int [optixGetExceptionInvalidSbtOffset](#) ()

- static `__forceinline__`
`__device__`
`OptixInvalidRayExceptionDetails` `optixGetExceptionInvalidRay ()`
- static `__forceinline__`
`__device__`
`OptixParameterMismatchExceptionDetails` `optixGetExceptionParameterMismatch ()`
- static `__forceinline__`
`__device__` `char *` `optixGetExceptionLineInfo ()`
- template<typename ReturnT , typename... ArgTypes>
static `__forceinline__`
`__device__` ReturnT `optixDirectCall` (unsigned int sbtIndex, ArgTypes...args)
- template<typename ReturnT , typename... ArgTypes>
static `__forceinline__`
`__device__` ReturnT `optixContinuationCall` (unsigned int sbtIndex, ArgTypes...args)

3.1.1 Detailed Description

OptiX Device API.

3.1.2 Function Documentation

3.1.2.1 `template<typename ReturnT , typename... ArgTypes> static __forceinline__`
`__device__ ReturnT optixContinuationCall (`
`unsigned int sbtIndex,`
`ArgTypes... args ) [static]`

Creates a call to the continuation callable program at the specified SBT entry.

This will call the program that was specified in the `OptixProgramGroupCallables::entryFunctionNameCC` in the module specified by `OptixProgramGroupCallables::moduleCC`. The address of the SBT entry is calculated by `OptixShaderBindingTable::callablesRecordBase + (OptixShaderBindingTable::callablesRecordStrideInBytes * sbtIndex)`. As opposed to direct callable programs, continuation callable programs are allowed to call `optixTrace` recursively.

Behavior is undefined if there is no continuation callable program at the specified SBT entry.

Behavior is undefined if the number of arguments that are being passed in does not match the number of parameters expected by the program that is called. In that case an exception of type `OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH` will be thrown if `OPTIX_EXCEPTION_FLAG_DEBUG` was specified for the `OptixPipelineCompileOptions::exceptionFlags`.

Parameters

in	<i>sbtIndex</i>	The offset of the SBT entry of the continuation callable program to call relative to <code>OptixShaderBindingTable::callablesRecordBase</code> .
in	<i>args</i>	The arguments to pass to the continuation callable program.

```

3.1.2.2  template<typename ReturnT , typename... ArgTypes> static __forceinline__
           __device__ ReturnT optixDirectCall (
               unsigned int sbtIndex,
               ArgTypes... args ) [static]

```

Creates a call to the direct callable program at the specified SBT entry.

This will call the program that was specified in the [OptixProgramGroupCallables::entryFunctionNameDC](#) in the module specified by [OptixProgramGroupCallables::moduleDC](#). The address of the SBT entry is calculated by [OptixShaderBindingTable::callablesRecordBase](#) + ([OptixShaderBindingTable::callablesRecordStrideInBytes](#) * *sbtIndex*).

Behavior is undefined if there is no direct callable program at the specified SBT entry.

Behavior is undefined if the number of arguments that are being passed in does not match the number of parameters expected by the program that is called. In that case an exception of type `OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH` will be thrown if `OPTIX_EXCEPTION_FLAG_DEBUG` was specified for the [OptixPipelineCompileOptions::exceptionFlags](#).

Parameters

in	<i>sbtIndex</i>	The offset of the SBT entry of the direct callable program to call relative to OptixShaderBindingTable::callablesRecordBase .
in	<i>args</i>	The arguments to pass to the direct callable program.

```

3.1.2.3  static __forceinline__ __device__ unsigned int optixGetAttribute_0 ( ) [static]

```

Returns the attribute at slot 0.

```

3.1.2.4  static __forceinline__ __device__ unsigned int optixGetAttribute_1 ( ) [static]

```

Returns the attribute at slot 1.

```

3.1.2.5  static __forceinline__ __device__ unsigned int optixGetAttribute_2 ( ) [static]

```

Returns the attribute at slot 2.

```

3.1.2.6  static __forceinline__ __device__ unsigned int optixGetAttribute_3 ( ) [static]

```

Returns the attribute at slot 3.

```

3.1.2.7  static __forceinline__ __device__ unsigned int optixGetAttribute_4 ( ) [static]

```

Returns the attribute at slot 4.

```

3.1.2.8  static __forceinline__ __device__ unsigned int optixGetAttribute_5 ( ) [static]

```

Returns the attribute at slot 5.

3.1.2.9 **static __forceinline__ __device__ unsigned int optixGetAttribute_6 () [static]**

Returns the attribute at slot 6.

3.1.2.10 **static __forceinline__ __device__ unsigned int optixGetAttribute_7 () [static]**

Returns the attribute at slot 7.

3.1.2.11 **static __forceinline__ __device__ void optixGetCubicBSplineVertexData (** **OptixTraversableHandle *gas*,** **unsigned int *primIdx*,** **unsigned int *sbtGASIndex*,** **float *time*,** **float4 *data[4]*) [static]**

Return the object space curve control vertex data of a cubic BSpline curve in a Geometry Acceleration Structure (GAS) at a given motion time.

$data[i] = \{x, y, z, w\}$ with $\{x, y, z\}$ the position and w the radius of control vertex i . If motion is disabled via [OptixPipelineCompileOptions::usesMotionBlur](#), or the GAS does not contain motion, the time parameter is ignored.

3.1.2.12 **static __forceinline__ __device__ float optixGetCurveParameter () [static]**

Convenience function that returns the curve parameter.

When using [OptixBuildInputCurveArray](#) objects, during intersection the curve parameter is stored into the first attribute register.

3.1.2.13 **static __forceinline__ __device__ int optixGetExceptionCode () [static]**

Returns the exception code.

Only available in EX.

3.1.2.14 **static __forceinline__ __device__ unsigned int optixGetExceptionDetail_0 ()** **[static]**

Returns the 32-bit exception detail at slot 0.

The behavior is undefined if the exception is not a user exception, or the used overload [optixThrowException\(\)](#) did not provide the queried exception detail.

Only available in EX.

3.1.2.15 **static __forceinline__ __device__ unsigned int optixGetExceptionDetail_1 ()** **[static]**

Returns the 32-bit exception detail at slot 1.

See Also

[optixGetExceptionDetail_0\(\)](#)

3.1.2.16 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_2 ()
[static]

Returns the 32-bit exception detail at slot 2.

See Also

[optixGetExceptionDetail_0\(\)](#)

3.1.2.17 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_3 ()
[static]

Returns the 32-bit exception detail at slot 3.

See Also

[optixGetExceptionDetail_0\(\)](#)

3.1.2.18 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_4 ()
[static]

Returns the 32-bit exception detail at slot 4.

See Also

[optixGetExceptionDetail_0\(\)](#)

3.1.2.19 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_5 ()
[static]

Returns the 32-bit exception detail at slot 5.

See Also

[optixGetExceptionDetail_0\(\)](#)

3.1.2.20 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_6 ()
[static]

Returns the 32-bit exception detail at slot 6.

See Also

[optixGetExceptionDetail_0\(\)](#)

3.1.2.21 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_7 () [static]

Returns the 32-bit exception detail at slot 7.

See Also

[optixGetExceptionDetail_0\(\)](#)

**3.1.2.22 static __forceinline__ __device__ OptixInvalidRayExceptionDetails
optixGetExceptionInvalidRay () [static]**

Returns the invalid ray for exceptions with exception code OPTIX_EXCEPTION_CODE_INVALID_RAY. Exceptions of type OPTIX_EXCEPTION_CODE_INVALID_RAY are thrown when one or more values that were passed into optixTrace are either inf or nan.

OptixInvalidRayExceptionDetails::rayTime will always be 0 if

[OptixPipelineCompileOptions::usesMotionBlur](#) is 0. Values in the returned struct are all zero for all other exception codes.

Only available in EX.

3.1.2.23 static __forceinline__ __device__ int optixGetExceptionInvalidSbtOffset () [static]

Returns the invalid sbt offset for exceptions with exception code OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_MISS_SBT and OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_HIT_SBT.

Returns zero for all other exception codes.

Only available in EX.

**3.1.2.24 static __forceinline__ __device__ OptixTraversableHandle
optixGetExceptionInvalidTraversable () [static]**

Returns the invalid traversable handle for exceptions with exception code OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_TRAVERSABLE.

Returns zero for all other exception codes.

Only available in EX.

3.1.2.25 static __forceinline__ __device__ char* optixGetExceptionLineInfo () [static]

Returns a string that includes information about the source location that caused the current exception.

The source location is only available for exceptions of type

OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH, OPTIX_EXCEPTION_CODE_UNSUPPORTED_PRIMITIVE_TYPE, OPTIX_EXCEPTION_CODE_INVALID_RAY, and for user exceptions. Line information needs to be present in the input PTX and [OptixModuleCompileOptions::debugLevel](#) may not be set to OPTIX_COMPILE_DEBUG_LEVEL_NONE.

Returns a NULL pointer if no line information is available.

Only available in EX.

3.1.2.26 **static __forceinline__ __device__ OptixParameterMismatchExceptionDetails optixGetExceptionParameterMismatch () [static]**

Returns information about an exception with code
OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH.

Exceptions of type OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH are called when the number of arguments that were passed into a call to `optixDirectCall` or `optixContinuationCall` does not match the number of parameters of the callable that is called. Note that the parameters are packed by OptiX into individual 32 bit values, so the number of expected and passed values may not correspond to the number of arguments passed into `optixDirectCall` or `optixContinuationCall`.

Values in the returned struct are all zero for all other exception codes.

Only available in EX.

3.1.2.27 **static __forceinline__ __device__ unsigned int optixGetGASMotionStepCount (OptixTraversableHandle *gas*) [static]**

Returns the number of motion steps of a GAS (see [OptixMotionOptions](#))

3.1.2.28 **static __forceinline__ __device__ float optixGetGASMotionTimeBegin (OptixTraversableHandle *gas*) [static]**

Returns the motion begin time of a GAS (see [OptixMotionOptions](#))

3.1.2.29 **static __forceinline__ __device__ float optixGetGASMotionTimeEnd (OptixTraversableHandle *gas*) [static]**

Returns the motion end time of a GAS (see [OptixMotionOptions](#))

3.1.2.30 **static __forceinline__ __device__ OptixTraversableHandle optixGetGASTraversableHandle () [static]**

Returns the traversable handle for the Geometry Acceleration Structure (GAS) containing the current hit. May be called from IS, AH and CH.

3.1.2.31 **static __forceinline__ __device__ unsigned int optixGetHitKind () [static]**

Returns the 8 bit hit kind associated with the current hit.

Use `optixGetPrimitiveType()` to interpret the hit kind. For custom intersections (primitive type OPTIX_PRIMITIVE_TYPE_CUSTOM), this is the 7-bit hitKind passed to `optixReportIntersection()`. Hit kinds greater than 127 are reserved for built-in primitives.

Available only in AH and CH.

3.1.2.32 **static __forceinline__ __device__ unsigned int optixGetInstanceId () [static]**

Returns the `OptixInstance::instanceId` of the instance within the top level acceleration structure associated with the current intersection.

When building an acceleration structure using `OptixBuildInputInstanceArray` each `OptixInstance` has a user supplied `instanceId`. `OptixInstance` objects reference another acceleration structure. During

traversal the acceleration structures are visited top down. In the IS and AH programs the `OptixInstance::instanceld` corresponding to the most recently visited `OptixInstance` is returned when calling `optixGetInstanceld()`. In CH `optixGetInstanceld()` returns the `OptixInstance::instanceld` when the hit was recorded with `optixReportIntersection`. In the case where there is no `OptixInstance` visited, `optixGetInstanceld` returns `~0u`

3.1.2.33 `static __forceinline__ __device__ unsigned int optixGetInstanceldFromHandle ( OptixTraversableHandle handle ) [static]`

Returns `instanceld` from an `OptixInstance` traversable.

Returns 0 if the traversable handle does not reference an `OptixInstance`.

3.1.2.34 `static __forceinline__ __device__ unsigned int optixGetInstanceIndex ( ) [static]`

Returns the zero-based index of the instance within its instance acceleration structure associated with the current intersection.

In the IS and AH programs the index corresponding to the most recently visited `OptixInstance` is returned when calling `optixGetInstanceIndex()`. In CH `optixGetInstanceIndex()` returns the index when the hit was recorded with `optixReportIntersection`. In the case where there is no `OptixInstance` visited, `optixGetInstanceIndex` returns 0

3.1.2.35 `static __forceinline__ __device__ const float4* optixGetInstanceInverseTransformFromHandle ( OptixTraversableHandle handle ) [static]`

Returns world-to-object transform from an `OptixInstance` traversable.

Returns 0 if the traversable handle does not reference an `OptixInstance`.

3.1.2.36 `static __forceinline__ __device__ const float4* optixGetInstanceTransformFromHandle ( OptixTraversableHandle handle ) [static]`

Returns object-to-world transform from an `OptixInstance` traversable.

Returns 0 if the traversable handle does not reference an `OptixInstance`.

3.1.2.37 `static __forceinline__ __device__ uint3 optixGetLaunchDimensions ( ) [static]`

Available in any program, it returns the dimensions of the current launch specified by `optixLaunch` on the host.

3.1.2.38 `static __forceinline__ __device__ uint3 optixGetLaunchIndex ( ) [static]`

Available in any program, it returns the current launch index within the launch dimensions specified by `optixLaunch` on the host.

The raygen program is typically only launched once per launch index.

3.1.2.39 `static __forceinline__ __device__ void optixGetLinearCurveVertexData ( OptixTraversableHandle gas,`

```

    unsigned int primIdx,
    unsigned int sbtGASIndex,
    float time,
    float4 data[2] ) [static]

```

Return the object space curve control vertex data of a linear curve in a Geometry Acceleration Structure (GAS) at a given motion time.

`data[i] = {x,y,z,w}` with `{x,y,z}` the position and `w` the radius of control vertex `i`. If motion is disabled via [OptixPipelineCompileOptions::usesMotionBlur](#), or the GAS does not contain motion, the time parameter is ignored.

```

3.1.2.40 static __forceinline__ __device__ const OptixMatrixMotionTransform*
optixGetMatrixMotionTransformFromHandle (
    OptixTraversableHandle handle ) [static]

```

Returns a pointer to a [OptixMatrixMotionTransform](#) from its traversable handle.

Returns 0 if the traversable is not of type

`OPTIX_TRANSFORM_TYPE_MATRIX_MOTION_TRANSFORM`.

```

3.1.2.41 static __forceinline__ __device__ float3 optixGetObjectRayDirection ( ) [static]

```

Returns the current object space ray direction based on the current transform stack.

Only available in IS and AH.

```

3.1.2.42 static __forceinline__ __device__ float3 optixGetObjectRayOrigin ( ) [static]

```

Returns the current object space ray origin based on the current transform stack.

Only available in IS and AH.

```

3.1.2.43 static __forceinline__ __device__ void optixGetObjectToWorldTransformMatrix (
    float m[12] ) [static]

```

Returns the object-to-world transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

```

3.1.2.44 static __forceinline__ __device__ unsigned int optixGetPayload_0 ( ) [static]

```

Reads the 32-bit payload value at slot 0.

```

3.1.2.45 static __forceinline__ __device__ unsigned int optixGetPayload_1 ( ) [static]

```

Reads the 32-bit payload value at slot 1.

```

3.1.2.46 static __forceinline__ __device__ unsigned int optixGetPayload_2 ( ) [static]

```

Reads the 32-bit payload value at slot 2.

3.1.2.47 `static __forceinline__ __device__ unsigned int optixGetPayload_3 ( ) [static]`

Reads the 32-bit payload value at slot 3.

3.1.2.48 `static __forceinline__ __device__ unsigned int optixGetPayload_4 ( ) [static]`

Reads the 32-bit payload value at slot 4.

3.1.2.49 `static __forceinline__ __device__ unsigned int optixGetPayload_5 ( ) [static]`

Reads the 32-bit payload value at slot 5.

3.1.2.50 `static __forceinline__ __device__ unsigned int optixGetPayload_6 ( ) [static]`

Reads the 32-bit payload value at slot 6.

3.1.2.51 `static __forceinline__ __device__ unsigned int optixGetPayload_7 ( ) [static]`

Reads the 32-bit payload value at slot 7.

3.1.2.52 `static __forceinline__ __device__ unsigned int optixGetPrimitiveIndex ( ) [static]`

For a given [OptixBuildInputTriangleArray](#) the number of primitives is defined as "(OptixBuildInputTriangleArray::indexBuffer == 0) ? OptixBuildInputTriangleArray::numVertices/3 : OptixBuildInputTriangleArray::numIndexTriplets;". For a given [OptixBuildInputCustomPrimitiveArray](#) the number of primitives is defined as numAabbs.

The primitive index returns the index into the array of primitives plus the primitiveIndexOffset.

In IS and AH this corresponds to the currently intersected primitive. In CH this corresponds to the primitive index of the closest intersected primitive.

3.1.2.53 `static __forceinline__ __device__ OptixPrimitiveType optixGetPrimitiveType ( unsigned int hitKind ) [static]`

Function interpreting the result of [optixGetHitKind\(\)](#).

3.1.2.54 `static __forceinline__ __device__ OptixPrimitiveType optixGetPrimitiveType ( ) [static]`

Function interpreting the hit kind associated with the current [optixReportIntersection](#).

3.1.2.55 `static __forceinline__ __device__ void optixGetQuadraticBSplineVertexData ( OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[3] ) [static]`

Return the object space curve control vertex data of a quadratic BSpline curve in a Geometry Acceleration Structure (GAS) at a given motion time.

$data[i] = \{x,y,z,w\}$ with $\{x,y,z\}$ the position and w the radius of control vertex i . If motion is disabled via [OptixPipelineCompileOptions::usesMotionBlur](#), or the GAS does not contain motion, the time parameter is ignored.

3.1.2.56 static __forceinline__ __device__ unsigned int optixGetRayFlags () [static]

Returns the rayFlags passed into optixTrace.

Only available in IS, AH, CH, MS

3.1.2.57 static __forceinline__ __device__ float optixGetRayTime () [static]

Returns the rayTime passed into optixTrace.

Will return 0 if motion is disabled. Only available in IS, AH, CH, MS

3.1.2.58 static __forceinline__ __device__ float optixGetRayTmax () [static]

In IS and CH returns the current smallest reported hitT or the tmax passed into optixTrace if no hit has been reported In AH returns the hitT value as passed in to optixReportIntersection In MS returns the tmax passed into optixTrace Only available in IS, AH, CH, MS.

3.1.2.59 static __forceinline__ __device__ float optixGetRayTmin () [static]

Returns the tmin passed into optixTrace.

Only available in IS, AH, CH, MS

3.1.2.60 static __forceinline__ __device__ unsigned int optixGetRayVisibilityMask () [static]

Returns the visibilityMask passed into optixTrace.

Only available in IS, AH, CH, MS

3.1.2.61 static __forceinline__ __device__ CUdeviceptr optixGetSbtDataPointer () [static]

Returns the generic memory space pointer to the data region (past the header) of the currently active SBT record corresponding to the current program.

3.1.2.62 static __forceinline__ __device__ unsigned int optixGetSbtGASIndex () [static]

Returns the Sbt GAS index of the primitive associated with the current intersection.

In IS and AH this corresponds to the currently intersected primitive. In CH this corresponds to the Sbt GAS index of the closest intersected primitive. In EX with exception code OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_HIT_SBT corresponds to the sbt index within the hit GAS. Returns zero for all other exceptions.

**3.1.2.63 static __forceinline__ __device__ const OptixSRTMotionTransform*
optixGetSRTMotionTransformFromHandle (
OptixTraversableHandle *handle*) [static]**

Returns a pointer to a [OptixSRTMotionTransform](#) from its traversable handle.

Returns 0 if the traversable is not of type `OPTIX_TRANSFORM_TYPE_SRT_MOTION_TRANSFORM`.

3.1.2.64 `static __forceinline__ __device__ const OptixStaticTransform*
optixGetStaticTransformFromHandle (
OptixTraversableHandle handle) [static]`

Returns a pointer to a [OptixStaticTransform](#) from its traversable handle.

Returns 0 if the traversable is not of type `OPTIX_TRANSFORM_TYPE_STATIC_TRANSFORM`.

3.1.2.65 `static __forceinline__ __device__ OptixTraversableHandle
optixGetTransformListHandle (
unsigned int index) [static]`

Returns the traversable handle for a transform on the current transform list.

Only available in IS, AH, CH, EX

3.1.2.66 `static __forceinline__ __device__ unsigned int optixGetTransformListSize ()
[static]`

Returns the number of transforms on the current transform list.

Only available in IS, AH, CH, EX

3.1.2.67 `static __forceinline__ __device__ OptixTransformType optixGetTransformType-
FromHandle (
OptixTraversableHandle handle) [static]`

Returns the transform type of a traversable handle from a transform list.

3.1.2.68 `static __forceinline__ __device__ float2 optixGetTriangleBarycentrics ( ) [static]`

Convenience function that returns the first two attributes as floats.

When using [OptixBuildInputTriangleArray](#) objects, during intersection the barycentric coordinates are stored into the first two attribute registers.

3.1.2.69 `static __forceinline__ __device__ void optixGetTriangleVertexData (
OptixTraversableHandle gas,
unsigned int primIdx,
unsigned int sbtGASIndex,
float time,
float3 data[3]) [static]`

Return the object space triangle vertex positions of a given triangle in a Geometry Acceleration Structure (GAS) at a given motion time.

If motion is disabled via [OptixPipelineCompileOptions::usesMotionBlur](#), or the GAS does not contain motion, the time parameter is ignored.

3.1.2.70 static __forceinline__ __device__ float3 optixGetWorldRayDirection () [static]

Returns the rayDirection passed into optixTrace.

May be more expensive to call in IS and AH than their object space counterparts, so effort should be made to use the object space ray in those programs. Only available in IS, AH, CH, MS

3.1.2.71 static __forceinline__ __device__ float3 optixGetWorldRayOrigin () [static]

Returns the rayOrigin passed into optixTrace.

May be more expensive to call in IS and AH than their object space counterparts, so effort should be made to use the object space ray in those programs. Only available in IS, AH, CH, MS

3.1.2.72 static __forceinline__ __device__ void optixGetWorldToObjectTransformMatrix (float m[12]) [static]

Returns the world-to-object transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

3.1.2.73 static __forceinline__ __device__ void optixIgnoreIntersection () [static]

Discards the hit, and returns control to the calling optixReportIntersection or built-in intersection routine.

Available only in AH.

3.1.2.74 static __forceinline__ __device__ bool optixIsBackFaceHit (unsigned int hitKind) [static]

Function interpreting the result of [optixGetHitKind\(\)](#).

3.1.2.75 static __forceinline__ __device__ bool optixIsBackFaceHit () [static]

Function interpreting the hit kind associated with the current optixReportIntersection.

3.1.2.76 static __forceinline__ __device__ bool optixIsFrontFaceHit (unsigned int hitKind) [static]

Function interpreting the result of [optixGetHitKind\(\)](#).

3.1.2.77 static __forceinline__ __device__ bool optixIsFrontFaceHit () [static]

Function interpreting the hit kind associated with the current optixReportIntersection.

3.1.2.78 static __forceinline__ __device__ bool optixIsTriangleBackFaceHit () [static]

Convenience function interpreting the result of [optixGetHitKind\(\)](#).

3.1.2.79 static __forceinline__ __device__ bool optixIsTriangleFrontFaceHit () [static]

Convenience function interpreting the result of [optixGetHitKind\(\)](#).

3.1.2.80 **static __forceinline__ __device__ bool optixIsTriangleHit () [static]**

Convenience function interpreting the result of [optixGetHitKind\(\)](#).

3.1.2.81 **static __forceinline__ __device__ bool optixReportIntersection (float *hitT*, unsigned int *hitKind*) [static]**

Reports an intersections (overload without attributes).

If [optixGetRayTmin\(\)](#) <= *hitT* <= [optixGetRayTmax\(\)](#), the any hit program associated with this intersection program (via the SBT entry) is called. The AH program can do one of three things:

1. call [optixIgnoreIntersection](#) - no hit is recorded, [optixReportIntersection](#) returns false
2. call [optixTerminateRay](#) - hit is recorded, [optixReportIntersection](#) does not return, no further traversal occurs, and the associated closest hit program is called
3. neither - hit is recorded, [optixReportIntersection](#) returns true *hitKind* - Only the 7 least significant bits should be written [0..127]. Any values above 127 are reserved for built in intersection. The value can be queried with [optixGetHitKind\(\)](#) in AH and CH.

The attributes specified with *a0*..*a7* are available in the AH and CH programs. Note that the attributes available in the CH program correspond to the closest recorded intersection. The number of attributes in registers and memory can be configured in the pipeline.

Parameters

in	<i>hitT</i>	
in	<i>hitKind</i>	

3.1.2.82 **static __forceinline__ __device__ bool optixReportIntersection (float *hitT*, unsigned int *hitKind*, unsigned int *a0*) [static]**

Reports an intersection (overload with 1 attribute register).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

3.1.2.83 **static __forceinline__ __device__ bool optixReportIntersection (float *hitT*, unsigned int *hitKind*, unsigned int *a0*, unsigned int *a1*) [static]**

Reports an intersection (overload with 2 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

```
3.1.2.84 static __forceinline__ __device__ bool optixReportIntersection (  
    float hitT,  
    unsigned int hitKind,  
    unsigned int a0,  
    unsigned int a1,  
    unsigned int a2 ) [static]
```

Reports an intersection (overload with 3 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

```
3.1.2.85 static __forceinline__ __device__ bool optixReportIntersection (  
    float hitT,  
    unsigned int hitKind,  
    unsigned int a0,  
    unsigned int a1,  
    unsigned int a2,  
    unsigned int a3 ) [static]
```

Reports an intersection (overload with 4 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

```
3.1.2.86 static __forceinline__ __device__ bool optixReportIntersection (  
    float hitT,  
    unsigned int hitKind,  
    unsigned int a0,  
    unsigned int a1,  
    unsigned int a2,  
    unsigned int a3,  
    unsigned int a4 ) [static]
```

Reports an intersection (overload with 5 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

3.1.2.87 `static __forceinline__ __device__ bool optixReportIntersection (`
`float hitT,`
`unsigned int hitKind,`
`unsigned int a0,`
`unsigned int a1,`
`unsigned int a2,`
`unsigned int a3,`
`unsigned int a4,`
`unsigned int a5 ) [static]`

Reports an intersection (overload with 6 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

3.1.2.88 `static __forceinline__ __device__ bool optixReportIntersection (`
`float hitT,`
`unsigned int hitKind,`
`unsigned int a0,`
`unsigned int a1,`
`unsigned int a2,`
`unsigned int a3,`
`unsigned int a4,`
`unsigned int a5,`
`unsigned int a6 ) [static]`

Reports an intersection (overload with 7 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

3.1.2.89 `static __forceinline__ __device__ bool optixReportIntersection (`
`float hitT,`
`unsigned int hitKind,`
`unsigned int a0,`
`unsigned int a1,`
`unsigned int a2,`
`unsigned int a3,`
`unsigned int a4,`
`unsigned int a5,`
`unsigned int a6,`
`unsigned int a7 ) [static]`

Reports an intersection (overload with 8 attribute registers).

See Also

[optixReportIntersection\(float,unsigned int\)](#)

3.1.2.90 `static __forceinline__ __device__ void optixSetPayload_0 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 0.

3.1.2.91 `static __forceinline__ __device__ void optixSetPayload_1 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 1.

3.1.2.92 `static __forceinline__ __device__ void optixSetPayload_2 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 2.

3.1.2.93 `static __forceinline__ __device__ void optixSetPayload_3 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 3.

3.1.2.94 `static __forceinline__ __device__ void optixSetPayload_4 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 4.

3.1.2.95 `static __forceinline__ __device__ void optixSetPayload_5 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 5.

3.1.2.96 `static __forceinline__ __device__ void optixSetPayload_6 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 6.

3.1.2.97 `static __forceinline__ __device__ void optixSetPayload_7 ( unsigned int p ) [static]`

Writes the 32-bit payload value at slot 7.

3.1.2.98 `static __forceinline__ __device__ void optixTerminateRay ( ) [static]`

Record the hit, stops traversal, and proceeds to CH.

Available only in AH.

3.1.2.99 `static __forceinline__ __device__ void optixThrowException (`

int *exceptionCode*) [static]

Throws a user exception with the given exception code (overload without exception details).

The exception code must be in the range from 0 to $2^{30} - 1$. Up to 8 optional exception details can be passed. They can be queried in the EX program using [optixGetExceptionDetail_0\(\)](#) to [..._8\(\)](#).

The exception details must not be used to encode pointers to the stack since the current stack is not preserved in the EX program.

Not available in EX.

Parameters

in	<i>exceptionCode</i>	The exception code to be thrown.
----	----------------------	----------------------------------

3.1.2.100 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*) [static]

Throws a user exception with the given exception code (overload with 1 exception detail).

See Also

[optixThrowException\(int\)](#)

3.1.2.101 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*) [static]

Throws a user exception with the given exception code (overload with 2 exception details).

See Also

[optixThrowException\(int\)](#)

3.1.2.102 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*,
unsigned int *exceptionDetail2*) [static]

Throws a user exception with the given exception code (overload with 3 exception details).

See Also

[optixThrowException\(int\)](#)

3.1.2.103 `static __forceinline__ __device__ void optixThrowException (`
 `int exceptionCode,`
 `unsigned int exceptionDetail0,`
 `unsigned int exceptionDetail1,`
 `unsigned int exceptionDetail2,`
 `unsigned int exceptionDetail3 ) [static]`

Throws a user exception with the given exception code (overload with 4 exception details).

See Also

[optixThrowException\(int\)](#)

3.1.2.104 `static __forceinline__ __device__ void optixThrowException (`
 `int exceptionCode,`
 `unsigned int exceptionDetail0,`
 `unsigned int exceptionDetail1,`
 `unsigned int exceptionDetail2,`
 `unsigned int exceptionDetail3,`
 `unsigned int exceptionDetail4 ) [static]`

Throws a user exception with the given exception code (overload with 5 exception details).

See Also

[optixThrowException\(int\)](#)

3.1.2.105 `static __forceinline__ __device__ void optixThrowException (`
 `int exceptionCode,`
 `unsigned int exceptionDetail0,`
 `unsigned int exceptionDetail1,`
 `unsigned int exceptionDetail2,`
 `unsigned int exceptionDetail3,`
 `unsigned int exceptionDetail4,`
 `unsigned int exceptionDetail5 ) [static]`

Throws a user exception with the given exception code (overload with 6 exception details).

See Also

[optixThrowException\(int\)](#)

3.1.2.106 `static __forceinline__ __device__ void optixThrowException (`
 `int exceptionCode,`
 `unsigned int exceptionDetail0,`
 `unsigned int exceptionDetail1,`
 `unsigned int exceptionDetail2,`

```

    unsigned int exceptionDetail3,
    unsigned int exceptionDetail4,
    unsigned int exceptionDetail5,
    unsigned int exceptionDetail6 ) [static]

```

Throws a user exception with the given exception code (overload with 7 exception details).

See Also

[optixThrowException\(int\)](#)

```

3.1.2.107 static __forceinline__ __device__ void optixThrowException (
    int exceptionCode,
    unsigned int exceptionDetail0,
    unsigned int exceptionDetail1,
    unsigned int exceptionDetail2,
    unsigned int exceptionDetail3,
    unsigned int exceptionDetail4,
    unsigned int exceptionDetail5,
    unsigned int exceptionDetail6,
    unsigned int exceptionDetail7 ) [static]

```

Throws a user exception with the given exception code (overload with 8 exception details).

See Also

[optixThrowException\(int\)](#)

```

3.1.2.108 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload without payload).

Parameters

in	<i>handle</i>	
----	---------------	--

Parameters

in	<i>rayOrigin</i>	
in	<i>rayDirection</i>	
in	<i>tmin</i>	
in	<i>tmax</i>	
in	<i>rayTime</i>	
in	<i>visibilityMask</i>	really only 8 bits
in	<i>rayFlags</i>	really only 8 bits, combination of OptixRayFlags
in	<i>SBToffset</i>	really only 8 bits
in	<i>SBTstride</i>	really only 8 bits
in	<i>missSBTIndex</i>	specifies the miss program invoked on a miss

```

3.1.2.109 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBToffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 1 payload register).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsigned int\)](#)

```

3.1.2.110 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,

```

```

    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 2 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsigned int\)](#)

```

3.1.2.111 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 3 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsigned int\)](#)

```

3.1.2.112 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,

```

```

    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 4 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsign](#)

```

3.1.2.113 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3,
    unsigned int & p4 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 5 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsign](#)

```

3.1.2.114 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,

```

```

float tmin,
float tmax,
float rayTime,
OptixVisibilityMask visibilityMask,
unsigned int rayFlags,
unsigned int SBTOffset,
unsigned int SBTstride,
unsigned int missSBTIndex,
unsigned int & p0,
unsigned int & p1,
unsigned int & p2,
unsigned int & p3,
unsigned int & p4,
unsigned int & p5 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 6 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsigned int,unsigned int,unsigned int,unsigned int\)](#)

```

3.1.2.115 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3,
    unsigned int & p4,
    unsigned int & p5,
    unsigned int & p6 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 7 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsigned int\)](#)

```

3.1.2.116 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3,
    unsigned int & p4,
    unsigned int & p5,
    unsigned int & p6,
    unsigned int & p7 ) [static]

```

Initiates a ray tracing query starting with the given traversable (overload with 8 payload registers).

See Also

[optixTrace\(OptixTraversableHandle,float3,float3,float,float,float,OptixVisibilityMask,unsigned int,unsigned int,unsigned int\)](#)

```

3.1.2.117 static __forceinline__ __device__ float3 optixTransformNormalFromObject-
    ToWorldSpace (
        float3 normal ) [static]

```

Transforms the normal using object-to-world transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

```

3.1.2.118 static __forceinline__ __device__ float3 optixTransformNormalFromWorldToOb-
    jectSpace (

```

float3 normal) [static]

Transforms the normal using world-to-object transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

3.1.2.119 static __forceinline__ __device__ float3 optixTransformPointFromObjectToWorldSpace (float3 point) [static]

Transforms the point using object-to-world transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

3.1.2.120 static __forceinline__ __device__ float3 optixTransformPointFromWorldToObjectSpace (float3 point) [static]

Transforms the point using world-to-object transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

3.1.2.121 static __forceinline__ __device__ float3 optixTransformVectorFromObjectToWorldSpace (float3 vec) [static]

Transforms the vector using object-to-world transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

3.1.2.122 static __forceinline__ __device__ float3 optixTransformVectorFromWorldToObjectSpace (float3 vec) [static]

Transforms the vector using world-to-object transformation matrix resulting from the current active transformation list.

The cost of this function may be proportional to the size of the transformation list.

3.1.2.123 static __forceinline__ __device__ unsigned int optixUndefinedValue () [static]

Returns an undefined value.

3.2 Host API

Modules

- [Error handling](#)
- [Device context](#)
- [Pipelines](#)
- [Modules](#)
- [Program groups](#)
- [Launches](#)
- [Acceleration structures](#)
- [Denoiser](#)

3.2.1 Detailed Description

OptiX Host API.

3.3 Error handling

Functions

- `const char * optixGetErrorName (OptixResult result)`
- `const char * optixGetErrorString (OptixResult result)`

3.3.1 Detailed Description

3.3.2 Function Documentation

3.3.2.1 `const char* optixGetErrorName (OptixResult result )`

Returns a string containing the name of an error code in the enum.

Output is a string representation of the enum. For example "OPTIX_SUCCESS" for OPTIX_SUCCESS and "OPTIX_ERROR_INVALID_VALUE" for OPTIX_ERROR_INVALID_VALUE.

If the error code is not recognized, "Unrecognized OptixResult code" is returned.

Parameters

in	<i>result</i>	OptixResult enum to generate string name for
----	---------------	--

See Also

[optixGetErrorString](#)

3.3.2.2 `const char* optixGetErrorString (OptixResult result )`

Returns the description string for an error code.

Output is a string description of the enum. For example "Success" for OPTIX_SUCCESS and "Invalid value" for OPTIX_ERROR_INVALID_VALUE.

If the error code is not recognized, "Unrecognized OptixResult code" is returned.

Parameters

in	<i>result</i>	OptixResult enum to generate string description for
----	---------------	---

See Also

[optixGetErrorName](#)

3.4 Device context

Functions

- `OptixResult optixDeviceContextCreate` (`CUcontext fromContext`, `const OptixDeviceContextOptions *options`, `OptixDeviceContext *context`)
- `OptixResult optixDeviceContextDestroy` (`OptixDeviceContext context`)
- `OptixResult optixDeviceContextGetProperty` (`OptixDeviceContext context`, `OptixDeviceProperty property`, `void *value`, `size_t sizeInBytes`)
- `OptixResult optixDeviceContextSetLogCallback` (`OptixDeviceContext context`, `OptixLogCallback callbackFunction`, `void *callbackData`, `unsigned int callbackLevel`)
- `OptixResult optixDeviceContextSetCacheEnabled` (`OptixDeviceContext context`, `int enabled`)
- `OptixResult optixDeviceContextSetCacheLocation` (`OptixDeviceContext context`, `const char *location`)
- `OptixResult optixDeviceContextSetCacheDatabaseSizes` (`OptixDeviceContext context`, `size_t lowWaterMark`, `size_t highWaterMark`)
- `OptixResult optixDeviceContextGetCacheEnabled` (`OptixDeviceContext context`, `int *enabled`)
- `OptixResult optixDeviceContextGetCacheLocation` (`OptixDeviceContext context`, `char *location`, `size_t locationSize`)
- `OptixResult optixDeviceContextGetCacheDatabaseSizes` (`OptixDeviceContext context`, `size_t *lowWaterMark`, `size_t *highWaterMark`)

3.4.1 Detailed Description

3.4.2 Function Documentation

3.4.2.1 `OptixResult optixDeviceContextCreate` (`CUcontext fromContext`, `const OptixDeviceContextOptions * options`, `OptixDeviceContext * context`)

Create a device context associated with the CUDA context specified with 'fromContext'.

If zero is specified for 'fromContext', OptiX will use the current CUDA context. The CUDA context should be initialized before calling `optixDeviceContextCreate`.

Parameters

in	<i>fromContext</i>	
in	<i>options</i>	
out	<i>context</i>	

Returns

- `OPTIX_ERROR_CUDA_NOT_INITIALIZED` If using zero for 'fromContext' and CUDA has not been initialized yet on the calling thread.
- `OPTIX_ERROR_CUDA_ERROR` CUDA operation failed.

- OPTIX_ERROR_HOST_OUT_OF_MEMORY Heap allocation failed.
- OPTIX_ERROR_INTERNAL_ERROR Internal error

3.4.2.2 OptixResult optixDeviceContextDestroy (OptixDeviceContext *context*)

Destroys all CPU and GPU state associated with the device.

It will attempt to block on CUDA streams that have launch work outstanding.

Any API objects, such as OptixModule and OptixPipeline, not already destroyed will be destroyed.

Thread safety: A device context must not be destroyed while it is still in use by concurrent API calls in other threads.

3.4.2.3 OptixResult optixDeviceContextGetCacheDatabaseSizes (OptixDeviceContext *context*, size_t * *lowWaterMark*, size_t * *highWaterMark*)

Returns the low and high water marks for disk cache garbage collection.

Parameters

in	<i>context</i>	the device context
out	<i>lowWaterMark</i>	the low water mark
out	<i>highWaterMark</i>	the high water mark

3.4.2.4 OptixResult optixDeviceContextGetCacheEnabled (OptixDeviceContext *context*, int * *enabled*)

Indicates whether the disk cache is enabled or disabled.

Parameters

in	<i>context</i>	the device context
out	<i>enabled</i>	1 if enabled, 0 if disabled

3.4.2.5 OptixResult optixDeviceContextGetCacheLocation (OptixDeviceContext *context*, char * *location*, size_t *locationSize*)

Returns the location of the disk cache.

Parameters

in	<i>context</i>	the device context
out	<i>location</i>	directory of disk cache, null terminated if locationSize > 0
in	<i>locationSize</i>	locationSize

3.4.2.6 OptixResult optixDeviceContextGetProperty (

OptixDeviceContext *context*,
OptixDeviceProperty *property*,
void * *value*,
size_t *sizeInBytes*)

Query properties of a device context.

Parameters

in	<i>context</i>	the device context to query the property for
in	<i>property</i>	the property to query
out	<i>value</i>	pointer to the returned
in	<i>sizeInBytes</i>	size of output

3.4.2.7 OptixResult optixDeviceContextSetCacheDatabaseSizes (

OptixDeviceContext *context*,
size_t *lowWaterMark*,
size_t *highWaterMark*)

Sets the low and high water marks for disk cache garbage collection.

Garbage collection is triggered when a new entry is written to the cache and the current cache data size plus the size of the cache entry that is about to be inserted exceeds the high water mark. Garbage collection proceeds until the size reaches the low water mark. Garbage collection will always free enough space to insert the new entry without exceeding the low water mark. Setting either limit to zero will disable garbage collection. An error will be returned if both limits are non-zero and the high water mark is smaller than the low water mark.

Note that garbage collection is performed only on writes to the disk cache. No garbage collection is triggered on disk cache initialization or immediately when calling this function, but on subsequent inserting of data into the database.

If the size of a compiled module exceeds the value configured for the high water mark and garbage collection is enabled, the module will not be added to the cache and a warning will be added to the log.

Parameters

in	<i>context</i>	the device context
in	<i>lowWaterMark</i>	the low water mark

Parameters

in	<i>highWaterMark</i>	the high water mark
----	----------------------	---------------------

3.4.2.8 **OptixResult optixDeviceContextSetCacheEnabled (** **OptixDeviceContext *context*,** **int *enabled*)**

Enables or disables the disk cache.

If caching was previously disabled, enabling it will attempt to initialize the disk cache database using the currently configured cache location. An error will be returned if initialization fails.

Note that no in-memory cache is used, so no caching behavior will be observed if the disk cache is disabled.

Parameters

in	<i>context</i>	the device context
in	<i>enabled</i>	1 to enabled, 0 to disable

3.4.2.9 **OptixResult optixDeviceContextSetCacheLocation (** **OptixDeviceContext *context*,** **const char * *location*)**

Sets the location of the disk cache.

The location is specified by a directory. This directory should not be used for other purposes and will be created if it does not exist. An error will be returned if it is not possible to create the disk cache at the specified location for any reason (e.g., the path is invalid or the directory is not writable). Caching will be disabled if the disk cache cannot be initialized in the new location. If caching is disabled, no error will be returned until caching is enabled. If the disk cache is located on a network file share, behavior is undefined.

The location of the disk cache can be overridden with the environment variable OPTIX_CACHE_PATH. The environment variable takes precedence over this setting.

The default location depends on the operating system:

- Windows: LOCALAPPDATA%\NVIDIA\OptixCache
- Linux: /var/tmp/OptixCache_<username> (or /tmp/OptixCache_<username> if the first choice is not usable), the underscore and username suffix are omitted if the username cannot be obtained
- MacOS X: /Library/Application Support/NVIDIA/OptixCache

Parameters

in	<i>context</i>	the device context
in	<i>location</i>	directory of disk cache

3.4.2.10 OptixResult optixDeviceContextSetLogCallback (

OptixDeviceContext *context*,
OptixLogCallback *callbackFunction*,
void * *callbackData*,
unsigned int *callbackLevel*)

Sets the current log callback method.

See [OptixLogCallback](#) for more details.

Thread safety: It is guaranteed that the callback itself (*callbackFunction* and *callbackData*) are updated atomically. It is not guaranteed that the callback itself (*callbackFunction* and *callbackData*) and the *callbackLevel* are updated atomically. It is unspecified when concurrent API calls using the same context start to make use of the new callback method.

Parameters

in	<i>context</i>	the device context
in	<i>callbackFunction</i>	the callback function to call
in	<i>callbackData</i>	pointer to data passed to callback function while invoking it
in	<i>callbackLevel</i>	callback level

3.5 Pipelines

Functions

- `OptixResult optixPipelineCreate` (`OptixDeviceContext` context, const `OptixPipelineCompileOptions` *pipelineCompileOptions, const `OptixPipelineLinkOptions` *pipelineLinkOptions, const `OptixProgramGroup` *programGroups, unsigned int numProgramGroups, char *logString, size_t *logStringSize, `OptixPipeline` *pipeline)
- `OptixResult optixPipelineDestroy` (`OptixPipeline` pipeline)
- `OptixResult optixPipelineSetStackSize` (`OptixPipeline` pipeline, unsigned int directCallableStackSizeFromTraversal, unsigned int directCallableStackSizeFromState, unsigned int continuationStackSize, unsigned int maxTraversableGraphDepth)

3.5.1 Detailed Description

3.5.2 Function Documentation

3.5.2.1 `OptixResult optixPipelineCreate` (

`OptixDeviceContext` context,
const `OptixPipelineCompileOptions` * pipelineCompileOptions,
const `OptixPipelineLinkOptions` * pipelineLinkOptions,
const `OptixProgramGroup` * programGroups,
unsigned int numProgramGroups,
char * logString,
size_t * logStringSize,
`OptixPipeline` * pipeline)

logString is an optional buffer that contains compiler feedback and errors. This information is also passed to the context logger (if enabled), however it may be difficult to correlate output to the logger to specific API invocations when using multiple threads. The output to logString will only contain feedback for this specific invocation of this API call.

logStringSize as input should be a pointer to the number of bytes backing logString. Upon return it contains the length of the log message (including the null terminator) which may be greater than the input value. In this case, the log message will be truncated to fit into logString.

If logString or logStringSize are NULL, no output is written to logString. If logStringSize points to a value that is zero, no output is written. This does not affect output to the context logger if enabled.

Parameters

in	context	
in	pipelineCompileOptions	
in	pipelineLinkOptions	
in	programGroups	array of ProgramGroup objects
in	numProgramGroups	number of ProgramGroup objects

Parameters

out	<i>logString</i>	Information will be written to this string. If <i>logStringSize</i> > 0 <i>logString</i> will be null terminated.
in,out	<i>logStringSize</i>	
out	<i>pipeline</i>	

3.5.2.2 OptixResult optixPipelineDestroy (OptixPipeline *pipeline*)

Thread safety: A pipeline must not be destroyed while it is still in use by concurrent API calls in other threads.

3.5.2.3 OptixResult optixPipelineSetStackSize (OptixPipeline *pipeline*, unsigned int *directCallableStackSizeFromTraversal*, unsigned int *directCallableStackSizeFromState*, unsigned int *continuationStackSize*, unsigned int *maxTraversableGraphDepth*)

Sets the stack sizes for a pipeline.

Users are encouraged to see the programming guide and the implementations of the helper functions to understand how to construct the stack sizes based on their particular needs.

If this method is not used, an internal default implementation is used. The default implementation is correct (but not necessarily optimal) as long as the maximum depth of call trees of CC and DC programs is at most 2 and no motion transforms are used.

The *maxTraversableGraphDepth* responds to the maximal number of traversables visited when calling trace. Every acceleration structure and motion transform count as one level of traversal. E.g., for a simple IAS (instance acceleration structure) -> GAS (geometry acceleration structure) traversal graph, the *maxTraversableGraphDepth* is two. For IAS -> MT (motion transform) -> GAS, the *maxTraversableGraphDepth* is three. Note that it does not matter whether a IAS or GAS has motion or not, it always counts as one. Launching optix with exceptions turned on (see [OPTIX_EXCEPTION_FLAG_TRACE_DEPTH](#)) will throw an exception if the specified *maxTraversableGraphDepth* is too small.

Parameters

in	<i>pipeline</i>	The pipeline to configure the stack size for.
in	<i>directCallableStackSizeFromTraversal</i>	The direct stack size requirement for direct callables invoked from IS or AH.
in	<i>directCallableStackSizeFromState</i>	The direct stack size requirement for direct callables invoked from RG, MS, or CH.
in	<i>continuationStackSize</i>	The continuation stack requirement.
in	<i>maxTraversableGraphDepth</i>	The maximum depth of a traversable graph passed to trace.

3.6 Modules

Functions

- `OptixResult optixModuleCreateFromPTX` (`OptixDeviceContext` context, const `OptixModuleCompileOptions` *moduleCompileOptions, const `OptixPipelineCompileOptions` *pipelineCompileOptions, const char *PTX, size_t PTXsize, char *logString, size_t *logStringSize, `OptixModule` *module)
- `OptixResult optixModuleDestroy` (`OptixModule` module)
- `OptixResult optixBuiltinISModuleGet` (`OptixDeviceContext` context, const `OptixModuleCompileOptions` *moduleCompileOptions, const `OptixPipelineCompileOptions` *pipelineCompileOptions, const `OptixBuiltinISOptions` *builtinISOptions, `OptixModule` *builtinModule)

3.6.1 Detailed Description

3.6.2 Function Documentation

3.6.2.1 `OptixResult optixBuiltinISModuleGet` (

`OptixDeviceContext` context,
const `OptixModuleCompileOptions` * *moduleCompileOptions*,
const `OptixPipelineCompileOptions` * *pipelineCompileOptions*,
const `OptixBuiltinISOptions` * *builtinISOptions*,
`OptixModule` * *builtinModule*)

Returns a module containing the intersection program for the built-in primitive type specified by the `builtinISOptions`. This module must be used as the moduleIS for the `OptixProgramGroupHitgroup` in any SBT record for that primitive type. (The `entryFunctionNameIS` should be null.)

3.6.2.2 `OptixResult optixModuleCreateFromPTX` (

`OptixDeviceContext` context,
const `OptixModuleCompileOptions` * *moduleCompileOptions*,
const `OptixPipelineCompileOptions` * *pipelineCompileOptions*,
const char * *PTX*,
size_t *PTXsize*,
char * *logString*,
size_t * *logStringSize*,
`OptixModule` * *module*)

`logString` is an optional buffer that contains compiler feedback and errors. This information is also passed to the context logger (if enabled), however it may be difficult to correlate output to the logger to specific API invocations when using multiple threads. The output to `logString` will only contain feedback for this specific invocation of this API call.

`logStringSize` as input should be a pointer to the number of bytes backing `logString`. Upon return it contains the length of the log message (including the null terminator) which may be greater than the input value. In this case, the log message will be truncated to fit into `logString`.

If `logString` or `logStringSize` are NULL, no output is written to `logString`. If `logStringSize` points to a value that is zero, no output is written. This does not affect output to the context logger if enabled.

Parameters

in	<i>context</i>	
in	<i>moduleCompileOptions</i>	
in	<i>pipelineCompileOptions</i>	All modules in a pipeline need to use the same values for the pipeline compile options.
in	<i>PTX</i>	Pointer to the PTX input string.
in	<i>PTXsize</i>	Parsing proceeds up to <code>PTXsize</code> characters, or the first NUL byte, whichever occurs first.
out	<i>logString</i>	Information will be written to this string. If <code>logStringSize > 0</code> <code>logString</code> will be null terminated.
in,out	<i>logStringSize</i>	
out	<i>module</i>	

Returns

OPTIX_ERROR_INVALID_VALUE - `context` is 0, `moduleCompileOptions` is 0, `pipelineCompileOptions` is 0, `PTX` is 0, `module` is 0.

3.6.2.3 OptixResult optixModuleDestroy (OptixModule *module*)

Call for `OptixModule` objects created with `optixModuleCreateFromPTX` and `optixModuleDeserialize`.

Modules must not be destroyed while they are still used by any program group.

Thread safety: A module must not be destroyed while it is still in use by concurrent API calls in other threads.

3.7 Program groups

Functions

- `OptixResult optixProgramGroupGetStackSize (OptixProgramGroup programGroup, OptixStackSizes *stackSizes)`
- `OptixResult optixProgramGroupCreate (OptixDeviceContext context, const OptixProgramGroupDesc *programDescriptions, unsigned int numProgramGroups, const OptixProgramGroupOptions *options, char *logString, size_t *logStringSize, OptixProgramGroup *programGroups)`
- `OptixResult optixProgramGroupDestroy (OptixProgramGroup programGroup)`

3.7.1 Detailed Description

3.7.2 Function Documentation

3.7.2.1 OptixResult optixProgramGroupCreate (
OptixDeviceContext *context*,
const OptixProgramGroupDesc * *programDescriptions*,
unsigned int *numProgramGroups*,
const OptixProgramGroupOptions * *options*,
char * *logString*,
size_t * *logStringSize*,
OptixProgramGroup * *programGroups*)

`logString` is an optional buffer that contains compiler feedback and errors. This information is also passed to the context logger (if enabled), however it may be difficult to correlate output to the logger to specific API invocations when using multiple threads. The output to `logString` will only contain feedback for this specific invocation of this API call.

`logStringSize` as input should be a pointer to the number of bytes backing `logString`. Upon return it contains the length of the log message (including the null terminator) which may be greater than the input value. In this case, the log message will be truncated to fit into `logString`.

If `logString` or `logStringSize` are NULL, no output is written to `logString`. If `logStringSize` points to a value that is zero, no output is written. This does not affect output to the context logger if enabled.

Creates `numProgramGroups` `OptiXProgramGroup` objects from the specified `OptixProgramGroupDesc` array. The size of the arrays must match.

Parameters

in	<i>context</i>	
in	<i>programDescriptions</i>	N * <code>OptixProgramGroupDesc</code>
in	<i>numProgramGroups</i>	N
in	<i>options</i>	
out	<i>logString</i>	Information will be written to this string. If <code>logStringSize > 0</code> <code>logString</code> will be null terminated.

Parameters

in, out	<i>logStringSize</i>	
out	<i>programGroups</i>	

3.7.2.2 OptixResult optixProgramGroupDestroy (
OptixProgramGroup *programGroup*)

Thread safety: A program group must not be destroyed while it is still in use by concurrent API calls in other threads.

3.7.2.3 OptixResult optixProgramGroupGetStackSize (
OptixProgramGroup *programGroup*,
OptixStackSizes * *stackSizes*)

Returns the stack sizes for the given program group.

Parameters

in	<i>programGroup</i>	the program group
out	<i>stackSizes</i>	the corresponding stack sizes

3.8 Launches

Functions

- [OptixResult optixLaunch](#) ([OptixPipeline](#) pipeline, [CUstream](#) stream, [CUdeviceptr](#) pipelineParams, [size_t](#) pipelineParamsSize, const [OptixShaderBindingTable](#) *sbt, unsigned int width, unsigned int height, unsigned int depth)
- [OptixResult optixSbtRecordPackHeader](#) ([OptixProgramGroup](#) programGroup, void *sbtRecordHeaderHostPointer)

3.8.1 Detailed Description

3.8.2 Function Documentation

3.8.2.1 OptixResult optixLaunch (
 [OptixPipeline](#) *pipeline*,
 [CUstream](#) *stream*,
 [CUdeviceptr](#) *pipelineParams*,
 [size_t](#) *pipelineParamsSize*,
 const [OptixShaderBindingTable](#) * *sbt*,
 unsigned int *width*,
 unsigned int *height*,
 unsigned int *depth*)

Where the magic happens.

The stream and pipeline must belong to the same device context. Multiple launches may be issues in parallel from multiple threads to different streams.

pipelineParamsSize number of bytes are copied from the device memory pointed to by pipelineParams before launch. It is an error if pipelineParamsSize is greater than the size of the variable declared in modules and identified by [OptixPipelineCompileOptions::pipelineLaunchParamsVariableName](#). If the launch params variable was optimized out or not found in the modules linked to the pipeline then the pipelineParams and pipelineParamsSize parameters are ignored.

sbt points to the shader binding table, which defines shader groupings and their resources. See the SBT spec.

Parameters

in	<i>pipeline</i>	
in	<i>stream</i>	
in	<i>pipelineParams</i>	
in	<i>pipelineParamsSize</i>	
in	<i>sbt</i>	
in	<i>width</i>	number of elements to compute
in	<i>height</i>	number of elements to compute

Parameters

in	<i>depth</i>	number of elements to compute
----	--------------	-------------------------------

Thread safety: In the current implementation concurrent launches to the same pipeline are not supported. Concurrent launches require separate OptixPipeline objects.

3.8.2.2 OptixResult optixSbtRecordPackHeader (
OptixProgramGroup *programGroup*,
void * *sbtRecordHeaderHostPointer*)**Parameters**

in	<i>programGroup</i>	the program group containing the program(s)
out	<i>sbtRecordHeaderHostPointer</i>	the result sbt record header

3.9 Acceleration structures

Functions

- `OptixResult optixAccelComputeMemoryUsage` (`OptixDeviceContext` context, const `OptixAccelBuildOptions` *accelOptions, const `OptixBuildInput` *buildInputs, unsigned int numBuildInputs, `OptixAccelBufferSizes` *bufferSizes)
- `OptixResult optixAccelBuild` (`OptixDeviceContext` context, `CUstream` stream, const `OptixAccelBuildOptions` *accelOptions, const `OptixBuildInput` *buildInputs, unsigned int numBuildInputs, `CUdeviceptr` tempBuffer, `size_t` tempBufferSizeInBytes, `CUdeviceptr` outputBuffer, `size_t` outputBufferSizeInBytes, `OptixTraversableHandle` *outputHandle, const `OptixAccelEmitDesc` *emittedProperties, unsigned int numEmittedProperties)
- `OptixResult optixAccelGetRelocationInfo` (`OptixDeviceContext` context, `OptixTraversableHandle` handle, `OptixAccelRelocationInfo` *info)
- `OptixResult optixAccelCheckRelocationCompatibility` (`OptixDeviceContext` context, const `OptixAccelRelocationInfo` *info, int *compatible)
- `OptixResult optixAccelRelocate` (`OptixDeviceContext` context, `CUstream` stream, const `OptixAccelRelocationInfo` *info, `CUdeviceptr` instanceTraversableHandles, `size_t` numInstanceTraversableHandles, `CUdeviceptr` targetAccel, `size_t` targetAccelSizeInBytes, `OptixTraversableHandle` *targetHandle)
- `OptixResult optixAccelCompact` (`OptixDeviceContext` context, `CUstream` stream, `OptixTraversableHandle` inputHandle, `CUdeviceptr` outputBuffer, `size_t` outputBufferSizeInBytes, `OptixTraversableHandle` *outputHandle)
- `OptixResult optixConvertPointerToTraversableHandle` (`OptixDeviceContext` onDevice, `CUdeviceptr` pointer, `OptixTraversableType` traversableType, `OptixTraversableHandle` *traversableHandle)

3.9.1 Detailed Description

3.9.2 Function Documentation

3.9.2.1 `OptixResult optixAccelBuild` (

`OptixDeviceContext` *context*,
`CUstream` *stream*,
const `OptixAccelBuildOptions` * *accelOptions*,
const `OptixBuildInput` * *buildInputs*,
unsigned int *numBuildInputs*,
`CUdeviceptr` *tempBuffer*,
`size_t` *tempBufferSizeInBytes*,
`CUdeviceptr` *outputBuffer*,
`size_t` *outputBufferSizeInBytes*,
`OptixTraversableHandle` * *outputHandle*,
const `OptixAccelEmitDesc` * *emittedProperties*,
unsigned int *numEmittedProperties*)

Parameters

in	<i>context</i>	
in	<i>stream</i>	
in	<i>accelOptions</i>	accel options
in	<i>buildInputs</i>	an array of OptixBuildInput objects
in	<i>numBuildInputs</i>	must be ≥ 1 for GAS, and $= 1$ for IAS
in	<i>tempBuffer</i>	must be a multiple of <code>OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT</code>
in	<i>tempBufferSizeInBytes</i>	
in	<i>outputBuffer</i>	must be a multiple of <code>OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT</code>
in	<i>outputBufferSizeInBytes</i>	
out	<i>outputHandle</i>	
out	<i>emittedProperties</i>	types of requested properties and output buffers
in	<i>numEmittedProperties</i>	number of post-build properties to populate (may be zero)

3.9.2.2 OptixResult optixAccelCheckRelocationCompatibility (

OptixDeviceContext *context*,
const OptixAccelRelocationInfo * *info*,
int * *compatible*)

Checks if an acceleration structure built using another OptixDeviceContext (that was used to fill in 'info') is compatible with the OptixDeviceContext specified in the 'context' parameter.

Any device is always compatible with itself.

Parameters

in	<i>context</i>	
in	<i>info</i>	
out	<i>compatible</i>	If <code>OPTIX_SUCCESS</code> is returned 'compatible' will have the value of either: • 0: This context is not compatible with acceleration structure data associated with 'info'. • 1: This context is compatible.

3.9.2.3 OptixResult optixAccelCompact (

OptixDeviceContext *context*,
CUstream *stream*,
OptixTraversableHandle *inputHandle*,
CUdeviceptr *outputBuffer*,
size_t *outputBufferSizeInBytes*,

OptixTraversableHandle * *outputHandle*)

After building an acceleration structure, it can be copied in a compacted form to reduce memory. In order to be compacted, `OPTIX_BUILD_FLAG_ALLOW_COMPACTION` must be supplied in `OptixAccelBuildOptions::buildFlags` passed to `optixAccelBuild`.

'outputBuffer' is the pointer to where the compacted acceleration structure will be written. This pointer must be a multiple of `OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT`.

The size of the memory specified in 'outputBufferSizeInBytes' should be at least the value computed using the `OPTIX_PROPERTY_TYPE_COMPACTED_SIZE` that was reported during `optixAccelBuild`.

Parameters

in	<i>context</i>	
in	<i>stream</i>	
in	<i>inputHandle</i>	
in	<i>outputBuffer</i>	
in	<i>outputBufferSizeInBytes</i>	
out	<i>outputHandle</i>	

3.9.2.4 OptixResult optixAccelComputeMemoryUsage (**OptixDeviceContext *context*,** **const OptixAccelBuildOptions * *accelOptions*,** **const OptixBuildInput * *buildInputs*,** **unsigned int *numBuildInputs*,** **OptixAccelBufferSizes * *bufferSizes*)**

Parameters

in	<i>context</i>	device context of the pipeline
in	<i>accelOptions</i>	accel options
in	<i>buildInputs</i>	an array of <code>OptixBuildInput</code> objects
in	<i>numBuildInputs</i>	number of elements in buildInputs (must be at least 1)
out	<i>bufferSizes</i>	fills in buffer sizes

3.9.2.5 OptixResult optixAccelGetRelocationInfo (**OptixDeviceContext *context*,** **OptixTraversableHandle *handle*,** **OptixAccelRelocationInfo * *info*)**

Obtain relocation information, stored in `OptixAccelRelocationInfo`, for a given context and acceleration structure's traversable handle.

The relocation information can be passed to `optixAccelCheckRelocationCompatibility` to determine if an

acceleration structure, referenced by 'handle', can be relocated to a different device's memory space (see [optixAccelCheckRelocationCompatibility](#)).

When used with `optixAccelRelocate`, it provides data necessary for doing the relocation.

If the acceleration structure data associated with 'handle' is copied multiple times, the same [OptixAccelRelocationInfo](#) can also be used on all copies.

Parameters

in	<i>context</i>	
in	<i>handle</i>	
out	<i>info</i>	

Returns

`OPTIX_ERROR_INVALID_VALUE` will be returned for traversable handles that are not from acceleration structure builds.

3.9.2.6 `OptixResult optixAccelRelocate (`
`OptixDeviceContext context,`
`CUstream stream,`
`const OptixAccelRelocationInfo * info,`
`CUdeviceptr instanceTraversableHandles,`
`size_t numInstanceTraversableHandles,`
`CUdeviceptr targetAccel,`
`size_t targetAccelSizeInBytes,`
`OptixTraversableHandle * targetHandle )`

`optixAccelRelocate` is called to update the acceleration structure after it has been relocated. Relocation is necessary when the acceleration structure's location in device memory has changed.

`optixAccelRelocate` does not copy the memory. This function only operates on the relocated memory whose new location is specified by 'targetAccel'. `optixAccelRelocate` also returns the new `OptixTraversableHandle` associated with 'targetAccel'. The original memory (source) is not required to be valid, only the [OptixAccelRelocationInfo](#).

Before copying the data and calling `optixAccelRelocate`, `optixAccelCheckRelocationCompatibility` should be called to ensure the copy will be compatible with the destination device context.

The memory pointed to by 'targetAccel' should be allocated with the same size as the source acceleration. Similar to the 'outputBuffer' used in `optixAccelBuild`, this pointer must be a multiple of `OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT`.

The memory in 'targetAccel' must be allocated as long as the accel is in use.

When relocating an accel that contains instances, 'instanceTraversableHandles' and 'numInstanceTraversableHandles' should be supplied. These are the traversable handles of the instances. These can be used when also relocating the instances. No updates to the bounds are performed. Use `optixAccelBuild` to update the bounds. 'instanceTraversableHandles' and 'numInstanceTraversableHandles' may be zero when relocating bottom level accel (i.e. an accel with no instances).

Parameters

in	<i>context</i>	
in	<i>stream</i>	
in	<i>info</i>	
in	<i>instanceTraversableHandles</i>	
in	<i>numInstanceTraversableHandles</i>	
in	<i>targetAccel</i>	
in	<i>targetAccelSizeInBytes</i>	
out	<i>targetHandle</i>	

3.9.2.7 OptixResult optixConvertPointerToTraversableHandle (
OptixDeviceContext *onDevice*,
CUdeviceptr *pointer*,
OptixTraversableType *traversableType*,
OptixTraversableHandle * *traversableHandle*)

Parameters

in	<i>onDevice</i>	
in	<i>pointer</i>	pointer to traversable allocated in OptixDeviceContext. This pointer must be a multiple of OPTIX_TRANSFORM_BYTE_ALIGNMENT
in	<i>traversableType</i>	Type of OptixTraversableHandle to create
out	<i>traversableHandle</i>	traversable handle. traversableHandle must be in host memory

3.10 Denoiser

Functions

- `OptixResult optixDenoiserCreate` (`OptixDeviceContext` context, const `OptixDenoiserOptions` *options, `OptixDenoiser` *denoiser)
- `OptixResult optixDenoiserSetModel` (`OptixDenoiser` denoiser, `OptixDenoiserModelKind` kind, void *data, size_t sizeInBytes)
- `OptixResult optixDenoiserDestroy` (`OptixDenoiser` denoiser)
- `OptixResult optixDenoiserComputeMemoryResources` (const `OptixDenoiser` denoiser, unsigned int outputWidth, unsigned int outputHeight, `OptixDenoiserSizes` *returnSizes)
- `OptixResult optixDenoiserSetup` (`OptixDenoiser` denoiser, `CUstream` stream, unsigned int inputWidth, unsigned int inputHeight, `CUdeviceptr` denoiserState, size_t denoiserStateSizeInBytes, `CUdeviceptr` scratch, size_t scratchSizeInBytes)
- `OptixResult optixDenoiserInvoke` (`OptixDenoiser` denoiser, `CUstream` stream, const `OptixDenoiserParams` *params, `CUdeviceptr` denoiserState, size_t denoiserStateSizeInBytes, const `OptixImage2D` *inputLayers, unsigned int numInputLayers, unsigned int inputOffsetX, unsigned int inputOffsetY, const `OptixImage2D` *outputLayer, `CUdeviceptr` scratch, size_t scratchSizeInBytes)
- `OptixResult optixDenoiserComputeIntensity` (`OptixDenoiser` denoiser, `CUstream` stream, const `OptixImage2D` *inputImage, `CUdeviceptr` outputIntensity, `CUdeviceptr` scratch, size_t scratchSizeInBytes)

3.10.1 Detailed Description

3.10.2 Function Documentation

3.10.2.1 `OptixResult optixDenoiserComputeIntensity` (

`OptixDenoiser` *denoiser*,
`CUstream` *stream*,
const `OptixImage2D` * *inputImage*,
`CUdeviceptr` *outputIntensity*,
`CUdeviceptr` *scratch*,
size_t *scratchSizeInBytes*)

Computes the logarithmic average intensity of the given image. The returned value 'outputIntensity' is multiplied with the RGB values of the input image/tile in `optixDenoiserInvoke` if given in the parameter `OptixDenoiserParams::hdrIntensity` (otherwise 'hdrIntensity' must be a null pointer). This is useful for denoising HDR images which are very dark or bright. When denoising tiles the intensity of the entire image should be computed, i.e. not per tile to get consistent results.

For each RGB pixel in the inputImage the intensity is calculated and summed if it is greater than 1e-8f: $\text{intensity} = \log(r * 0.212586f + g * 0.715170f + b * 0.072200f)$. The function returns $0.18 / \exp(\text{sum of intensities} / \text{number of summed pixels})$. More details could be found in the Reinhard tonemapping paper: http://www.cmap.polytechnique.fr/~peyre/cours/x2005signal/hdr_photographic.pdf

This function needs scratch memory with a size of at least `sizeof( int ) * ( 2 + inputImage::width * inputImage::height )`. When denoising entire images (no tiling) the same scratch memory as passed to

optixDenoiserInvoke could be used.

Parameters

in	<i>denoiser</i>	
in	<i>stream</i>	
in	<i>inputImage</i>	
out	<i>outputIntensity</i>	single float
in	<i>scratch</i>	
in	<i>scratchSizeInBytes</i>	

3.10.2.2 OptixResult optixDenoiserComputeMemoryResources (

const OptixDenoiser *denoiser*,
unsigned int *outputWidth*,
unsigned int *outputHeight*,
OptixDenoiserSizes * *returnSizes*)

Computes the GPU memory resources required to execute the denoiser.

Memory for state and scratch buffers must be allocated with the sizes in 'returnSizes' and scratch memory passed to optixDenoiserSetup, optixDenoiserInvoke and optixDenoiserComputeIntensity. For tiled denoising an overlap area must be added to each tile on all sides which increases the amount of memory needed to denoise a tile. In case of tiling use withOverlapScratchSizeInBytes. If only full resolution images are denoised, withoutOverlapScratchSizeInBytes can be used which is always smaller than withOverlapScratchSizeInBytes.

'outputWidth' and 'outputHeight' is the dimension of the image to be denoised (without overlap in case tiling is being used). 'outputWidth' and 'outputHeight' must be greater than or equal to the dimensions passed to optixDenoiserSetup.

Parameters

in	<i>denoiser</i>	
in	<i>outputWidth</i>	
in	<i>outputHeight</i>	
out	<i>returnSizes</i>	

3.10.2.3 OptixResult optixDenoiserCreate (

OptixDeviceContext *context*,
const OptixDenoiserOptions * *options*,
OptixDenoiser * *denoiser*)

Creates a denoiser object with the given options.

Parameters

in	<i>context</i>	
in	<i>options</i>	
out	<i>denoiser</i>	

3.10.2.4 OptixResult optixDenoiserDestroy (
OptixDenoiser *denoiser*)

Destroys the denoiser object and any associated host resources.

3.10.2.5 OptixResult optixDenoiserInvoke (
OptixDenoiser *denoiser*,
CUstream *stream*,
const OptixDenoiserParams * *params*,
CUdeviceptr *denoiserState*,
size_t *denoiserStateSizeInBytes*,
const OptixImage2D * *inputLayers*,
unsigned int *numInputLayers*,
unsigned int *inputOffsetX*,
unsigned int *inputOffsetY*,
const OptixImage2D * *outputLayer*,
CUdeviceptr *scratch*,
size_t *scratchSizeInBytes*)

Invokes denoiser on a set of input data and produces one output image. State memory must be available during the execution of the denoiser (or until `optixDenoiserSetup` is called with a new state memory pointer). Scratch memory passed is used only for the duration of this function. Scratch and state memory sizes must have a size greater than or equal to the sizes as returned by `optixDenoiserComputeMemoryResources`.

'inputOffsetX' and 'inputOffsetY' are pixel offsets in the 'inputLayers' image specifying the beginning of the image without overlap. When denoising an entire image without tiling there is no overlap and 'inputOffsetX' and 'inputOffsetY' must be zero. When denoising a tile which is adjacent to one of the four sides of the entire image the corresponding offsets must also be zero since there is no overlap at the side adjacent to the image border.

Parameters

in	<i>denoiser</i>	
in	<i>stream</i>	
in	<i>params</i>	
in	<i>denoiserState</i>	
in	<i>denoiserStateSizeInBytes</i>	
in	<i>inputLayers</i>	

Parameters

in	<i>numInputLayers</i>	
in	<i>inputOffsetX</i>	
in	<i>inputOffsetY</i>	
in	<i>outputLayer</i>	
in	<i>scratch</i>	
in	<i>scratchSizeInBytes</i>	

3.10.2.6 OptixResult optixDenoiserSetModel (

OptixDenoiser *denoiser*,
OptixDenoiserModelKind *kind*,
void * *data*,
size_t *sizeInBytes*)

Sets the model of the denoiser.

If the kind is OPTIX_DENOISER_MODEL_KIND_USER, then the data and sizeInBytes must not be null and zero respectively. For other kinds, these parameters must be zero.

Parameters

in	<i>denoiser</i>	
in	<i>kind</i>	
in	<i>data</i>	
in	<i>sizeInBytes</i>	

3.10.2.7 OptixResult optixDenoiserSetup (

OptixDenoiser *denoiser*,
CUstream *stream*,
unsigned int *inputWidth*,
unsigned int *inputHeight*,
CUdeviceptr *denoiserState*,
size_t *denoiserStateSizeInBytes*,
CUdeviceptr *scratch*,
size_t *scratchSizeInBytes*)

Initializes the state required by the denoiser.

'inputWidth' and 'inputHeight' must include overlap on both sides of the image if tiling is being used. The overlap is returned by [optixDenoiserComputeMemoryResources](#). For subsequent calls to [optixDenoiserInvoke](#) 'inputWidth' and 'inputHeight' are the maximum dimensions of the input layers. Dimensions of the input layers passed to [optixDenoiserInvoke](#) may be different in each invocation however they always must be smaller than 'inputWidth' and 'inputHeight' passed to [optixDenoiserSetup](#).

Parameters

in	<i>denoiser</i>	
in	<i>stream</i>	
in	<i>inputWidth</i>	
in	<i>inputHeight</i>	
in	<i>denoiserState</i>	
in	<i>denoiserStateSizeInBytes</i>	
in	<i>scratch</i>	
in	<i>scratchSizeInBytes</i>	

3.11 Types

Classes

- struct OptixDeviceContextOptions
- struct OptixBuildInputTriangleArray
- struct OptixBuildInputCurveArray
- struct OptixAabb
- struct OptixBuildInputCustomPrimitiveArray
- struct OptixBuildInputInstanceArray
- struct OptixBuildInput
- struct OptixInstance
- struct OptixMotionOptions
- struct OptixAccelBuildOptions
- struct OptixAccelBufferSizes
- struct OptixAccelEmitDesc
- struct OptixAccelRelocationInfo
- struct OptixStaticTransform
- struct OptixMatrixMotionTransform
- struct OptixSRTData
- struct OptixSRTMotionTransform
- struct OptixImage2D
- struct OptixDenoiserOptions
- struct OptixDenoiserParams
- struct OptixDenoiserSizes
- struct OptixModuleCompileOptions
- struct OptixProgramGroupSingleModule
- struct OptixProgramGroupHitgroup
- struct OptixProgramGroupCallables
- struct OptixProgramGroupDesc
- struct OptixProgramGroupOptions
- struct OptixPipelineCompileOptions
- struct OptixPipelineLinkOptions
- struct OptixShaderBindingTable
- struct OptixStackSizes
- struct OptixBuiltinISOOptions

Macros

- #define OPTIX_SBT_RECORD_HEADER_SIZE ((size_t)32)
- #define OPTIX_SBT_RECORD_ALIGNMENT 16ull
- #define OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT 128ull
- #define OPTIX_INSTANCE_BYTE_ALIGNMENT 16ull
- #define OPTIX_AABB_BUFFER_BYTE_ALIGNMENT 8ull
- #define OPTIX_GEOMETRY_TRANSFORM_BYTE_ALIGNMENT 16ull
- #define OPTIX_TRANSFORM_BYTE_ALIGNMENT 64ull
- #define OPTIX_COMPILE_DEFAULT_MAX_REGISTER_COUNT 0

Typedefs

- typedef unsigned long long CUdeviceptr
- typedef struct
OptixDeviceContext_t * OptixDeviceContext
- typedef struct OptixModule_t * OptixModule
- typedef struct
OptixProgramGroup_t * OptixProgramGroup
- typedef struct OptixPipeline_t * OptixPipeline
- typedef struct OptixDenoiser_t * OptixDenoiser
- typedef unsigned long long OptixTraversableHandle
- typedef unsigned int OptixVisibilityMask
- typedef enum OptixResult OptixResult
- typedef enum OptixDeviceProperty OptixDeviceProperty
- typedef void(* OptixLogCallback)(unsigned int level, const char *tag, const char *message, void *cbdata)
- typedef struct
OptixDeviceContextOptions OptixDeviceContextOptions
- typedef enum OptixGeometryFlags OptixGeometryFlags
- typedef enum OptixHitKind OptixHitKind
- typedef enum OptixIndicesFormat OptixIndicesFormat
- typedef enum OptixVertexFormat OptixVertexFormat
- typedef enum OptixTransformFormat OptixTransformFormat
- typedef struct
OptixBuildInputTriangleArray OptixBuildInputTriangleArray
- typedef enum OptixPrimitiveType OptixPrimitiveType
- typedef enum
OptixPrimitiveTypeFlags OptixPrimitiveTypeFlags
- typedef struct
OptixBuildInputCurveArray OptixBuildInputCurveArray
- typedef struct OptixAabb OptixAabb
- typedef struct
OptixBuildInputCustomPrimitiveArray OptixBuildInputCustomPrimitiveArray
- typedef struct
OptixBuildInputInstanceArray OptixBuildInputInstanceArray
- typedef enum OptixBuildInputType OptixBuildInputType
- typedef struct OptixBuildInput OptixBuildInput
- typedef enum OptixInstanceFlags OptixInstanceFlags
- typedef struct OptixInstance OptixInstance
- typedef enum OptixBuildFlags OptixBuildFlags
- typedef enum OptixBuildOperation OptixBuildOperation
- typedef enum OptixMotionFlags OptixMotionFlags
- typedef struct OptixMotionOptions OptixMotionOptions
- typedef struct
OptixAccelBuildOptions OptixAccelBuildOptions
- typedef struct
OptixAccelBufferSizes OptixAccelBufferSizes

- typedef enum `OptixAccelPropertyType` `OptixAccelPropertyType`
- typedef struct `OptixAccelEmitDesc` `OptixAccelEmitDesc`
- typedef struct
 `OptixAccelRelocationInfo` `OptixAccelRelocationInfo`
- typedef struct `OptixStaticTransform` `OptixStaticTransform`
- typedef struct
 `OptixMatrixMotionTransform` `OptixMatrixMotionTransform`
- typedef struct `OptixSRTData` `OptixSRTData`
- typedef struct
 `OptixSRTMotionTransform` `OptixSRTMotionTransform`
- typedef enum `OptixTraversableType` `OptixTraversableType`
- typedef enum `OptixPixelFormat` `OptixPixelFormat`
- typedef struct `OptixImage2D` `OptixImage2D`
- typedef enum `OptixDenoiserInputKind` `OptixDenoiserInputKind`
- typedef enum `OptixDenoiserModelKind` `OptixDenoiserModelKind`
- typedef struct `OptixDenoiserOptions` `OptixDenoiserOptions`
- typedef struct `OptixDenoiserParams` `OptixDenoiserParams`
- typedef struct `OptixDenoiserSizes` `OptixDenoiserSizes`
- typedef enum `OptixRayFlags` `OptixRayFlags`
- typedef enum `OptixTransformType` `OptixTransformType`
- typedef enum
 `OptixTraversableGraphFlags` `OptixTraversableGraphFlags`
- typedef enum
 `OptixCompileOptimizationLevel` `OptixCompileOptimizationLevel`
- typedef enum `OptixCompileDebugLevel` `OptixCompileDebugLevel`
- typedef struct
 `OptixModuleCompileOptions` `OptixModuleCompileOptions`
- typedef enum `OptixProgramGroupKind` `OptixProgramGroupKind`
- typedef enum `OptixProgramGroupFlags` `OptixProgramGroupFlags`
- typedef struct
 `OptixProgramGroupSingleModule` `OptixProgramGroupSingleModule`
- typedef struct
 `OptixProgramGroupHitgroup` `OptixProgramGroupHitgroup`
- typedef struct
 `OptixProgramGroupCallables` `OptixProgramGroupCallables`
- typedef struct
 `OptixProgramGroupDesc` `OptixProgramGroupDesc`
- typedef struct
 `OptixProgramGroupOptions` `OptixProgramGroupOptions`
- typedef enum `OptixExceptionCodes` `OptixExceptionCodes`
- typedef enum `OptixExceptionFlags` `OptixExceptionFlags`
- typedef struct
 `OptixPipelineCompileOptions` `OptixPipelineCompileOptions`
- typedef struct
 `OptixPipelineLinkOptions` `OptixPipelineLinkOptions`
- typedef struct
 `OptixShaderBindingTable` `OptixShaderBindingTable`

- typedef struct OptixStackSizes OptixStackSizes
- typedef enum
OptixQueryFunctionTableOptions OptixQueryFunctionTableOptions
- typedef OptixResult(OptixQueryFunctionTable_t)(int abild, unsigned int numOptions,
OptixQueryFunctionTableOptions *, const void **, void *functionTable, size_t sizeOfTable)
- typedef struct
OptixBuiltinISOOptions OptixBuiltinISOOptions

Enumerations

- enum OptixResult {
OPTIX_SUCCESS = 0,
OPTIX_ERROR_INVALID_VALUE = 7001,
OPTIX_ERROR_HOST_OUT_OF_MEMORY = 7002,
OPTIX_ERROR_INVALID_OPERATION = 7003,
OPTIX_ERROR_FILE_IO_ERROR = 7004,
OPTIX_ERROR_INVALID_FILE_FORMAT = 7005,
OPTIX_ERROR_DISK_CACHE_INVALID_PATH = 7010,
OPTIX_ERROR_DISK_CACHE_PERMISSION_ERROR = 7011,
OPTIX_ERROR_DISK_CACHE_DATABASE_ERROR = 7012,
OPTIX_ERROR_DISK_CACHE_INVALID_DATA = 7013,
OPTIX_ERROR_LAUNCH_FAILURE = 7050,
OPTIX_ERROR_INVALID_DEVICE_CONTEXT = 7051,
OPTIX_ERROR_CUDA_NOT_INITIALIZED = 7052,
OPTIX_ERROR_INVALID_PTX = 7200,
OPTIX_ERROR_INVALID_LAUNCH_PARAMETER = 7201,
OPTIX_ERROR_INVALID_PAYLOAD_ACCESS = 7202,
OPTIX_ERROR_INVALID_ATTRIBUTE_ACCESS = 7203,
OPTIX_ERROR_INVALID_FUNCTION_USE = 7204,
OPTIX_ERROR_INVALID_FUNCTION_ARGUMENTS = 7205,
OPTIX_ERROR_PIPELINE_OUT_OF_CONSTANT_MEMORY = 7250,
OPTIX_ERROR_PIPELINE_LINK_ERROR = 7251,
OPTIX_ERROR_INTERNAL_COMPILER_ERROR = 7299,
OPTIX_ERROR_DENOISER_MODEL_NOT_SET = 7300,
OPTIX_ERROR_DENOISER_NOT_INITIALIZED = 7301,
OPTIX_ERROR_ACCEL_NOT_COMPATIBLE = 7400,
OPTIX_ERROR_NOT_SUPPORTED = 7800,
OPTIX_ERROR_UNSUPPORTED_ABI_VERSION = 7801,
OPTIX_ERROR_FUNCTION_TABLE_SIZE_MISMATCH = 7802,
OPTIX_ERROR_INVALID_ENTRY_FUNCTION_OPTIONS = 7803,
OPTIX_ERROR_LIBRARY_NOT_FOUND = 7804,
OPTIX_ERROR_ENTRY_SYMBOL_NOT_FOUND = 7805,
OPTIX_ERROR_CUDA_ERROR = 7900,
OPTIX_ERROR_INTERNAL_ERROR = 7990,
OPTIX_ERROR_UNKNOWN = 7999 }
- enum OptixDeviceProperty {
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_TRACE_DEPTH = 0x2001,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_TRAVERSABLE_GRAPH_DEPTH = 0x2002,

```

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_PRIMITIVES_PER_GAS = 0x2003,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCES_PER_IAS = 0x2004,
OPTIX_DEVICE_PROPERTY_RTCORE_VERSION = 0x2005,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCE_ID = 0x2006,
OPTIX_DEVICE_PROPERTY_LIMIT_NUM_BITS_INSTANCE_VISIBILITY_MASK = 0x2007,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_RECORDS_PER_GAS = 0x2008,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_OFFSET = 0x2009 }

• enum OptixGeometryFlags {
    OPTIX_GEOMETRY_FLAG_NONE = 0,
    OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT = 1u << 0,
    OPTIX_GEOMETRY_FLAG_REQUIRE_SINGLE_ANYHIT_CALL = 1u << 1 }

• enum OptixHitKind {
    OPTIX_HIT_KIND_TRIANGLE_FRONT_FACE = 0xFE,
    OPTIX_HIT_KIND_TRIANGLE_BACK_FACE = 0xFF }

• enum OptixIndicesFormat {
    OPTIX_INDICES_FORMAT_NONE = 0,
    OPTIX_INDICES_FORMAT_UNSIGNED_SHORT3 = 0x2102,
    OPTIX_INDICES_FORMAT_UNSIGNED_INT3 = 0x2103 }

• enum OptixVertexFormat {
    OPTIX_VERTEX_FORMAT_NONE = 0,
    OPTIX_VERTEX_FORMAT_FLOAT3 = 0x2121,
    OPTIX_VERTEX_FORMAT_FLOAT2 = 0x2122,
    OPTIX_VERTEX_FORMAT_HALF3 = 0x2123,
    OPTIX_VERTEX_FORMAT_HALF2 = 0x2124,
    OPTIX_VERTEX_FORMAT_SNORM16_3 = 0x2125,
    OPTIX_VERTEX_FORMAT_SNORM16_2 = 0x2126 }

• enum OptixTransformFormat {
    OPTIX_TRANSFORM_FORMAT_NONE = 0,
    OPTIX_TRANSFORM_FORMAT_MATRIX_FLOAT12 = 0x21E1 }

• enum OptixPrimitiveType {
    OPTIX_PRIMITIVE_TYPE_CUSTOM = 0x2500,
    OPTIX_PRIMITIVE_TYPE_ROUND_QUADRATIC_BSPLINE = 0x2501,
    OPTIX_PRIMITIVE_TYPE_ROUND_CUBIC_BSPLINE = 0x2502,
    OPTIX_PRIMITIVE_TYPE_ROUND_LINEAR = 0x2503,
    OPTIX_PRIMITIVE_TYPE_TRIANGLE = 0x2531 }

• enum OptixPrimitiveTypeFlags {
    OPTIX_PRIMITIVE_TYPE_FLAGS_CUSTOM = 1 << 0,
    OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_QUADRATIC_BSPLINE = 1 << 1,
    OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_CUBIC_BSPLINE = 1 << 2,
    OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_LINEAR = 1 << 3,
    OPTIX_PRIMITIVE_TYPE_FLAGS_TRIANGLE = 1 << 31 }

• enum OptixBuildInputType {
    OPTIX_BUILD_INPUT_TYPE_TRIANGLES = 0x2141,
    OPTIX_BUILD_INPUT_TYPE_CUSTOM_PRIMITIVES = 0x2142,
    OPTIX_BUILD_INPUT_TYPE_INSTANCES = 0x2143,
    OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS = 0x2144,
    OPTIX_BUILD_INPUT_TYPE_CURVES = 0x2145 }

• enum OptixInstanceFlags {

```

```

OPTIX_INSTANCE_FLAG_NONE = 0,
OPTIX_INSTANCE_FLAG_DISABLE_TRIANGLE_FACE_CULLING = 1u << 0,
OPTIX_INSTANCE_FLAG_FLIP_TRIANGLE_FACING = 1u << 1,
OPTIX_INSTANCE_FLAG_DISABLE_ANYHIT = 1u << 2,
OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT = 1u << 3,
OPTIX_INSTANCE_FLAG_DISABLE_TRANSFORM = 1u << 6 }
• enum OptixBuildFlags {
    OPTIX_BUILD_FLAG_NONE = 0,
    OPTIX_BUILD_FLAG_ALLOW_UPDATE = 1u << 0,
    OPTIX_BUILD_FLAG_ALLOW_COMPACTION = 1u << 1,
    OPTIX_BUILD_FLAG_PREFER_FAST_TRACE = 1u << 2,
    OPTIX_BUILD_FLAG_PREFER_FAST_BUILD = 1u << 3,
    OPTIX_BUILD_FLAG_ALLOW_RANDOM_VERTEX_ACCESS = 1u << 4 }
• enum OptixBuildOperation {
    OPTIX_BUILD_OPERATION_BUILD = 0x2161,
    OPTIX_BUILD_OPERATION_UPDATE = 0x2162 }
• enum OptixMotionFlags {
    OPTIX_MOTION_FLAG_NONE = 0,
    OPTIX_MOTION_FLAG_START_VANISH = 1u << 0,
    OPTIX_MOTION_FLAG_END_VANISH = 1u << 1 }
• enum OptixAccelPropertyType {
    OPTIX_PROPERTY_TYPE_COMPACTED_SIZE = 0x2181,
    OPTIX_PROPERTY_TYPE_AABBS = 0x2182 }
• enum OptixTraversableType {
    OPTIX_TRAVERSABLE_TYPE_STATIC_TRANSFORM = 0x21C1,
    OPTIX_TRAVERSABLE_TYPE_MATRIX_MOTION_TRANSFORM = 0x21C2,
    OPTIX_TRAVERSABLE_TYPE_SRT_MOTION_TRANSFORM = 0x21C3 }
• enum OptixPixelFormat {
    OPTIX_PIXEL_FORMAT_HALF3 = 0x2201,
    OPTIX_PIXEL_FORMAT_HALF4 = 0x2202,
    OPTIX_PIXEL_FORMAT_FLOAT3 = 0x2203,
    OPTIX_PIXEL_FORMAT_FLOAT4 = 0x2204,
    OPTIX_PIXEL_FORMAT_UCHAR3 = 0x2205,
    OPTIX_PIXEL_FORMAT_UCHAR4 = 0x2206 }
• enum OptixDenoiserInputKind {
    OPTIX_DENOISER_INPUT_RGB = 0x2301,
    OPTIX_DENOISER_INPUT_RGB_ALBEDO = 0x2302,
    OPTIX_DENOISER_INPUT_RGB_ALBEDO_NORMAL = 0x2303 }
• enum OptixDenoiserModelKind {
    OPTIX_DENOISER_MODEL_KIND_USER = 0x2321,
    OPTIX_DENOISER_MODEL_KIND_LDR = 0x2322,
    OPTIX_DENOISER_MODEL_KIND_HDR = 0x2323 }
• enum OptixRayFlags {
    OPTIX_RAY_FLAG_NONE = 0u,
    OPTIX_RAY_FLAG_DISABLE_ANYHIT = 1u << 0,
    OPTIX_RAY_FLAG_ENFORCE_ANYHIT = 1u << 1,
    OPTIX_RAY_FLAG_TERMINATE_ON_FIRST_HIT = 1u << 2,
    OPTIX_RAY_FLAG_DISABLE_CLOSESTHIT = 1u << 3,

```

```

OPTIX_RAY_FLAG_CULL_BACK_FACING_TRIANGLES = 1u << 4,
OPTIX_RAY_FLAG_CULL_FRONT_FACING_TRIANGLES = 1u << 5,
OPTIX_RAY_FLAG_CULL_DISABLED_ANYHIT = 1u << 6,
OPTIX_RAY_FLAG_CULL_ENFORCED_ANYHIT = 1u << 7 }
• enum OptixTransformType {
    OPTIX_TRANSFORM_TYPE_NONE = 0,
    OPTIX_TRANSFORM_TYPE_STATIC_TRANSFORM = 1,
    OPTIX_TRANSFORM_TYPE_MATRIX_MOTION_TRANSFORM = 2,
    OPTIX_TRANSFORM_TYPE_SRT_MOTION_TRANSFORM = 3,
    OPTIX_TRANSFORM_TYPE_INSTANCE = 4 }
• enum OptixTraversableGraphFlags {
    OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_ANY = 0,
    OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_GAS = 1u << 0,
    OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_LEVEL_INSTANCING = 1u << 1 }
• enum OptixCompileOptimizationLevel {
    OPTIX_COMPILE_OPTIMIZATION_DEFAULT = 0,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_0 = 0x2340,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_1 = 0x2341,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_2 = 0x2342,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_3 = 0x2343 }
• enum OptixCompileDebugLevel {
    OPTIX_COMPILE_DEBUG_LEVEL_DEFAULT = 0,
    OPTIX_COMPILE_DEBUG_LEVEL_NONE = 0x2350,
    OPTIX_COMPILE_DEBUG_LEVEL_LINEINFO = 0x2351,
    OPTIX_COMPILE_DEBUG_LEVEL_FULL = 0x2352 }
• enum OptixProgramGroupKind {
    OPTIX_PROGRAM_GROUP_KIND_RAYGEN = 0x2421,
    OPTIX_PROGRAM_GROUP_KIND_MISS = 0x2422,
    OPTIX_PROGRAM_GROUP_KIND_EXCEPTION = 0x2423,
    OPTIX_PROGRAM_GROUP_KIND_HITGROUP = 0x2424,
    OPTIX_PROGRAM_GROUP_KIND_CALLABLES = 0x2425 }
• enum OptixProgramGroupFlags { OPTIX_PROGRAM_GROUP_FLAGS_NONE = 0 }
• enum OptixExceptionCodes {
    OPTIX_EXCEPTION_CODE_STACK_OVERFLOW = -1,
    OPTIX_EXCEPTION_CODE_TRACE_DEPTH_EXCEEDED = -2,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_DEPTH_EXCEEDED = -3,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_TRAVERSABLE = -5,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_MISS_SBT = -6,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_HIT_SBT = -7,
    OPTIX_EXCEPTION_CODE_UNSUPPORTED_PRIMITIVE_TYPE = -8,
    OPTIX_EXCEPTION_CODE_INVALID_RAY = -9,
    OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH = -10,
    OPTIX_EXCEPTION_CODE_BUILTIN_IS_MISMATCH = -11,
    OPTIX_EXCEPTION_CODE_UNSUPPORTED_SINGLE_LEVEL_GAS = -15 }
• enum OptixExceptionFlags {
    OPTIX_EXCEPTION_FLAG_NONE = 0,
    OPTIX_EXCEPTION_FLAG_STACK_OVERFLOW = 1u << 0,
    OPTIX_EXCEPTION_FLAG_TRACE_DEPTH = 1u << 1,
    OPTIX_EXCEPTION_FLAG_USER = 1u << 2,

```

```

    OPTIX_EXCEPTION_FLAG_DEBUG = 1u << 3 }
• enum OptixQueryFunctionTableOptions {
    OPTIX_QUERY_FUNCTION_TABLE_OPTION_DUMMY = 0 }

```

3.11.1 Detailed Description

OptiX Types.

3.11.2 Macro Definition Documentation

3.11.2.1 **#define OPTIX_AABB_BUFFER_BYTE_ALIGNMENT 8ull**

Alignment requirement for [OptixBuildInputCustomPrimitiveArray::aabbBuffers](#) and [OptixBuildInputInstanceArray::aabbs](#).

3.11.2.2 **#define OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT 128ull**

Alignment requirement for output and temporary buffers for acceleration structures.

3.11.2.3 **#define OPTIX_COMPILE_DEFAULT_MAX_REGISTER_COUNT 0**

Maximum number of registers allowed. Defaults to no explicit limit.

3.11.2.4 **#define OPTIX_GEOMETRY_TRANSFORM_BYTE_ALIGNMENT 16ull**

Alignment requirement for [OptixBuildInputTriangleArray::preTransform](#).

3.11.2.5 **#define OPTIX_INSTANCE_BYTE_ALIGNMENT 16ull**

Alignment requirement for [OptixBuildInputInstanceArray::instances](#).

3.11.2.6 **#define OPTIX_SBT_RECORD_ALIGNMENT 16ull**

Alignment requirement for device pointers in [OptixShaderBindingTable](#).

3.11.2.7 **#define OPTIX_SBT_RECORD_HEADER_SIZE ((size_t)32)**

Size of the SBT record headers.

3.11.2.8 **#define OPTIX_TRANSFORM_BYTE_ALIGNMENT 64ull**

Alignment requirement for [OptixStaticTransform](#), [OptixMatrixMotionTransform](#), [OptixSRTMotionTransform](#).

3.11.3 Typedef Documentation

3.11.3.1 **typedef unsigned long long CUdeviceptr**

CUDA device pointer.

3.11.3.2 typedef struct OptixAabb OptixAabb

AABB inputs.

3.11.3.3 typedef struct OptixAccelBufferSizes OptixAccelBufferSizes

Struct for querying builder allocation requirements.

Once queried the sizes should be used to allocate device memory of at least these sizes.

See Also

[optixAccelComputeMemoryUsage\(\)](#)

3.11.3.4 typedef struct OptixAccelBuildOptions OptixAccelBuildOptions

Build options for acceleration structures.

See Also

[optixAccelComputeMemoryUsage\(\)](#), [optixAccelBuild\(\)](#)

3.11.3.5 typedef struct OptixAccelEmitDesc OptixAccelEmitDesc

Specifies a type and output destination for emitted post-build properties.

See Also

[optixAccelBuild\(\)](#)

3.11.3.6 typedef enum OptixAccelPropertyType OptixAccelPropertyType

Properties which can be emitted during acceleration structure build.

See Also

[OptixAccelEmitDesc::type](#).

3.11.3.7 typedef struct OptixAccelRelocationInfo OptixAccelRelocationInfo

Used to store information related to relocation of acceleration structures.

See Also

[optixAccelGetRelocationInfo\(\)](#), [optixAccelCheckRelocationCompatibility\(\)](#), [optixAccelRelocate\(\)](#)

3.11.3.8 typedef enum OptixBuildFlags OptixBuildFlags

Builder Options.

Used for [OptixAccelBuildOptions::buildFlags](#). Can be or'ed together.

3.11.3.9 `typedef struct OptixBuildInput OptixBuildInput`

Build inputs.

All of them support motion and the size of the data arrays needs to match the number of motion steps

See Also

[optixAccelComputeMemoryUsage\(\)](#), [optixAccelBuild\(\)](#)

3.11.3.10 `typedef struct OptixBuildInputCurveArray OptixBuildInputCurveArray`

Curve inputs.

A curve is a swept surface defined by a 3D spline curve and a varying width (radius). A curve (or "strand") of degree d (3=cubic, 2=quadratic, 1=linear) is represented by $N > d$ vertices and N width values, and comprises $N - d$ segments. Each segment is defined by $d+1$ consecutive vertices. Each curve may have a different number of vertices.

OptiX describes the curve array as a list of curve segments. The primitive id is the segment number. It is the user's responsibility to maintain a mapping between curves and curve segments. Each index buffer entry $i = \text{indexBuffer}[\text{primid}]$ specifies the start of a curve segment, represented by $d+1$ consecutive vertices in the vertex buffer, and $d+1$ consecutive widths in the width buffer. Width is interpolated the same way vertices are interpolated, that is, using the curve basis.

Each curves build input has only one SBT record. To create curves with different materials in the same BVH, use multiple build inputs.

See Also

[OptixBuildInput::curveArray](#)

3.11.3.11 `typedef struct OptixBuildInputCustomPrimitiveArray OptixBuildInputCustomPrimitiveArray`

Custom primitive inputs.

See Also

[OptixBuildInput::customPrimitiveArray](#)

3.11.3.12 `typedef struct OptixBuildInputInstanceArray OptixBuildInputInstanceArray`

Instance and instance pointer inputs.

See Also

[OptixBuildInput::instanceArray](#)

3.11.3.13 `typedef struct OptixBuildInputTriangleArray OptixBuildInputTriangleArray`

Triangle inputs.

See Also

[OptixBuildInput::triangleArray](#)

3.11.3.14 **typedef enum OptixBuildInputType OptixBuildInputType**

Enum to distinguish the different build input types.

See Also

[OptixBuildInput::type](#)

3.11.3.15 **typedef enum OptixBuildOperation OptixBuildOperation**

Enum to specify the acceleration build operation.

Used in [OptixAccelBuildOptions](#), which is then passed to `optixAccelBuild` and `optixAccelComputeMemoryUsage`, this enum indicates whether to do a build or an update of the acceleration structure.

Acceleration structure updates utilize the same acceleration structure, but with updated bounds. Updates are typically much faster than builds, however, large perturbations can degrade the quality of the acceleration structure.

See Also

[optixAccelComputeMemoryUsage\(\)](#), [optixAccelBuild\(\)](#), [OptixAccelBuildOptions](#)

3.11.3.16 **typedef struct OptixBuiltinISOptions OptixBuiltinISOptions**

Specifies the options for retrieving an intersection program for a built-in primitive type. The primitive type must not be `OPTIX_PRIMITIVE_TYPE_CUSTOM`.

See Also

[optixBuiltinISModuleGet\(\)](#)

3.11.3.17 **typedef enum OptixCompileDebugLevel OptixCompileDebugLevel**

Debug levels.

See Also

[OptixModuleCompileOptions::debugLevel](#)

3.11.3.18 **typedef enum OptixCompileOptimizationLevel OptixCompileOptimizationLevel**

Optimization levels.

See Also

[OptixModuleCompileOptions::optLevel](#)

3.11.3.19 typedef struct OptixDenoiser_t* OptixDenoiser

Opaque type representing a denoiser instance.

3.11.3.20 typedef enum OptixDenoiserInputKind OptixDenoiserInputKind

Input kinds used by the denoiser.

RGB(A) values less than zero will be clamped to zero. Albedo values must be in the range [0..1] (values less than zero will be clamped to zero). The normals must be transformed into screen space. The z component is not used.

See Also

[OptixDenoiserOptions::inputKind](#)

3.11.3.21 typedef enum OptixDenoiserModelKind OptixDenoiserModelKind

Model kind used by the denoiser.

See Also

[optixDenoiserSetModel\(\)](#)

3.11.3.22 typedef struct OptixDenoiserOptions OptixDenoiserOptions

Options used by the denoiser.

See Also

[optixDenoiserCreate\(\)](#)

3.11.3.23 typedef struct OptixDenoiserParams OptixDenoiserParams

Various parameters used by the denoiser.

See Also

[optixDenoiserInvoke\(\)](#)

3.11.3.24 typedef struct OptixDenoiserSizes OptixDenoiserSizes

Various sizes related to the denoiser.

See Also

[optixDenoiserComputeMemoryResources\(\)](#)

3.11.3.25 typedef struct OptixDeviceContext_t* OptixDeviceContext

Opaque type representing a device context.

3.11.3.26 **typedef struct OptixDeviceContextOptions OptixDeviceContextOptions**

Parameters used for [optixDeviceContextCreate\(\)](#)

See Also

[optixDeviceContextCreate\(\)](#)

3.11.3.27 **typedef enum OptixDeviceProperty OptixDeviceProperty**

Parameters used for [optixDeviceContextGetProperty\(\)](#)

See Also

[optixDeviceContextGetProperty\(\)](#)

3.11.3.28 **typedef enum OptixExceptionCodes OptixExceptionCodes**

The following values are used to indicate which exception was thrown.

3.11.3.29 **typedef enum OptixExceptionFlags OptixExceptionFlags**

Exception flags.

See Also

[OptixPipelineCompileOptions::exceptionFlags](#), [OptixExceptionCodes](#)

3.11.3.30 **typedef enum OptixGeometryFlags OptixGeometryFlags**

Flags used by [OptixBuildInputTriangleArray::flags](#) and [#OptixBuildInputCurveArray::flags](#) and [OptixBuildInputCustomPrimitiveArray::flags](#).

3.11.3.31 **typedef enum OptixHitKind OptixHitKind**

Legacy type: A subset of the hit kinds for built-in primitive intersections. It is preferred to use [optixGetPrimitiveType\(\)](#), together with [optixIsFrontFaceHit\(\)](#) or [optixIsBackFaceHit\(\)](#).

See Also

[optixGetHitKind\(\)](#)

3.11.3.32 **typedef struct OptixImage2D OptixImage2D**

Image descriptor used by the denoiser.

See Also

[optixDenoiserInvoke\(\)](#), [optixDenoiserComputeIntensity\(\)](#)

3.11.3.33 **typedef enum OptixIndicesFormat OptixIndicesFormat**

Format of indices used in [OptixBuildInputTriangleArray::indexFormat](#).

3.11.3.34 typedef struct OptixInstance OptixInstance

Instances.

See Also

[OptixBuildInputInstanceArray::instances](#)

3.11.3.35 typedef enum OptixInstanceFlags OptixInstanceFlags

Flags set on the [OptixInstance::flags](#).

These can be or'ed together to combine multiple flags.

3.11.3.36 typedef void(* OptixLogCallback)(unsigned int level, const char *tag, const char *message, void *cbdata)

Type of the callback function used for log messages.

Parameters

in	<i>level</i>	The log level indicates the severity of the message. See below for possible values.
in	<i>tag</i>	A terse message category description (e.g., 'SCENE STAT').
in	<i>message</i>	Null terminated log message (without newline at the end).
in	<i>cbdata</i>	Callback data that was provided with the callback pointer.

It is the users responsibility to ensure thread safety within this function.

The following log levels are defined.

0 disable Setting the callback level will disable all messages. The callback function will not be called in this case. 1 fatal A non-recoverable error. The context and/or OptiX itself might no longer be in a usable state. 2 error A recoverable error, e.g., when passing invalid call parameters. 3 warning Hints that OptiX might not behave exactly as requested by the user or may perform slower than expected. 4 print Status or progress messages.

Higher levels might occur.

See Also

[optixDeviceContextSetLogCallback\(\)](#), [OptixDeviceContextOptions](#)

3.11.3.37 typedef struct OptixMatrixMotionTransform OptixMatrixMotionTransform

Represents a matrix motion transformation.

The device address of instances of this type must be a multiple of OPTIX_TRANSFORM_BYTE_ALIGNMENT.

This struct, as defined here, handles only N=2 motion keys due to the fixed array length of its transform member. The following example shows how to create instances for an arbitrary number N of motion keys:

```

float matrixData[N][12];
... // setup matrixData

size_t transformSizeInBytes = sizeof( OptixMatrixMotionTransform ) + ( N-2 ) * 12
    * sizeof( float );
OptixMatrixMotionTransform* matrixMoptionTransform = (
    OptixMatrixMotionTransform*) malloc( transformSizeInBytes );
memset( matrixMoptionTransform, 0, transformSizeInBytes );

... // setup other members of matrixMoptionTransform
matrixMoptionTransform->motionOptions.numKeys
memcpy( matrixMoptionTransform->transform, matrixData, N * 12 * sizeof( float ) );

... // copy matrixMoptionTransform to device memory
free( matrixMoptionTransform )

```

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

3.11.3.38 typedef struct OptixModule_t* OptixModule

Opaque type representing a module.

3.11.3.39 typedef struct OptixModuleCompileOptions OptixModuleCompileOptions

Compilation options for module.

See Also

[optixModuleCreateFromPTX\(\)](#)

3.11.3.40 typedef enum OptixMotionFlags OptixMotionFlags

Enum to specify motion flags.

See Also

[OptixMotionOptions::flags.](#)

3.11.3.41 typedef struct OptixMotionOptions OptixMotionOptions

Motion options.

See Also

[OptixAccelBuildOptions::motionOptions](#), [OptixMatrixMotionTransform::motionOptions](#),
[OptixSRTMotionTransform::motionOptions](#)

3.11.3.42 typedef struct OptixPipeline_t* OptixPipeline

Opaque type representing a pipeline.

3.11.3.43 typedef struct OptixPipelineCompileOptions OptixPipelineCompileOptions

Compilation options for all modules of a pipeline.

Similar to [OptixModuleCompileOptions](#), but these options here need to be equal for all modules of a pipeline.

See Also

[optixModuleCreateFromPTX\(\)](#), [optixPipelineCreate\(\)](#)

3.11.3.44 typedef struct OptixPipelineLinkOptions OptixPipelineLinkOptions

Link options for a pipeline.

See Also

[optixPipelineCreate\(\)](#)

3.11.3.45 typedef enum OptixPixelFormat OptixPixelFormat

Pixel formats used by the denoiser.

See Also

[OptixImage2D::format](#)

3.11.3.46 typedef enum OptixPrimitiveType OptixPrimitiveType

Builtin primitive types.

3.11.3.47 typedef enum OptixPrimitiveTypeFlags OptixPrimitiveTypeFlags

Builtin flags may be bitwise combined.

See Also

[OptixPipelineCompileOptions::usesPrimitiveTypeFlags](#)

3.11.3.48 typedef struct OptixProgramGroup_t* OptixProgramGroup

Opaque type representing a program group.

3.11.3.49 typedef struct OptixProgramGroupCallables OptixProgramGroupCallables

Program group representing callables.

Module and entry function name need to be valid for at least one of the two callables.

See Also

[#OptixProgramGroupDesc::callables](#)

3.11.3.50 **typedef struct OptixProgramGroupDesc OptixProgramGroupDesc**

Descriptor for program groups.

3.11.3.51 **typedef enum OptixProgramGroupFlags OptixProgramGroupFlags**

Flags for program groups.

3.11.3.52 **typedef struct OptixProgramGroupHitgroup OptixProgramGroupHitgroup**

Program group representing the hitgroup.

For each of the three program types, module and entry function name might both be nullptr.

See Also

[OptixProgramGroupDesc::hitgroup](#)

3.11.3.53 **typedef enum OptixProgramGroupKind OptixProgramGroupKind**

Distinguishes different kinds of program groups.

3.11.3.54 **typedef struct OptixProgramGroupOptions OptixProgramGroupOptions**

Program group options.

See Also

[optixProgramGroupCreate\(\)](#)

3.11.3.55 **typedef struct OptixProgramGroupSingleModule OptixProgramGroupSingleModule**

Program group representing a single module.

Used for raygen, miss, and exception programs. In case of raygen and exception programs, module and entry function name need to be valid. For miss programs, module and entry function name might both be nullptr.

See Also

[OptixProgramGroupDesc::raygen](#), [OptixProgramGroupDesc::miss](#),
[OptixProgramGroupDesc::exception](#)

3.11.3.56 **typedef OptixResult(OptixQueryFunctionTable_t)(int abild, unsigned int numOptions, OptixQueryFunctionTableOptions *, const void **, void *functionTable, size_t sizeOfTable)**

Type of the function `optixQueryFunctionTable()`

3.11.3.57 typedef enum OptixQueryFunctionTableOptions OptixQueryFunctionTableOptions

Options that can be passed to `optixQueryFunctionTable()`

3.11.3.58 typedef enum OptixRayFlags OptixRayFlags

Ray flags passed to the device function `optixTrace()`. These affect the behavior of traversal per invocation.

See Also

`optixTrace()`

3.11.3.59 typedef enum OptixResult OptixResult

Result codes returned from API functions.

All host side API functions return `OptixResult` with the exception of `optixGetErrorName` and `optixGetErrorString`. When successful `OPTIX_SUCCESS` is returned. All return codes except for `OPTIX_SUCCESS` should be assumed to be errors as opposed to a warning.

See Also

`optixGetErrorName()`, `optixGetErrorString()`

3.11.3.60 typedef struct OptixShaderBindingTable OptixShaderBindingTable

Describes the shader binding table (SBT)

See Also

`optixLaunch()`

3.11.3.61 typedef struct OptixSRTData OptixSRTData

Represents an SRT transformation.

An SRT transformation can represent a smooth rotation with fewer motion keys than a matrix transformation. Each motion key is constructed from elements taken from a matrix *S*, a quaternion *R*, and a translation *T*.

$$\begin{bmatrix} s_x & a & b & p_{vx} \end{bmatrix}$$

The scaling matrix $S = \begin{bmatrix} 0 & s_y & c & p_{vy} \end{bmatrix}$ defines an affine transformation that can include scale, shear, and a $\begin{bmatrix} 0 & 0 & s_z & p_{vz} \end{bmatrix}$

translation. The translation allows to define the pivot point for the subsequent rotation.

The quaternion $R = \begin{bmatrix} q_x, q_y, q_z, q_w \end{bmatrix}$ describes a rotation with angular component $q_w = \cos(\theta/2)$ and other components $\begin{bmatrix} q_x, q_y, q_z \end{bmatrix} = \sin(\theta/2) * \begin{bmatrix} a_x, a_y, a_z \end{bmatrix}$ where the axis $\begin{bmatrix} a_x, a_y, a_z \end{bmatrix}$ is normalized.

$$\begin{bmatrix} 1 & 0 & 0 & t_x \end{bmatrix}$$

The translation $T = \begin{bmatrix} 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \end{bmatrix}$ defines another translation that is applied after the rotation. Typically, this

translation includes the inverse translation from the matrix S to reverse its effect.

To obtain the effective transformation at time t , the elements of the components of S , R , and T will be interpolated linearly. The components are then multiplied to obtain the combined transformation $C = T * R * S$. The transformation C is the effective object-to-world transformations at time t , and C^{-1} is the effective world-to-object transformation at time t .

See Also

[OptixSRTMotionTransform::srtData](#), [optixConvertPointerToTraversableHandle\(\)](#)

3.11.3.62 typedef struct OptixSRTMotionTransform OptixSRTMotionTransform

Represents an SRT motion transformation.

The device address of instances of this type must be a multiple of `OPTIX_TRANSFORM_BYTE_ALIGNMENT`.

This struct, as defined here, handles only $N=2$ motion keys due to the fixed array length of its `srtData` member. The following example shows how to create instances for an arbitrary number N of motion keys:

```
OptixSRTData srtData[N];
... // setup srtData

size_t transformSizeInBytes = sizeof( OptixSRTMotionTransform ) + ( N-2 ) * sizeof(
    OptixSRTData );
OptixSRTMotionTransform* srtMotionTransform = (
    OptixSRTMotionTransform*) malloc( transformSizeInBytes );
memset( srtMotionTransform, 0, transformSizeInBytes );

... // setup other members of srtMotionTransform
srtMotionTransform->motionOptions.numKeys = N;
memcpy( srtMotionTransform->srtData, srtData, N * sizeof( OptixSRTData ) );

... // copy srtMotionTransform to device memory
free( srtMotionTransform )
```

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

3.11.3.63 typedef struct OptixStackSizes OptixStackSizes

Describes the stack size requirements of a program group.

See Also

[optixProgramGroupGetStackSize\(\)](#)

3.11.3.64 typedef struct OptixStaticTransform OptixStaticTransform

Static transform.

The device address of instances of this type must be a multiple of `OPTIX_TRANSFORM_BYTE_ALIGNMENT`.

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

3.11.3.65 typedef enum OptixTransformFormat OptixTransformFormat

Format of transform used in [OptixBuildInputTriangleArray::transformFormat](#).

3.11.3.66 typedef enum OptixTransformType OptixTransformType

Transform.

`OptixTransformType` is used by the device function [optixGetTransformTypeFromHandle\(\)](#) to determine the type of the `OptixTraversableHandle` returned from [optixGetTransformListHandle\(\)](#).

3.11.3.67 typedef enum OptixTraversableGraphFlags OptixTraversableGraphFlags

Specifies the set of valid traversable graphs that may be passed to invocation of [optixTrace\(\)](#). Flags may be bitwise combined.

3.11.3.68 typedef unsigned long long OptixTraversableHandle

Traversable handle.

3.11.3.69 typedef enum OptixTraversableType OptixTraversableType

Traversable Handles.

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

3.11.3.70 typedef enum OptixVertexFormat OptixVertexFormat

Format of vertices used in [OptixBuildInputTriangleArray::vertexFormat](#).

3.11.3.71 typedef unsigned int OptixVisibilityMask

Visibility mask.

3.11.4 Enumeration Type Documentation

3.11.4.1 enum OptixAccelPropertyType

Properties which can be emitted during acceleration structure build.

See Also

[OptixAccelEmitDesc::type](#).

Enumerator

OPTIX_PROPERTY_TYPE_COMPACTED_SIZE Size of a compacted acceleration structure.

The device pointer points to a uint64.

OPTIX_PROPERTY_TYPE_AABBS [OptixAabb](#) * numMotionSteps.

3.11.4.2 enum OptixBuildFlags

Builder Options.

Used for [OptixAccelBuildOptions::buildFlags](#). Can be or'ed together.

Enumerator

OPTIX_BUILD_FLAG_NONE No special flags set.

OPTIX_BUILD_FLAG_ALLOW_UPDATE Allow updating the build with new vertex positions with subsequent calls to `optixAccelBuild`.

OPTIX_BUILD_FLAG_ALLOW_COMPACTION

OPTIX_BUILD_FLAG_PREFER_FAST_TRACE

OPTIX_BUILD_FLAG_PREFER_FAST_BUILD

OPTIX_BUILD_FLAG_ALLOW_RANDOM_VERTEX_ACCESS Allow access to random baked vertex in `closesthit`.

3.11.4.3 enum OptixBuildInputType

Enum to distinguish the different build input types.

See Also

[OptixBuildInput::type](#)

Enumerator

OPTIX_BUILD_INPUT_TYPE_TRIANGLES Triangle inputs.

See Also

[OptixBuildInputTriangleArray](#)

OPTIX_BUILD_INPUT_TYPE_CUSTOM_PRIMITIVES Custom primitive inputs.

See Also

[OptixBuildInputCustomPrimitiveArray](#)

OPTIX_BUILD_INPUT_TYPE_INSTANCES Instance inputs.

See Also

[OptixBuildInputInstanceArray](#)

OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS Instance pointer inputs.

See Also

[OptixBuildInputInstanceArray](#)

OPTIX_BUILD_INPUT_TYPE_CURVES Curve inputs.

See Also

[OptixBuildInputCurveArray](#)

3.11.4.4 enum OptixBuildOperation

Enum to specify the acceleration build operation.

Used in [OptixAccelBuildOptions](#), which is then passed to `optixAccelBuild` and `optixAccelComputeMemoryUsage`, this enum indicates whether to do a build or an update of the acceleration structure.

Acceleration structure updates utilize the same acceleration structure, but with updated bounds. Updates are typically much faster than builds, however, large perturbations can degrade the quality of the acceleration structure.

See Also

[optixAccelComputeMemoryUsage\(\)](#), [optixAccelBuild\(\)](#), [OptixAccelBuildOptions](#)

Enumerator

OPTIX_BUILD_OPERATION_BUILD Perform a full build operation.

OPTIX_BUILD_OPERATION_UPDATE Perform an update using new bounds.

3.11.4.5 enum OptixCompileDebugLevel

Debug levels.

See Also

[OptixModuleCompileOptions::debugLevel](#)

Enumerator

OPTIX_COMPILE_DEBUG_LEVEL_DEFAULT Default currently is to add line info.

OPTIX_COMPILE_DEBUG_LEVEL_NONE No debug information.

OPTIX_COMPILE_DEBUG_LEVEL_LINEINFO Generate lineinfo only.

OPTIX_COMPILE_DEBUG_LEVEL_FULL Generate dwarf debug information.

3.11.4.6 enum OptixCompileOptimizationLevel

Optimization levels.

See Also

[OptixModuleCompileOptions::optLevel](#)

Enumerator

OPTIX_COMPILE_OPTIMIZATION_DEFAULT Default is to run all optimizations.

OPTIX_COMPILE_OPTIMIZATION_LEVEL_0 No optimizations.

OPTIX_COMPILE_OPTIMIZATION_LEVEL_1 Some optimizations.

OPTIX_COMPILE_OPTIMIZATION_LEVEL_2 Most optimizations.

OPTIX_COMPILE_OPTIMIZATION_LEVEL_3 All optimizations.

3.11.4.7 enum OptixDenoiserInputKind

Input kinds used by the denoiser.

RGB(A) values less than zero will be clamped to zero. Albedo values must be in the range [0..1] (values less than zero will be clamped to zero). The normals must be transformed into screen space. The z component is not used.

See Also

[OptixDenoiserOptions::inputKind](#)

Enumerator

OPTIX_DENOISER_INPUT_RGB

OPTIX_DENOISER_INPUT_RGB_ALBEDO

OPTIX_DENOISER_INPUT_RGB_ALBEDO_NORMAL

3.11.4.8 enum OptixDenoiserModelKind

Model kind used by the denoiser.

See Also

[optixDenoiserSetModel\(\)](#)

Enumerator

OPTIX_DENOISER_MODEL_KIND_USER Use the model provided by the associated pointer.
See the programming guide for a description of how to format the data.

OPTIX_DENOISER_MODEL_KIND_LDR Use the built-in model appropriate for low dynamic range input.

OPTIX_DENOISER_MODEL_KIND_HDR Use the built-in model appropriate for high dynamic range input.

3.11.4.9 enum OptixDeviceProperty

Parameters used for [optixDeviceContextGetProperty\(\)](#)

See Also

[optixDeviceContextGetProperty\(\)](#)

Enumerator

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_TRACE_DEPTH Maximum value for [OptixPipelineLinkOptions::maxTraceDepth](#). sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_TRAVERSABLE_GRAPH_DEPTH Maximum value to pass into [optixPipelineSetStackSize](#) for parameter [maxTraversableGraphDepth.v](#) sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_PRIMITIVES_PER_GAS The maximum number of primitives (over all build inputs) as input to a single Geometry Acceleration Structure (GAS). sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCES_PER_IAS The maximum number of instances (over all build inputs) as input to a single Instance Acceleration Structure (IAS). sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_RTCORE_VERSION The RT core version supported by the device (0 for no support, 10 for version 1.0). sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCE_ID The maximum value for [OptixInstance::instanceId](#). sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_NUM_BITS_INSTANCE_VISIBILITY_MASK The number of bits available for the [OptixInstance::visibilityMask](#). Higher bits must be set to zero. sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_RECORDS_PER_GAS The maximum number of instances that can be added to a single Instance Acceleration Structure (IAS). sizeof(unsigned int)

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_OFFSET The maximum value for [OptixInstance::sbtOffset](#). sizeof(unsigned int)

3.11.4.10 enum OptixExceptionCodes

The following values are used to indicate which exception was thrown.

Enumerator

OPTIX_EXCEPTION_CODE_STACK_OVERFLOW Stack overflow of the continuation stack. no exception details.

OPTIX_EXCEPTION_CODE_TRACE_DEPTH_EXCEEDED The trace depth is exceeded. no exception details.

OPTIX_EXCEPTION_CODE_TRAVERSAL_DEPTH_EXCEEDED The traversal depth is exceeded. Exception details: [optixGetTransformListSize\(\)](#) [optixGetTransformListHandle\(\)](#)

OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_TRAVERSABLE Traversal encountered an invalid traversable type. Exception details: [optixGetTransformListSize\(\)](#) [optixGetTransformListHandle\(\)](#) [optixGetExceptionInvalidTraversable\(\)](#)

OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_MISS_SBT The miss SBT record index is out of bounds A miss SBT record index is valid within the range [0, [OptixShaderBindingTable::missRecordCount](#)) (See [optixLaunch](#)) Exception details: [optixGetExceptionInvalidSbtOffset\(\)](#)

OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_HIT_SBT The traversal hit SBT record index out of bounds. A traversal hit SBT record index is valid within the range [0, [OptixShaderBindingTable::hitgroupRecordCount](#)) (See [optixLaunch](#)) The following formula relates the sbt-geometry-acceleration-structure-index (See [optixGetSbtGASIndex](#)), sbt-stride-from-trace-call and sbt-offset-from-trace-call (See [optixTrace](#))

$$\text{sbt-index} = \text{sbt-instance-offset} + (\text{sbt-geometry-acceleration-structure-index} * \text{sbt-stride-from-trace-call}) + \text{sbt-offset-from-trace-call}$$

Exception details: [optixGetTransformListSize\(\)](#) [optixGetTransformListHandle\(\)](#)
[optixGetExceptionInvalidSbtOffset\(\)](#) [optixGetSbtGASIndex\(\)](#)

OPTIX_EXCEPTION_CODE_UNSUPPORTED_PRIMITIVE_TYPE The shader encountered an unsupported primitive type (See [OptixPipelineCompileOptions::usesPrimitiveTypeFlags](#)). no exception details.

OPTIX_EXCEPTION_CODE_INVALID_RAY The shader encountered a call to [optixTrace](#) with at least one of the float arguments being inf or nan. Exception details:
[optixGetExceptionInvalidRay\(\)](#)

OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH The shader encountered a call to either [optixDirectCall](#) or [optixCallableCall](#) where the argument count does not match the parameter count of the callable program which is called. Exception details:
[optixGetExceptionParameterMismatch](#).

OPTIX_EXCEPTION_CODE_BUILTIN_IS_MISMATCH The invoked builtin IS does not match the current GAS.

OPTIX_EXCEPTION_CODE_UNSUPPORTED_SINGLE_LEVEL_GAS Tried to directly traverse a single gas while single gas traversable graphs are not enabled. Exception details:
[optixGetTransformListSize\(\)](#) [optixGetTransformListHandle\(\)](#)
[optixGetExceptionInvalidTraversable\(\)](#)

3.11.4.11 enum OptixExceptionFlags

Exception flags.

See Also

[OptixPipelineCompileOptions::exceptionFlags](#), [OptixExceptionCodes](#)

Enumerator

OPTIX_EXCEPTION_FLAG_NONE No exception are enabled.

OPTIX_EXCEPTION_FLAG_STACK_OVERFLOW Enables exceptions check related to the continuation stack.

OPTIX_EXCEPTION_FLAG_TRACE_DEPTH Enables exceptions check related to trace depth.

OPTIX_EXCEPTION_FLAG_USER Enables user exceptions via [optixThrowException\(\)](#). This flag must be specified for all modules in a pipeline if any module calls [optixThrowException\(\)](#).

OPTIX_EXCEPTION_FLAG_DEBUG Enables various exceptions check related to traversal.

3.11.4.12 enum OptixGeometryFlags

Flags used by `OptixBuildInputTriangleArray::flags` and `#OptixBuildInputCurveArray::flags` and `OptixBuildInputCustomPrimitiveArray::flags`.

Enumerator

OPTIX_GEOMETRY_FLAG_NONE No flags set.

OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT Disables the invocation of the anyhit program.
Can be overridden by `OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT` and `OPTIX_RAY_FLAG_ENFORCE_ANYHIT`.

OPTIX_GEOMETRY_FLAG_REQUIRE_SINGLE_ANYHIT_CALL If set, an intersection with the primitive will trigger one and only one invocation of the anyhit program. Otherwise, the anyhit program may be invoked more than once.

3.11.4.13 enum OptixHitKind

Legacy type: A subset of the hit kinds for built-in primitive intersections. It is preferred to use `optixGetPrimitiveType()`, together with `optixIsFrontFaceHit()` or `optixIsBackFaceHit()`.

See Also

`optixGetHitKind()`

Enumerator

OPTIX_HIT_KIND_TRIANGLE_FRONT_FACE Ray hit the triangle on the front face.

OPTIX_HIT_KIND_TRIANGLE_BACK_FACE Ray hit the triangle on the back face.

3.11.4.14 enum OptixIndicesFormat

Format of indices used in `OptixBuildInputTriangleArray::indexFormat`.

Enumerator

OPTIX_INDICES_FORMAT_NONE No indices, this format must only be used in combination with triangle soups, i.e., `numIndexTriplets` must be zero.

OPTIX_INDICES_FORMAT_UNSIGNED_SHORT3 Three shorts.

OPTIX_INDICES_FORMAT_UNSIGNED_INT3 Three ints.

3.11.4.15 enum OptixInstanceFlags

Flags set on the `OptixInstance::flags`.

These can be or'ed together to combine multiple flags.

Enumerator

OPTIX_INSTANCE_FLAG_NONE No special flag set.

OPTIX_INSTANCE_FLAG_DISABLE_TRIANGLE_FACE_CULLING Prevent triangles from getting culled due to their orientation. Effectively ignores ray flags `OPTIX_RAY_FLAG_CULL_BACK_FACING_TRIANGLES` and `OPTIX_RAY_FLAG_CULL_FRONT_FACING_TRIANGLES`.

OPTIX_INSTANCE_FLAG_FLIP_TRIANGLE_FACING Flip triangle orientation. This affects front/backface culling as well as the reported face in case of a hit.

OPTIX_INSTANCE_FLAG_DISABLE_ANYHIT Disable anyhit programs for all geometries of the instance. Can be overridden by **OPTIX_RAY_FLAG_ENFORCE_ANYHIT**. This flag is mutually exclusive with **OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT**.

OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT Enables anyhit programs for all geometries of the instance. Overrides **OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT**. Can be overridden by **OPTIX_RAY_FLAG_DISABLE_ANYHIT**. This flag is mutually exclusive with **OPTIX_INSTANCE_FLAG_DISABLE_ANYHIT**.

OPTIX_INSTANCE_FLAG_DISABLE_TRANSFORM Disable the instance transformation.

3.11.4.16 enum OptixMotionFlags

Enum to specify motion flags.

See Also

[OptixMotionOptions::flags](#).

Enumerator

OPTIX_MOTION_FLAG_NONE

OPTIX_MOTION_FLAG_START_VANISH

OPTIX_MOTION_FLAG_END_VANISH

3.11.4.17 enum OptixPixelFormat

Pixel formats used by the denoiser.

See Also

[OptixImage2D::format](#)

Enumerator

OPTIX_PIXEL_FORMAT_HALF3 three halves, RGB

OPTIX_PIXEL_FORMAT_HALF4 four halves, RGBA

OPTIX_PIXEL_FORMAT_FLOAT3 three floats, RGB

OPTIX_PIXEL_FORMAT_FLOAT4 four floats, RGBA

OPTIX_PIXEL_FORMAT_UCHAR3 three unsigned chars, RGB

OPTIX_PIXEL_FORMAT_UCHAR4 four unsigned chars, RGBA

3.11.4.18 enum OptixPrimitiveType

Builtin primitive types.

Enumerator

OPTIX_PRIMITIVE_TYPE_CUSTOM Custom primitive.

OPTIX_PRIMITIVE_TYPE_ROUND_QUADRATIC_BSPLINE B-spline curve of degree 2 with circular cross-section.

OPTIX_PRIMITIVE_TYPE_ROUND_CUBIC_BSPLINE B-spline curve of degree 3 with circular cross-section.

OPTIX_PRIMITIVE_TYPE_ROUND_LINEAR Piecewise linear curve with circular cross-section.

OPTIX_PRIMITIVE_TYPE_TRIANGLE Triangle.

3.11.4.19 enum OptixPrimitiveTypeFlags

Builtin flags may be bitwise combined.

See Also

[OptixPipelineCompileOptions::usesPrimitiveTypeFlags](#)

Enumerator

OPTIX_PRIMITIVE_TYPE_FLAGS_CUSTOM Custom primitive.

OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_QUADRATIC_BSPLINE B-spline curve of degree 2 with circular cross-section.

OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_CUBIC_BSPLINE B-spline curve of degree 3 with circular cross-section.

OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_LINEAR Piecewise linear curve with circular cross-section.

OPTIX_PRIMITIVE_TYPE_FLAGS_TRIANGLE Triangle.

3.11.4.20 enum OptixProgramGroupFlags

Flags for program groups.

Enumerator

OPTIX_PROGRAM_GROUP_FLAGS_NONE Currently there are no flags.

3.11.4.21 enum OptixProgramGroupKind

Distinguishes different kinds of program groups.

Enumerator

OPTIX_PROGRAM_GROUP_KIND_RAYGEN Program group containing a raygen (RG) program.

See Also

[OptixProgramGroupSingleModule](#), [OptixProgramGroupDesc::raygen](#)

OPTIX_PROGRAM_GROUP_KIND_MISS Program group containing a miss (MS) program.

See Also

[OptixProgramGroupSingleModule](#), [OptixProgramGroupDesc::miss](#)

OPTIX_PROGRAM_GROUP_KIND_EXCEPTION Program group containing an exception (EX) program.

See Also

[OptixProgramGroupHitgroup](#), [OptixProgramGroupDesc::exception](#)

OPTIX_PROGRAM_GROUP_KIND_HITGROUP Program group containing an intersection (IS), any hit (AH), and/or closest hit (CH) program.

See Also

[OptixProgramGroupSingleModule](#), [OptixProgramGroupDesc::hitgroup](#)

OPTIX_PROGRAM_GROUP_KIND_CALLABLES Program group containing a direct (DC) or continuation (CC) callable program.

See Also

[OptixProgramGroupCallables](#), [OptixProgramGroupDesc::callables](#)

3.11.4.22 enum OptixQueryFunctionTableOptions

Options that can be passed to [optixQueryFunctionTable\(\)](#)

Enumerator

OPTIX_QUERY_FUNCTION_TABLE_OPTION_DUMMY Placeholder (there are no options yet)

3.11.4.23 enum OptixRayFlags

Ray flags passed to the device function [optixTrace\(\)](#). These affect the behavior of traversal per invocation.

See Also

[optixTrace\(\)](#)

Enumerator

OPTIX_RAY_FLAG_NONE No change from the behavior configured for the individual AS.

OPTIX_RAY_FLAG_DISABLE_ANYHIT Disables anyhit programs for the ray. Overrides [OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT](#). This flag is mutually exclusive with [OPTIX_RAY_FLAG_ENFORCE_ANYHIT](#), [OPTIX_RAY_FLAG_CULL_DISABLED_ANYHIT](#), [OPTIX_RAY_FLAG_CULL_ENFORCED_ANYHIT](#).

OPTIX_RAY_FLAG_ENFORCE_ANYHIT Forces anyhit program execution for the ray. Overrides [OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT](#) as well as [OPTIX_INSTANCE_FLAG_DISABLE_ANYHIT](#). This flag is mutually exclusive with [OPTIX_RAY_FLAG_DISABLE_ANYHIT](#), [OPTIX_RAY_FLAG_CULL_DISABLED_ANYHIT](#), [OPTIX_RAY_FLAG_CULL_ENFORCED_ANYHIT](#).

OPTIX_RAY_FLAG_TERMINATE_ON_FIRST_HIT Terminates the ray after the first hit and executes the closesthit program of that hit.

OPTIX_RAY_FLAG_DISABLE_CLOSESTHIT Disables closesthit programs for the ray, but still executes miss program in case of a miss.

OPTIX_RAY_FLAG_CULL_BACK_FACING_TRIANGLES Do not intersect triangle back faces (respects a possible face change due to instance flag OPTIX_INSTANCE_FLAG_FLIP_TRIANGLE_FACING). This flag is mutually exclusive with OPTIX_RAY_FLAG_CULL_FRONT_FACING_TRIANGLES.

OPTIX_RAY_FLAG_CULL_FRONT_FACING_TRIANGLES Do not intersect triangle front faces (respects a possible face change due to instance flag OPTIX_INSTANCE_FLAG_FLIP_TRIANGLE_FACING). This flag is mutually exclusive with OPTIX_RAY_FLAG_CULL_BACK_FACING_TRIANGLES.

OPTIX_RAY_FLAG_CULL_DISABLED_ANYHIT Do not intersect geometry which disables anyhit programs (due to setting geometry flag OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT or instance flag OPTIX_INSTANCE_FLAG_DISABLE_ANYHIT). This flag is mutually exclusive with OPTIX_RAY_FLAG_CULL_ENFORCED_ANYHIT, OPTIX_RAY_FLAG_ENFORCE_ANYHIT, OPTIX_RAY_FLAG_DISABLE_ANYHIT.

OPTIX_RAY_FLAG_CULL_ENFORCED_ANYHIT Do not intersect geometry which have an enabled anyhit program (due to not setting geometry flag OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT or setting instance flag OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT). This flag is mutually exclusive with OPTIX_RAY_FLAG_CULL_DISABLED_ANYHIT, OPTIX_RAY_FLAG_ENFORCE_ANYHIT, OPTIX_RAY_FLAG_DISABLE_ANYHIT.

3.11.4.24 enum OptixResult

Result codes returned from API functions.

All host side API functions return OptixResult with the exception of optixGetErrorName and optixGetErrorString. When successful OPTIX_SUCCESS is returned. All return codes except for OPTIX_SUCCESS should be assumed to be errors as opposed to a warning.

See Also

[optixGetErrorName\(\)](#), [optixGetErrorString\(\)](#)

Enumerator

OPTIX_SUCCESS
OPTIX_ERROR_INVALID_VALUE
OPTIX_ERROR_HOST_OUT_OF_MEMORY
OPTIX_ERROR_INVALID_OPERATION
OPTIX_ERROR_FILE_IO_ERROR
OPTIX_ERROR_INVALID_FILE_FORMAT
OPTIX_ERROR_DISK_CACHE_INVALID_PATH
OPTIX_ERROR_DISK_CACHE_PERMISSION_ERROR
OPTIX_ERROR_DISK_CACHE_DATABASE_ERROR
OPTIX_ERROR_DISK_CACHE_INVALID_DATA

OPTIX_ERROR_LAUNCH_FAILURE
OPTIX_ERROR_INVALID_DEVICE_CONTEXT
OPTIX_ERROR_CUDA_NOT_INITIALIZED
OPTIX_ERROR_INVALID_PTX
OPTIX_ERROR_INVALID_LAUNCH_PARAMETER
OPTIX_ERROR_INVALID_PAYLOAD_ACCESS
OPTIX_ERROR_INVALID_ATTRIBUTE_ACCESS
OPTIX_ERROR_INVALID_FUNCTION_USE
OPTIX_ERROR_INVALID_FUNCTION_ARGUMENTS
OPTIX_ERROR_PIPELINE_OUT_OF_CONSTANT_MEMORY
OPTIX_ERROR_PIPELINE_LINK_ERROR
OPTIX_ERROR_INTERNAL_COMPILER_ERROR
OPTIX_ERROR_DENOISER_MODEL_NOT_SET
OPTIX_ERROR_DENOISER_NOT_INITIALIZED
OPTIX_ERROR_ACCEL_NOT_COMPATIBLE
OPTIX_ERROR_NOT_SUPPORTED
OPTIX_ERROR_UNSUPPORTED_ABI_VERSION
OPTIX_ERROR_FUNCTION_TABLE_SIZE_MISMATCH
OPTIX_ERROR_INVALID_ENTRY_FUNCTION_OPTIONS
OPTIX_ERROR_LIBRARY_NOT_FOUND
OPTIX_ERROR_ENTRY_SYMBOL_NOT_FOUND
OPTIX_ERROR_CUDA_ERROR
OPTIX_ERROR_INTERNAL_ERROR
OPTIX_ERROR_UNKNOWN

3.11.4.25 enum OptixTransformFormat

Format of transform used in [OptixBuildInputTriangleArray::transformFormat](#).

Enumerator

OPTIX_TRANSFORM_FORMAT_NONE no transform, default for zero initialization
OPTIX_TRANSFORM_FORMAT_MATRIX_FLOAT12 3x4 row major affine matrix

3.11.4.26 enum OptixTransformType

Transform.

OptixTransformType is used by the device function [optixGetTransformTypeFromHandle\(\)](#) to determine the type of the OptixTraversableHandle returned from [optixGetTransformListHandle\(\)](#).

Enumerator

OPTIX_TRANSFORM_TYPE_NONE Not a transformation.

See Also

OPTIX_TRANSFORM_TYPE_STATIC_TRANSFORM[OptixStaticTransform](#)

See Also

OPTIX_TRANSFORM_TYPE_MATRIX_MOTION_TRANSFORM[OptixMatrixMotionTransform](#)

See Also

OPTIX_TRANSFORM_TYPE_SRT_MOTION_TRANSFORM[OptixSRTMotionTransform](#)

See Also

OPTIX_TRANSFORM_TYPE_INSTANCE[OptixInstance](#)

3.11.4.27 enum OptixTraversableGraphFlags

Specifies the set of valid traversable graphs that may be passed to invocation of [optixTrace\(\)](#). Flags may be bitwise combined.

Enumerator

OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_ANY Used to signal that any traversable graphs is valid. This flag is mutually exclusive with all other flags.

OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_GAS Used to signal that a traversable graph of a single Geometry Acceleration Structure (GAS) without any transforms is valid. This flag may be combined with other flags except for **OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_ANY**.

OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_LEVEL_INSTANCING Used to signal that a traversable graph of a single Instance Acceleration Structure (IAS) directly connected to Geometry Acceleration Structure (GAS) traversables without transform traversables in between is valid. This flag may be combined with other flags except for **OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_ANY**.

3.11.4.28 enum OptixTraversableType

Traversable Handles.

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

Enumerator

OPTIX_TRAVERSABLE_TYPE_STATIC_TRANSFORM Static transforms.

See Also

[OptixStaticTransform](#)

OPTIX_TRAVERSABLE_TYPE_MATRIX_MOTION_TRANSFORM Matrix motion transform.

See Also

[OptixMatrixMotionTransform](#)

OPTIX_TRAVERSABLE_TYPE_SRT_MOTION_TRANSFORM SRT motion transform.

See Also

[OptixSRTMotionTransform](#)

3.11.4.29 enum OptixVertexFormat

Format of vertices used in [OptixBuildInputTriangleArray::vertexFormat](#).

Enumerator

OPTIX_VERTEX_FORMAT_NONE No vertices.

OPTIX_VERTEX_FORMAT_FLOAT3 Vertices are represented by three floats.

OPTIX_VERTEX_FORMAT_FLOAT2 Vertices are represented by two floats.

OPTIX_VERTEX_FORMAT_HALF3 Vertices are represented by three halves.

OPTIX_VERTEX_FORMAT_HALF2 Vertices are represented by two halves.

OPTIX_VERTEX_FORMAT_SNORM16_3

OPTIX_VERTEX_FORMAT_SNORM16_2

3.12 Function Table

Classes

- struct [OptixFunctionTable](#)

Typedefs

- typedef struct [OptixFunctionTable](#) [OptixFunctionTable](#)

Variables

- [OptixFunctionTable](#) [g_optixFunctionTable](#)

3.12.1 Detailed Description

OptiX Function Table.

3.12.2 Typedef Documentation

3.12.2.1 typedef struct [OptixFunctionTable](#) [OptixFunctionTable](#)

The function table containing all API functions.

See [optixInit\(\)](#) and [optixInitWithHandle\(\)](#).

3.12.3 Variable Documentation

3.12.3.1 [OptixFunctionTable](#) [g_optixFunctionTable](#)

If the stubs in [optix_stubs.h](#) are used, then the function table needs to be defined in exactly one translation unit. This can be achieved by including this header file in that translation unit.

3.13 Utilities

Classes

- struct [OptixUtilDenoiserImageTile](#)

Functions

- [OptixResult optixUtilAccumulateStackSizes](#) ([OptixProgramGroup](#) programGroup, [OptixStackSizes](#) *stackSizes)
- [OptixResult optixUtilComputeStackSizes](#) (const [OptixStackSizes](#) *stackSizes, unsigned int maxTraceDepth, unsigned int maxCCDepth, unsigned int maxDCDepth, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)
- [OptixResult optixUtilComputeStackSizesDCSplit](#) (const [OptixStackSizes](#) *stackSizes, unsigned int dssDCFromTraversal, unsigned int dssDCFromState, unsigned int maxTraceDepth, unsigned int maxCCDepth, unsigned int maxDCDepthFromTraversal, unsigned int maxDCDepthFromState, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)
- [OptixResult optixUtilComputeStackSizesCssCCTree](#) (const [OptixStackSizes](#) *stackSizes, unsigned int cssCCTree, unsigned int maxTraceDepth, unsigned int maxDCDepth, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)
- [OptixResult optixUtilComputeStackSizesSimplePathTracer](#) ([OptixProgramGroup](#) programGroupRG, [OptixProgramGroup](#) programGroupMS1, const [OptixProgramGroup](#) *programGroupCH1, unsigned int programGroupCH1Count, [OptixProgramGroup](#) programGroupMS2, const [OptixProgramGroup](#) *programGroupCH2, unsigned int programGroupCH2Count, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)
- unsigned int [optixUtilGetPixelStride](#) (const [OptixImage2D](#) &image)
- [OptixResult optixUtilDenoiserSplitImage](#) (const [OptixImage2D](#) &input, const [OptixImage2D](#) &output, unsigned int overlapWindowSizeInPixels, unsigned int tileWidth, unsigned int tileHeight, std::vector< [OptixUtilDenoiserImageTile](#) > &tiles)
- [OptixResult optixUtilDenoiserInvokeTiled](#) ([OptixDenoiser](#) &denoiser, CUstream stream, const [OptixDenoiserParams](#) *params, CUdeviceptr denoiserState, size_t denoiserStateSizeInBytes, const [OptixImage2D](#) *inputLayers, unsigned int numInputLayers, const [OptixImage2D](#) *outputLayer, CUdeviceptr scratch, size_t scratchSizeInBytes, unsigned int overlapWindowSizeInPixels, unsigned int tileWidth, unsigned int tileHeight)
- [OptixResult optixInitWithHandle](#) (void **handlePtr)
- [OptixResult optixInit](#) (void)

3.13.1 Detailed Description

OptiX Utilities.

3.13.2 Function Documentation

3.13.2.1 OptixResult optixInit (void) [inline]

Loads the OptiX library and initializes the function table used by the stubs below.

A variant of [optixInitWithHandle\(\)](#) that does not make the handle to the loaded library available.

3.13.2.2 OptixResult optixInitWithHandle (void ** *handlePtr*) [inline]

Loads the OptiX library and initializes the function table used by the stubs below.

If *handlePtr* is not nullptr, an OS-specific handle to the library will be returned in **handlePtr*.

3.13.2.3 OptixResult optixUtilAccumulateStackSizes (OptixProgramGroup *programGroup*, OptixStackSizes * *stackSizes*)

Retrieves direct and continuation stack sizes for each program in the program group and accumulates the upper bounds in the corresponding output variables based on the semantic type of the program.

Before the first invocation of this function with a given instance of [OptixStackSizes](#), the members of that instance should be set to 0.

3.13.2.4 OptixResult optixUtilComputeStackSizes (const OptixStackSizes * *stackSizes*, unsigned int *maxTraceDepth*, unsigned int *maxCCDepth*, unsigned int *maxDCDepth*, unsigned int * *directCallableStackSizeFromTraversal*, unsigned int * *directCallableStackSizeFromState*, unsigned int * *continuationStackSize*)

Computes the stack size values needed to configure a pipeline.

See the programming guide for an explanation of the formula.

Parameters

in	<i>stackSizes</i>	Accumulated stack sizes of all programs in the call graph.
in	<i>maxTraceDepth</i>	Maximum depth of optixTrace() calls.
in	<i>maxCCDepth</i>	Maximum depth of calls trees of continuation callables.
in	<i>maxDCDepth</i>	Maximum depth of calls trees of direct callables.
out	<i>directCallableStackSizeFromTraversal</i>	Direct stack size requirement for direct callables invoked from IS or AH.

Parameters

out	<i>directCallableStackSizeFromState</i>	Direct stack size requirement for direct callables invoked from RG, MS, or CH.
out	<i>continuationStackSize</i>	Continuation stack requirement.

3.13.2.5 OptixResult optixUtilComputeStackSizesCssCCTree (

```

    const OptixStackSizes * stackSizes,
    unsigned int cssCCTree,
    unsigned int maxTraceDepth,
    unsigned int maxDCDepth,
    unsigned int * directCallableStackSizeFromTraversal,
    unsigned int * directCallableStackSizeFromState,
    unsigned int * continuationStackSize )

```

Computes the stack size values needed to configure a pipeline.

This variant is similar to [optixUtilComputeStackSizes\(\)](#), except that it expects the value *cssCCTree* instead of *cssCC* and *maxCCDepth*.

See programming guide for an explanation of the formula.

Parameters

in	<i>stackSizes</i>	Accumulated stack sizes of all programs in the call graph.
in	<i>cssCCTree</i>	Maximum stack size used by calls trees of continuation callables.
in	<i>maxTraceDepth</i>	Maximum depth of optixTrace() calls.
in	<i>maxDCDepth</i>	Maximum depth of calls trees of direct callables.
out	<i>directCallableStackSizeFromTraversal</i>	Direct stack size requirement for direct callables invoked from IS or AH.
out	<i>directCallableStackSizeFromState</i>	Direct stack size requirement for direct callables invoked from RG, MS, or CH.
out	<i>continuationStackSize</i>	Continuation stack requirement.

3.13.2.6 OptixResult optixUtilComputeStackSizesDCSplit (

```

    const OptixStackSizes * stackSizes,
    unsigned int dssDCFromTraversal,
    unsigned int dssDCFromState,
    unsigned int maxTraceDepth,
    unsigned int maxCCDepth,
    unsigned int maxDCDepthFromTraversal,
    unsigned int maxDCDepthFromState,

```

```

unsigned int * directCallableStackSizeFromTraversal,
unsigned int * directCallableStackSizeFromState,
unsigned int * continuationStackSize )

```

Computes the stack size values needed to configure a pipeline.

This variant is similar to `optixUtilComputeStackSizes()`, except that it expects the values `dssDC` and `maxDCDepth` split by call site semantic.

See programming guide for an explanation of the formula.

Parameters

in	<i>stackSizes</i>	Accumulated stack sizes of all programs in the call graph.
in	<i>dssDCFromTraversal</i>	Accumulated direct stack size of all DC programs invoked from IS or AH.
in	<i>dssDCFromState</i>	Accumulated direct stack size of all DC programs invoked from RG, MS, or CH.
in	<i>maxTraceDepth</i>	Maximum depth of <code>optixTrace()</code> calls.
in	<i>maxCCDepth</i>	Maximum depth of calls trees of continuation callables.
in	<i>maxDCDepthFromTraversal</i>	Maximum depth of calls trees of direct callables invoked from IS or AH.
in	<i>maxDCDepthFromState</i>	Maximum depth of calls trees of direct callables invoked from RG, MS, or CH.
out	<i>directCallableStackSizeFromTraversal</i>	Direct stack size requirement for direct callables invoked from IS or AH.
out	<i>directCallableStackSizeFromState</i>	Direct stack size requirement for direct callables invoked from RG, MS, or CH.
out	<i>continuationStackSize</i>	Continuation stack requirement.

3.13.2.7 OptixResult optixUtilComputeStackSizesSimplePathTracer (

```

OptixProgramGroup programGroupRG,
OptixProgramGroup programGroupMS1,
const OptixProgramGroup * programGroupCH1,
unsigned int programGroupCH1Count,
OptixProgramGroup programGroupMS2,
const OptixProgramGroup * programGroupCH2,
unsigned int programGroupCH2Count,
unsigned int * directCallableStackSizeFromTraversal,
unsigned int * directCallableStackSizeFromState,
unsigned int * continuationStackSize )

```

Computes the stack size values needed to configure a pipeline.

This variant is a specialization of [optixUtilComputeStackSizes\(\)](#) for a simple path tracer with the following assumptions: There are only two ray types, camera rays and shadow rays. There are only RG, MS, and CH programs, and no AH, IS, CC, or DC programs. The camera rays invoke only the miss and closest hit programs MS1 and CH1, respectively. The CH1 program might trace shadow rays, which invoke only the miss and closest hit programs MS2 and CH2, respectively.

For flexibility, we allow for each of CH1 and CH2 not just one single program group, but an array of programs groups, and compute the maximas of the stack size requirements per array.

See programming guide for an explanation of the formula.

3.13.2.8 OptixResult optixUtilDenoiserInvokeTiled (

```

    OptixDenoiser & denoiser,
    CUstream stream,
    const OptixDenoiserParams * params,
    CUdeviceptr denoiserState,
    size_t denoiserStateSizeInBytes,
    const OptixImage2D * inputLayers,
    unsigned int numInputLayers,
    const OptixImage2D * outputLayer,
    CUdeviceptr scratch,
    size_t scratchSizeInBytes,
    unsigned int overlapWindowSizeInPixels,
    unsigned int tileWidth,
    unsigned int tileHeight ) [inline]

```

Run denoiser on input layers see [optixDenoiserInvoke](#) additional parameters:

Runs the denoiser on the input layers on a single GPU and stream using [optixDenoiserInvoke](#). If the input layers' dimensions are larger than the specified tile size, the image is divided into tiles using [optixUtilDenoiserSplitImage](#), and multiple back-to-back invocations are performed in order to reuse the scratch space. Multiple tiles can be invoked concurrently if [optixUtilDenoiserSplitImage](#) is used directly and multiple scratch allocations for each concurrent invocation are used. The input parameters are the same as [optixDenoiserInvoke](#) except for the addition of the maximum tile size.

Parameters

in	<i>denoiser</i>	
in	<i>stream</i>	
in	<i>params</i>	
in	<i>denoiserState</i>	
in	<i>denoiserStateSizeInBytes</i>	
in	<i>inputLayers</i>	
in	<i>numInputLayers</i>	
in	<i>inputOffsetX</i>	
in	<i>inputOffsetY</i>	

Parameters

in	<i>outputLayer</i>	
in	<i>scratch</i>	
in	<i>scratchSizeInBytes</i>	
in	<i>overlapWindowSizeInPixels</i>	
in	<i>tileWidth</i>	
in	<i>tileHeight</i>	

3.13.2.9 OptixResult optixUtilDenoiserSplitImage (
const OptixImage2D & *input*,
const OptixImage2D & *output*,
unsigned int *overlapWindowSizeInPixels*,
unsigned int *tileWidth*,
unsigned int *tileHeight*,
std::vector< OptixUtilDenoiserImageTile > & *tiles*) [inline]

Split image into 2D tiles given horizontal and vertical tile size.

Parameters

in	<i>input</i>	full resolution input image to be split
in	<i>output</i>	full resolution output image
in	<i>overlapWindowSizeInPixels</i>	see OptixDenoiserSizes , optixDenoiserComputeMemoryResources
in	<i>tileWidth</i>	maximum width of tiles
in	<i>tileHeight</i>	maximum height of tiles
out	<i>tiles</i>	list of tiles covering the input image

3.13.2.10 unsigned int optixUtilGetPixelStride (
const OptixImage2D & *image*) [inline]

Return pixel stride in bytes for the given pixel format if the pixelStrideInBytes member of the image is zero. Otherwise return pixelStrideInBytes from the image.

Parameters

in	<i>image</i>	Image containing the pixel stride
----	--------------	-----------------------------------

4 Namespace Documentation

4.1 optix_impl Namespace Reference

Functions

- static __forceinline__
__device__ void [optixDumpStaticTransformFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__
__device__ void [optixDumpMotionMatrixTransformFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__
__device__ void [optixDumpSrtMatrixTransformFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__
__device__ void [optixDumpInstanceFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__
__device__ void [optixDumpTransform](#) (OptixTraversableHandle handle)
- static __forceinline__
__device__ void [optixDumpTransformList](#) ()
- static __forceinline__
__device__ void [optixDumpExceptionDetails](#) ()
- static __forceinline__
__device__ float4 [optixAddFloat4](#) (const float4 &a, const float4 &b)
- static __forceinline__
__device__ float4 [optixMulFloat4](#) (const float4 &a, float b)
- static __forceinline__
__device__ uint4 [optixLdg](#) (unsigned long long addr)
- template<class T >
static __forceinline__ __device__ T [optixLoadReadOnlyAlign16](#) (const T *ptr)
- static __forceinline__
__device__ float4 [optixMultiplyRowMatrix](#) (const float4 vec, const float4 m0, const float4 m1, const float4 m2)
- static __forceinline__
__device__ void [optixGetMatrixFromSrt](#) (float4 &m0, float4 &m1, float4 &m2, const [OptixSRTData](#) &srt)
- static __forceinline__
__device__ void [optixInvertMatrix](#) (float4 &m0, float4 &m1, float4 &m2)
- static __forceinline__
__device__ void [optixLoadInterpolatedMatrixKey](#) (float4 &m0, float4 &m1, float4 &m2, const float4 *matrix, const float t1)
- static __forceinline__
__device__ void [optixLoadInterpolatedSrtKey](#) (float4 &srt0, float4 &srt1, float4 &srt2, float4 &srt3, const float4 *srt, const float t1)
- static __forceinline__
__device__ void [optixResolveMotionKey](#) (float &localt, int &key, const [OptixMotionOptions](#) &options, const float globalt)
- static __forceinline__
__device__ void [optixGetInterpolatedTransformation](#) (float4 &trf0, float4 &trf1, float4 &trf2, const [OptixMatrixMotionTransform](#) *transformData, const float time)

- static `__forceinline__`
`__device__ void optixGetInterpolatedTransformation` (float4 &trf0, float4 &trf1, float4 &trf2, const `OptixSRTMotionTransform` *transformData, const float time)
- static `__forceinline__`
`__device__ void optixGetInterpolatedTransformationFromHandle` (float4 &trf0, float4 &trf1, float4 &trf2, const `OptixTraversableHandle` handle, const float time, const bool objectToWorld)
- static `__forceinline__`
`__device__ void optixGetWorldToObjectTransformMatrix` (float4 &m0, float4 &m1, float4 &m2)
- static `__forceinline__`
`__device__ void optixGetObjectToWorldTransformMatrix` (float4 &m0, float4 &m1, float4 &m2)
- static `__forceinline__`
`__device__ float3 optixTransformPoint` (const float4 &m0, const float4 &m1, const float4 &m2, const float3 &p)
- static `__forceinline__`
`__device__ float3 optixTransformVector` (const float4 &m0, const float4 &m1, const float4 &m2, const float3 &v)
- static `__forceinline__`
`__device__ float3 optixTransformNormal` (const float4 &m0, const float4 &m1, const float4 &m2, const float3 &n)

4.1.1 Function Documentation

- 4.1.1.1** static `__forceinline__` `__device__` float4 optix_impl::optixAddFloat4 (const float4 & *a*, const float4 & *b*) [static]
- 4.1.1.2** static `__forceinline__` `__device__` void optix_impl::optixDumpExceptionDetails () [static]
- 4.1.1.3** static `__forceinline__` `__device__` void optix_impl::optixDumpInstanceFromHandle (`OptixTraversableHandle` *handle*) [static]
- 4.1.1.4** static `__forceinline__` `__device__` void optix_impl::optixDumpMotionMatrixTransformFromHandle (`OptixTraversableHandle` *handle*) [static]
- 4.1.1.5** static `__forceinline__` `__device__` void optix_impl::optixDumpSrtMatrixTransformFromHandle (`OptixTraversableHandle` *handle*) [static]
- 4.1.1.6** static `__forceinline__` `__device__` void optix_impl::optixDumpStaticTransformFromHandle (`OptixTraversableHandle` *handle*) [static]
- 4.1.1.7** static `__forceinline__` `__device__` void optix_impl::optixDumpTransform (

OptixTraversableHandle *handle*) [static]

4.1.1.8 static __forceinline__ __device__ void optix_impl::optixDumpTransformList ()
[static]

4.1.1.9 static __forceinline__ __device__ void optix_impl::optixGetInterpolatedTransformation
(
 float4 & *trf0*,
 float4 & *trf1*,
 float4 & *trf2*,
 const OptixMatrixMotionTransform * *transformData*,
 const float *time*) [static]

4.1.1.10 static __forceinline__ __device__ void optix_impl::optixGetInterpolatedTransformation
(
 float4 & *trf0*,
 float4 & *trf1*,
 float4 & *trf2*,
 const OptixSRTMotionTransform * *transformData*,
 const float *time*) [static]

4.1.1.11 static __forceinline__ __device__ void optix_impl::optixGetInterpolatedTransformationFromHandle
(
 float4 & *trf0*,
 float4 & *trf1*,
 float4 & *trf2*,
 const OptixTraversableHandle *handle*,
 const float *time*,
 const bool *objectToWorld*) [static]

4.1.1.12 static __forceinline__ __device__ void optix_impl::optixGetMatrixFromSrt (
 float4 & *m0*,
 float4 & *m1*,
 float4 & *m2*,
 const OptixSRTData & *srt*) [static]

4.1.1.13 static __forceinline__ __device__ void optix_impl::optixGetObjectToWorldTransformMatrix (
 float4 & *m0*,
 float4 & *m1*,

float4 & *m2*) [static]

4.1.1.14 static __forceinline__ __device__ void optix_impl::optixGetWorldToObjectTransformMatrix (
float4 & *m0*,
float4 & *m1*,
float4 & *m2*) [static]

4.1.1.15 static __forceinline__ __device__ void optix_impl::optixInvertMatrix (
float4 & *m0*,
float4 & *m1*,
float4 & *m2*) [static]

4.1.1.16 static __forceinline__ __device__ uint4 optix_impl::optixLdg (
unsigned long long *addr*) [static]

4.1.1.17 static __forceinline__ __device__ void optix_impl::optixLoadInterpolatedMatrixKey (
float4 & *m0*,
float4 & *m1*,
float4 & *m2*,
const float4 * *matrix*,
const float *t1*) [static]

4.1.1.18 static __forceinline__ __device__ void optix_impl::optixLoadInterpolatedSrtKey (
float4 & *srt0*,
float4 & *srt1*,
float4 & *srt2*,
float4 & *srt3*,
const float4 * *srt*,
const float *t1*) [static]

4.1.1.19 template<class T > static __forceinline__ __device__ T
optix_impl::optixLoadReadOnlyAlign16 (
const T * *ptr*) [static]

4.1.1.20 static __forceinline__ __device__ float4 optix_impl::optixMulFloat4 (
const float4 & *a*,
float *b*) [static]

4.1.1.21 static __forceinline__ __device__ float4 optix_impl::optixMultiplyRowMatrix (
const float4 *vec*,
const float4 *m0*,
const float4 *m1*,

```
const float4 m2 ) [static]
```

```
4.1.1.22 static __forceinline__ __device__ void optix_impl::optixResolveMotionKey (
    float & localt,
    int & key,
    const OptixMotionOptions & options,
    const float globalt ) [static]
```

```
4.1.1.23 static __forceinline__ __device__ float3 optix_impl::optixTransformNormal (
    const float4 & m0,
    const float4 & m1,
    const float4 & m2,
    const float3 & n ) [static]
```

```
4.1.1.24 static __forceinline__ __device__ float3 optix_impl::optixTransformPoint (
    const float4 & m0,
    const float4 & m1,
    const float4 & m2,
    const float3 & p ) [static]
```

```
4.1.1.25 static __forceinline__ __device__ float3 optix_impl::optixTransformVector (
    const float4 & m0,
    const float4 & m1,
    const float4 & m2,
    const float3 & v ) [static]
```

5 Class Documentation

5.1 OptixAabb Struct Reference

Public Attributes

- float [minX](#)
- float [minY](#)
- float [minZ](#)
- float [maxX](#)
- float [maxY](#)
- float [maxZ](#)

5.1.1 Detailed Description

AABB inputs.

5.1.2 Member Data Documentation

5.1.2.1 float OptixAabb::maxX

Upper extent in X direction.

5.1.2.2 float OptixAabb::maxY

Upper extent in Y direction.

5.1.2.3 float OptixAabb::maxZ

Upper extent in Z direction.

5.1.2.4 float OptixAabb::minX

Lower extent in X direction.

5.1.2.5 float OptixAabb::minY

Lower extent in Y direction.

5.1.2.6 float OptixAabb::minZ

Lower extent in Z direction.

5.2 OptixAccelBufferSizes Struct Reference

Public Attributes

- [size_t outputSizeInBytes](#)
- [size_t tempSizeInBytes](#)
- [size_t tempUpdateSizeInBytes](#)

5.2.1 Detailed Description

Struct for querying builder allocation requirements.

Once queried the sizes should be used to allocate device memory of at least these sizes.

See Also

[optixAccelComputeMemoryUsage\(\)](#)

5.2.2 Member Data Documentation

5.2.2.1 size_t OptixAccelBufferSizes::outputSizeInBytes

The size in bytes required for the outputBuffer parameter to optixAccelBuild when doing a build (OPTIX_BUILD_OPERATION_BUILD).

5.2.2.2 `size_t OptixAccelBufferSizes::tempSizeInBytes`

The size in bytes required for the tempBuffer parameter to optixAccelBuild when doing a build (OPTIX_BUILD_OPERATION_BUILD).

5.2.2.3 `size_t OptixAccelBufferSizes::tempUpdateSizeInBytes`

The size in bytes required for the tempBuffer parameter to optixAccelBuild when doing an update (OPTIX_BUILD_OPERATION_UPDATE). This value can be different than tempSizeInBytes used for a full build. Only non-zero if OPTIX_BUILD_FLAG_ALLOW_UPDATE flag is set in [OptixAccelBuildOptions](#).

5.3 OptixAccelBuildOptions Struct Reference

Public Attributes

- unsigned int [buildFlags](#)
- [OptixBuildOperation](#) operation
- [OptixMotionOptions](#) motionOptions

5.3.1 Detailed Description

Build options for acceleration structures.

See Also

[optixAccelComputeMemoryUsage\(\)](#), [optixAccelBuild\(\)](#)

5.3.2 Member Data Documentation

5.3.2.1 unsigned int `OptixAccelBuildOptions::buildFlags`

Combinations of OptixBuildFlags.

5.3.2.2 `OptixMotionOptions OptixAccelBuildOptions::motionOptions`

Options for motion.

5.3.2.3 `OptixBuildOperation OptixAccelBuildOptions::operation`

If OPTIX_BUILD_OPERATION_UPDATE the output buffer is assumed to contain the result of a full build with OPTIX_BUILD_FLAG_ALLOW_UPDATE set and using the same number of primitives. It is updated incrementally to reflect the current position of the primitives.

5.4 OptixAccelEmitDesc Struct Reference

Public Attributes

- [CUdeviceptr](#) result
- [OptixAccelPropertyType](#) type

5.4.1 Detailed Description

Specifies a type and output destination for emitted post-build properties.

See Also

[optixAccelBuild\(\)](#)

5.4.2 Member Data Documentation

5.4.2.1 CUdeviceptr OptixAccelEmitDesc::result

Output buffer for the properties.

5.4.2.2 OptixAccelPropertyType OptixAccelEmitDesc::type

Requested property.

5.5 OptixAccelRelocationInfo Struct Reference

Public Attributes

- unsigned long long [info](#) [4]

5.5.1 Detailed Description

Used to store information related to relocation of acceleration structures.

See Also

[optixAccelGetRelocationInfo\(\)](#), [optixAccelCheckRelocationCompatibility\(\)](#), [optixAccelRelocate\(\)](#)

5.5.2 Member Data Documentation

5.5.2.1 unsigned long long OptixAccelRelocationInfo::info[4]

Opaque data, used internally, should not be modified.

5.6 OptixBuildInput Struct Reference

Public Attributes

- [OptixBuildInputType](#) type

```

• union {
    OptixBuildInputTriangleArray triangleArray
    OptixBuildInputCurveArray curveArray
    OptixBuildInputCustomPrimitiveArray customPrimitiveArray
    OptixBuildInputInstanceArray instanceArray
    char pad [1024]
};

```

5.6.1 Detailed Description

Build inputs.

All of them support motion and the size of the data arrays needs to match the number of motion steps

See Also

[optixAccelComputeMemoryUsage\(\)](#), [optixAccelBuild\(\)](#)

5.6.2 Member Data Documentation

5.6.2.1 union { ... }

5.6.2.2 OptixBuildInputCurveArray OptixBuildInput::curveArray

Curve inputs.

5.6.2.3 OptixBuildInputCustomPrimitiveArray OptixBuildInput::customPrimitiveArray

Custom primitive inputs.

5.6.2.4 OptixBuildInputInstanceArray OptixBuildInput::instanceArray

Instance and instance pointer inputs.

5.6.2.5 char OptixBuildInput::pad[1024]

5.6.2.6 OptixBuildInputTriangleArray OptixBuildInput::triangleArray

Triangle inputs.

5.6.2.7 OptixBuildInputType OptixBuildInput::type

The type of the build input.

5.7 OptixBuildInputCurveArray Struct Reference

Public Attributes

```

• OptixPrimitiveType curveType

```

- unsigned int `numPrimitives`
- const `CUdeviceptr` * `vertexBuffers`
- unsigned int `numVertices`
- unsigned int `vertexStrideInBytes`
- const `CUdeviceptr` * `widthBuffers`
- unsigned int `widthStrideInBytes`
- const `CUdeviceptr` * `normalBuffers`
- unsigned int `normalStrideInBytes`
- `CUdeviceptr` `indexBuffer`
- unsigned int `indexStrideInBytes`
- unsigned int `flag`
- unsigned int `primitiveIndexOffset`

5.7.1 Detailed Description

Curve inputs.

A curve is a swept surface defined by a 3D spline curve and a varying width (radius). A curve (or "strand") of degree d (3=cubic, 2=quadratic, 1=linear) is represented by $N > d$ vertices and N width values, and comprises $N - d$ segments. Each segment is defined by $d+1$ consecutive vertices. Each curve may have a different number of vertices.

OptiX describes the curve array as a list of curve segments. The primitive id is the segment number. It is the user's responsibility to maintain a mapping between curves and curve segments. Each index buffer entry $i = \text{indexBuffer}[\text{primid}]$ specifies the start of a curve segment, represented by $d+1$ consecutive vertices in the vertex buffer, and $d+1$ consecutive widths in the width buffer. Width is interpolated the same way vertices are interpolated, that is, using the curve basis.

Each curves build input has only one SBT record. To create curves with different materials in the same BVH, use multiple build inputs.

See Also

[OptixBuildInput::curveArray](#)

5.7.2 Member Data Documentation

5.7.2.1 `OptixPrimitiveType` `OptixBuildInputCurveArray::curveType`

Curve degree and basis.

See Also

[OptixPrimitiveType](#)

5.7.2.2 `unsigned int` `OptixBuildInputCurveArray::flag`

Combination of `OptixGeometryFlags` describing the primitive behavior.

5.7.2.3 CUdeviceptr OptixBuildInputCurveArray::indexBuffer

Device pointer to array of unsigned ints, one per curve segment. This buffer is required (unlike for [OptixBuildInputTriangleArray](#)). Each index is the start of degree+1 consecutive vertices in vertexBuffers, and corresponding widths in widthBuffers and normals in normalBuffers. These define a single segment. Size of array is numPrimitives.

5.7.2.4 unsigned int OptixBuildInputCurveArray::indexStrideInBytes

Stride between indices. If set to zero, indices are assumed to be tightly packed and stride is sizeof(unsigned int).

5.7.2.5 const CUdeviceptr* OptixBuildInputCurveArray::normalBuffers

Reserved for future use.

5.7.2.6 unsigned int OptixBuildInputCurveArray::normalStrideInBytes

Reserved for future use.

5.7.2.7 unsigned int OptixBuildInputCurveArray::numPrimitives

Number of primitives. Each primitive is a polynomial curve segment.

5.7.2.8 unsigned int OptixBuildInputCurveArray::numVertices

Number of vertices in each buffer in vertexBuffers.

5.7.2.9 unsigned int OptixBuildInputCurveArray::primitiveIndexOffset

Primitive index bias, applied in [optixGetPrimitiveIndex\(\)](#). Sum of primitiveIndexOffset and number of primitives must not overflow 32bits.

5.7.2.10 const CUdeviceptr* OptixBuildInputCurveArray::vertexBuffers

Pointer to host array of device pointers, one per motion step. Host array size must match number of motion keys as set in [OptixMotionOptions](#) (or an array of size 1 if [OptixMotionOptions::numKeys](#) is set to 1). Each per-motion-key device pointer must point to an array of floats (the vertices of the curves).

5.7.2.11 unsigned int OptixBuildInputCurveArray::vertexStrideInBytes

Stride between vertices. If set to zero, vertices are assumed to be tightly packed and stride is sizeof(float3).

5.7.2.12 const CUdeviceptr* OptixBuildInputCurveArray::widthBuffers

Parallel to vertexBuffers: a device pointer per motion step, each with numVertices float values, specifying the curve width (radius) corresponding to each vertex.

5.7.2.13 unsigned int OptixBuildInputCurveArray::widthStrideInBytes

Stride between widths. If set to zero, widths are assumed to be tightly packed and stride is sizeof(float).

5.8 OptixBuildInputCustomPrimitiveArray Struct Reference

Public Attributes

- const [CUdeviceptr](#) * aabbBuffers
- unsigned int numPrimitives
- unsigned int strideInBytes
- const unsigned int * flags
- unsigned int numSbtRecords
- [CUdeviceptr](#) sbtIndexOffsetBuffer
- unsigned int sbtIndexOffsetSizeInBytes
- unsigned int sbtIndexOffsetStrideInBytes
- unsigned int primitiveIndexOffset

5.8.1 Detailed Description

Custom primitive inputs.

See Also

[OptixBuildInput::customPrimitiveArray](#)

5.8.2 Member Data Documentation

5.8.2.1 const [CUdeviceptr](#)* OptixBuildInputCustomPrimitiveArray::aabbBuffers

Points to host array of device pointers to AABBs (type [OptixAabb](#)), one per motion step. Host array size must match number of motion keys as set in [OptixMotionOptions](#) (or an array of size 1 if [OptixMotionOptions::numKeys](#) is set to 1). Each device pointer must be a multiple of OPTIX_AABB_BUFFER_BYTE_ALIGNMENT.

5.8.2.2 const unsigned int* OptixBuildInputCustomPrimitiveArray::flags

Array of flags, to specify flags per sbt record, combinations of [OptixGeometryFlags](#) describing the primitive behavior, size must match numSbtRecords.

5.8.2.3 unsigned int OptixBuildInputCustomPrimitiveArray::numPrimitives

Number of primitives in each buffer (i.e., per motion step) in [OptixBuildInputCustomPrimitiveArray::aabbBuffers](#).

5.8.2.4 unsigned int OptixBuildInputCustomPrimitiveArray::numSbtRecords

Number of sbt records available to the sbt index offset override.

5.8.2.5 unsigned int OptixBuildInputCustomPrimitiveArray::primitiveIndexOffset

Primitive index bias, applied in [optixGetPrimitiveIndex\(\)](#). Sum of primitiveIndexOffset and number of primitive must not overflow 32bits.

5.8.2.6 CUdeviceptr OptixBuildInputCustomPrimitiveArray::sbtIndexOffsetBuffer

Device pointer to per-primitive local sbt index offset buffer. May be NULL. Every entry must be in range [0,numSbtRecords-1]. Size needs to be the number of primitives.

5.8.2.7 unsigned int OptixBuildInputCustomPrimitiveArray::sbtIndexOffsetSizeInBytes

Size of type of the sbt index offset. Needs to be 0, 1, 2 or 4 (8, 16 or 32 bit).

5.8.2.8 unsigned int OptixBuildInputCustomPrimitiveArray::sbtIndexOffsetStrideInBytes

Stride between the index offsets. If set to zero, the offsets are assumed to be tightly packed and the stride matches the size of the type (sbtIndexOffsetSizeInBytes).

5.8.2.9 unsigned int OptixBuildInputCustomPrimitiveArray::strideInBytes

Stride between AABBs (per motion key). If set to zero, the aabbs are assumed to be tightly packed and the stride is assumed to be sizeof(OptixAabb). If non-zero, the value must be a multiple of OPTIX_AABB_BUFFER_BYTE_ALIGNMENT.

5.9 OptixBuildInputInstanceArray Struct Reference

Public Attributes

- [CUdeviceptr](#) instances
- unsigned int numInstances
- [CUdeviceptr](#) aabbs
- unsigned int numAabbs

5.9.1 Detailed Description

Instance and instance pointer inputs.

See Also

[OptixBuildInput::instanceArray](#)

5.9.2 Member Data Documentation

5.9.2.1 CUdeviceptr OptixBuildInputInstanceArray::aabbs

Optional AABBs. In [OptixAabb](#) format.

Required for traversables ([OptixMatrixMotionTransform](#), [OptixSRTMotionTransform](#), [OptixStaticTransform](#)) and certain configurations of motion AS as instance. Will be ignored for non-motion AS, since no AABBs are required. May be NULL in that case.

The following table illustrates this (IAS is Instance Acceleration Structure) instance type | traversable | motion AS | static AS building a motion IAS | required | required | ignored building a static IAS | required | ignored | ignored

The AABBs must enclose the space spanned by the referenced handle of the instance. I.e., the AABBs are in 'object space' and must NOT include the transformation of the instance itself. If the instance's handle references a traversable the AABBs must contain the transformation of the traversable.

If [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS` the unused pointers for unused aabbs may be set to NULL.

If [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCES` this pointer must be a multiple of `OPTIX_AABB_BUFFER_BYTE_ALIGNMENT`. If [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS` the array elements must be a multiple of `OPTIX_AABB_BUFFER_BYTE_ALIGNMENT`.

Motion:

In case of motion ([OptixMotionOptions::numKeys](#) >= 2), [OptixMotionOptions::numKeys](#) aabbs are expected per instance, e.g., for N instances and M motion keys: `aabb[inst0][t0]`, `aabb[inst0][t1]`, ..., `aabb[inst0][tM-1]`, ..., `aabb[instN-1][t0]`, `aabb[instN-1][t1]`, ..., `aabb[instN-1][tM-1]`.

If [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCES` aabbs must be a device pointer to an array of $N * M * 6$ floats.

If [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS` aabbs must be a device pointer to an array of N device pointers, each pointing to an array of $M * 6$ floats in [OptixAabb](#) format. Pointers may be NULL if the aabbs are not required. Hence, if the second instance (inst1) points to a static GAS, aabbs are not required for that instance. While being ignored, aabbs must still be a device pointer to an array of N elements.

In case of `OPTIX_BUILD_INPUT_TYPE_INSTANCES`, the second element (with a size of $M * 6$ floats) will be ignored. In case of `OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS`, the second element (with a size of pointer to $M * 6$ floats) can be NULL.

5.9.2.2 CUdeviceptr OptixBuildInputInstanceArray::instances

If [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS` instances and aabbs should be interpreted as arrays of pointers instead of arrays of structs.

This pointer must be a multiple of `OPTIX_INSTANCE_BYTE_ALIGNMENT` if [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCES`. The array elements must be a multiple of `OPTIX_INSTANCE_BYTE_ALIGNMENT` if [OptixBuildInput::type](#) is `OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS`.

5.9.2.3 unsigned int OptixBuildInputInstanceArray::numAabbs

number of aabbs, in case of motion, this needs to match numInstances multiplied with [OptixMotionOptions::numKeys](#)

5.9.2.4 unsigned int OptixBuildInputInstanceArray::numInstances

Number of elements in [OptixBuildInputInstanceArray::instances](#).

5.10 OptixBuildInputTriangleArray Struct Reference

Public Attributes

- [const CUdeviceptr * vertexBuffers](#)
- [unsigned int numVertices](#)
- [OptixVertexFormat vertexFormat](#)
- [unsigned int vertexStrideInBytes](#)
- [CUdeviceptr indexBuffer](#)
- [unsigned int numIndexTriplets](#)
- [OptixIndicesFormat indexFormat](#)
- [unsigned int indexStrideInBytes](#)
- [CUdeviceptr preTransform](#)
- [const unsigned int * flags](#)
- [unsigned int numSbtRecords](#)
- [CUdeviceptr sbtIndexOffsetBuffer](#)
- [unsigned int sbtIndexOffsetSizeInBytes](#)
- [unsigned int sbtIndexOffsetStrideInBytes](#)
- [unsigned int primitiveIndexOffset](#)
- [OptixTransformFormat transformFormat](#)

5.10.1 Detailed Description

Triangle inputs.

See Also

[OptixBuildInput::triangleArray](#)

5.10.2 Member Data Documentation

5.10.2.1 const unsigned int* OptixBuildInputTriangleArray::flags

Array of flags, to specify flags per sbt record, combinations of [OptixGeometryFlags](#) describing the primitive behavior, size must match [numSbtRecords](#).

5.10.2.2 CUdeviceptr OptixBuildInputTriangleArray::indexBuffer

Optional pointer to array of 16 or 32-bit int triplets, one triplet per triangle. The minimum alignment must match the natural alignment of the type as specified in the [indexFormat](#), i.e., for [OPTIX_INDICES_FORMAT_UNSIGNED_INT3](#) 4-byte and for [OPTIX_INDICES_FORMAT_UNSIGNED_SHORT3](#) a 2-byte alignment.

5.10.2.3 OptixIndicesFormat OptixBuildInputTriangleArray::indexFormat

See Also

[OptixIndicesFormat](#)

5.10.2.4 unsigned int OptixBuildInputTriangleArray::indexStrideInBytes

Stride between triplets of indices. If set to zero, indices are assumed to be tightly packed and stride is inferred from indexFormat.

5.10.2.5 unsigned int OptixBuildInputTriangleArray::numIndexTriplets

Size of array in [OptixBuildInputTriangleArray::indexBuffer](#). For build, needs to be zero if indexBuffer is nullptr.

5.10.2.6 unsigned int OptixBuildInputTriangleArray::numSbtRecords

Number of sbt records available to the sbt index offset override.

5.10.2.7 unsigned int OptixBuildInputTriangleArray::numVertices

Number of vertices in each of buffer in [OptixBuildInputTriangleArray::vertexBuffers](#).

5.10.2.8 CUdeviceptr OptixBuildInputTriangleArray::preTransform

Optional pointer to array of floats representing a 3x4 row major affine transformation matrix. This pointer must be a multiple of OPTIX_GEOMETRY_TRANSFORM_BYTE_ALIGNMENT.

5.10.2.9 unsigned int OptixBuildInputTriangleArray::primitiveIndexOffset

Primitive index bias, applied in [optixGetPrimitiveIndex\(\)](#). Sum of primitiveIndexOffset and number of triangles must not overflow 32bits.

5.10.2.10 CUdeviceptr OptixBuildInputTriangleArray::sbtIndexOffsetBuffer

Device pointer to per-primitive local sbt index offset buffer. May be NULL. Every entry must be in range [0,numSbtRecords-1]. Size needs to be the number of primitives.

5.10.2.11 unsigned int OptixBuildInputTriangleArray::sbtIndexOffsetSizeInBytes

Size of type of the sbt index offset. Needs to be 0, 1, 2 or 4 (8, 16 or 32 bit).

5.10.2.12 unsigned int OptixBuildInputTriangleArray::sbtIndexOffsetStrideInBytes

Stride between the index offsets. If set to zero, the offsets are assumed to be tightly packed and the stride matches the size of the type (sbtIndexOffsetSizeInBytes).

5.10.2.13 OptixTransformFormat OptixBuildInputTriangleArray::transformFormat

See Also

[OptixTransformFormat](#)

5.10.2.14 `const CUdeviceptr* OptixBuildInputTriangleArray::vertexBuffers`

Points to host array of device pointers, one per motion step. Host array size must match the number of motion keys as set in [OptixMotionOptions](#) (or an array of size 1 if [OptixMotionOptions::numKeys](#) is set to 0 or 1). Each per motion key device pointer must point to an array of vertices of the triangles in the format as described by `vertexFormat`. The minimum alignment must match the natural alignment of the type as specified in the `vertexFormat`, i.e., for `OPTIX_VERTEX_FORMAT_FLOATX` 4-byte, for all others a 2-byte alignment. However, an 16-byte stride (and buffer alignment) is recommended for vertices of format `OPTIX_VERTEX_FORMAT_FLOAT3` for GAS build performance.

5.10.2.15 `OptixVertexFormat OptixBuildInputTriangleArray::vertexFormat`

See Also

[OptixVertexFormat](#)

5.10.2.16 `unsigned int OptixBuildInputTriangleArray::vertexStrideInBytes`

Stride between vertices. If set to zero, vertices are assumed to be tightly packed and stride is inferred from `vertexFormat`.

5.11 OptixBuiltinISOptions Struct Reference

Public Attributes

- [OptixPrimitiveType builtinISModuleType](#)
- `int usesMotionBlur`

5.11.1 Detailed Description

Specifies the options for retrieving an intersection program for a built-in primitive type. The primitive type must not be `OPTIX_PRIMITIVE_TYPE_CUSTOM`.

See Also

[optixBuiltinISModuleGet\(\)](#)

5.11.2 Member Data Documentation

5.11.2.1 OptixPrimitiveType OptixBuiltinISOptions::builtinISModuleType

5.11.2.2 int OptixBuiltinISOptions::usesMotionBlur

5.12 OptixDenoiserOptions Struct Reference

Public Attributes

- [OptixDenoiserInputKind](#) inputKind

5.12.1 Detailed Description

Options used by the denoiser.

See Also

[optixDenoiserCreate\(\)](#)

5.12.2 Member Data Documentation

5.12.2.1 OptixDenoiserInputKind OptixDenoiserOptions::inputKind

The kind of denoiser input.

5.13 OptixDenoiserParams Struct Reference

Public Attributes

- unsigned int [denoiseAlpha](#)
- [CUdeviceptr](#) [hdrIntensity](#)
- float [blendFactor](#)

5.13.1 Detailed Description

Various parameters used by the denoiser.

See Also

[optixDenoiserInvoke\(\)](#)

5.13.2 Member Data Documentation

5.13.2.1 float OptixDenoiserParams::blendFactor

5.13.2.2 unsigned int OptixDenoiserParams::denoiseAlpha

5.13.2.3 CUdeviceptr OptixDenoiserParams::hdrIntensity

5.14 OptixDenoiserSizes Struct Reference

Public Attributes

- [size_t stateSizeInBytes](#)
- [size_t withOverlapScratchSizeInBytes](#)
- [size_t withoutOverlapScratchSizeInBytes](#)
- unsigned int [overlapWindowSizeInPixels](#)

5.14.1 Detailed Description

Various sizes related to the denoiser.

See Also

[optixDenoiserComputeMemoryResources\(\)](#)

5.14.2 Member Data Documentation

5.14.2.1 unsigned int OptixDenoiserSizes::overlapWindowSizeInPixels

5.14.2.2 size_t OptixDenoiserSizes::stateSizeInBytes

5.14.2.3 size_t OptixDenoiserSizes::withoutOverlapScratchSizeInBytes

5.14.2.4 size_t OptixDenoiserSizes::withOverlapScratchSizeInBytes

5.15 OptixDeviceContextOptions Struct Reference

Public Attributes

- [OptixLogCallback logCallbackFunction](#)
- void * [logCallbackData](#)
- int [logCallbackLevel](#)

5.15.1 Detailed Description

Parameters used for [optixDeviceContextCreate\(\)](#)

See Also

[optixDeviceContextCreate\(\)](#)

5.15.2 Member Data Documentation

5.15.2.1 void* OptixDeviceContextOptions::logCallbackData

Pointer stored and passed to logCallbackFunction when a message is generated.

5.15.2.2 OptixLogCallback OptixDeviceContextOptions::logCallbackFunction

Function pointer used when OptiX wishes to generate messages.

5.15.2.3 int OptixDeviceContextOptions::logCallbackLevel

Maximum callback level to generate message for (see [OptixLogCallback](#))

5.16 OptixFunctionTable Struct Reference

Public Attributes

Error handling

- [const char *\(* optixGetErrorName \)\(OptixResult result\)](#)
- [const char *\(* optixGetErrorString \)\(OptixResult result\)](#)

Device context

- [OptixResult\(* optixDeviceContextCreate \)\(CUcontext fromContext, const OptixDeviceContextOptions *options, OptixDeviceContext *context\)](#)
- [OptixResult\(* optixDeviceContextDestroy \)\(OptixDeviceContext context\)](#)
- [OptixResult\(* optixDeviceContextGetProperty \)\(OptixDeviceContext context, OptixDeviceProperty property, void *value, size_t sizeInBytes\)](#)
- [OptixResult\(* optixDeviceContextSetLogCallback \)\(OptixDeviceContext context, OptixLogCallback callbackFunction, void *callbackData, unsigned int callbackLevel\)](#)
- [OptixResult\(* optixDeviceContextSetCacheEnabled \)\(OptixDeviceContext context, int enabled\)](#)
- [OptixResult\(* optixDeviceContextSetCacheLocation \)\(OptixDeviceContext context, const char *location\)](#)
- [OptixResult\(* optixDeviceContextSetCacheDatabaseSizes \)\(OptixDeviceContext context, size_t lowWaterMark, size_t highWaterMark\)](#)
- [OptixResult\(* optixDeviceContextGetCacheEnabled \)\(OptixDeviceContext context, int *enabled\)](#)
- [OptixResult\(* optixDeviceContextGetCacheLocation \)\(OptixDeviceContext context, char *location, size_t locationSize\)](#)
- [OptixResult\(* optixDeviceContextGetCacheDatabaseSizes \)\(OptixDeviceContext context, size_t *lowWaterMark, size_t *highWaterMark\)](#)

Modules

- `OptixResult(* optixModuleCreateFromPTX )(OptixDeviceContext context, const OptixModuleCompileOptions *moduleCompileOptions, const OptixPipelineCompileOptions *pipelineCompileOptions, const char *PTX, size_t PTXsize, char *logString, size_t *logStringSize, OptixModule *module)`
- `OptixResult(* optixModuleDestroy )(OptixModule module)`
- `OptixResult(* optixBuiltinISModuleGet )(OptixDeviceContext context, const OptixModuleCompileOptions *moduleCompileOptions, const OptixPipelineCompileOptions *pipelineCompileOptions, const OptixBuiltinISOOptions *builtinISOOptions, OptixModule *builtinModule)`

Program groups

- `OptixResult(* optixProgramGroupCreate )(OptixDeviceContext context, const OptixProgramGroupDesc *programDescriptions, unsigned int numProgramGroups, const OptixProgramGroupOptions *options, char *logString, size_t *logStringSize, OptixProgramGroup *programGroups)`
- `OptixResult(* optixProgramGroupDestroy )(OptixProgramGroup programGroup)`
- `OptixResult(* optixProgramGroupGetStackSize )(OptixProgramGroup programGroup, OptixStackSizes *stackSizes)`

Pipeline

- `OptixResult(* optixPipelineCreate)(OptixDeviceContext context, const OptixPipelineCompileOptions *pipelineCompileOptions, const OptixPipelineLinkOptions *pipelineLinkOptions, const OptixProgramGroup *programGroups, unsigned int numProgramGroups, char *logString, size_t *logStringSize, OptixPipeline *pipeline)`
- `OptixResult(* optixPipelineDestroy )(OptixPipeline pipeline)`
- `OptixResult(* optixPipelineSetStackSize )(OptixPipeline pipeline, unsigned int directCallableStackSizeFromTraversal, unsigned int directCallableStackSizeFromState, unsigned int continuationStackSize, unsigned int maxTraversableGraphDepth)`

Acceleration structures

- `OptixResult(* optixAccelComputeMemoryUsage )(OptixDeviceContext context, const OptixAccelBuildOptions *accelOptions, const OptixBuildInput *buildInputs, unsigned int numBuildInputs, OptixAccelBufferSizes *bufferSizes)`
- `OptixResult(* optixAccelBuild)(OptixDeviceContext context, CUstream stream, const OptixAccelBuildOptions *accelOptions, const OptixBuildInput *buildInputs, unsigned int numBuildInputs, CUdeviceptr tempBuffer, size_t tempBufferSizeInBytes, CUdeviceptr outputBuffer, size_t outputBufferSizeInBytes, OptixTraversableHandle *outputHandle, const OptixAccelEmitDesc *emittedProperties, unsigned int numEmittedProperties)`
- `OptixResult(* optixAccelGetRelocationInfo )(OptixDeviceContext context, OptixTraversableHandle handle, OptixAccelRelocationInfo *info)`
- `OptixResult(* optixAccelCheckRelocationCompatibility )(OptixDeviceContext context, const OptixAccelRelocationInfo *info, int *compatible)`
- `OptixResult(* optixAccelRelocate )(OptixDeviceContext context, CUstream stream, const OptixAccelRelocationInfo *info, CUdeviceptr instanceTraversableHandles, size_t numInstanceTraversableHandles, CUdeviceptr targetAccel, size_t targetAccelSizeInBytes, OptixTraversableHandle *targetHandle)`
- `OptixResult(* optixAccelCompact )(OptixDeviceContext context, CUstream stream, OptixTraversableHandle inputHandle, CUdeviceptr outputBuffer, size_t outputBufferSizeInBytes, OptixTraversableHandle *outputHandle)`

- [OptixResult](#)(* [optixConvertPointerToTraversableHandle](#))(OptixDeviceContext onDevice, CUdeviceptr pointer, [OptixTraversableType](#) traversableType, [OptixTraversableHandle](#) *traversableHandle)

Launch

- [OptixResult](#)(* [optixSbtRecordPackHeader](#))(OptixProgramGroup programGroup, void *sbtRecordHeaderHostPointer)
- [OptixResult](#)(* [optixLaunch](#))(OptixPipeline pipeline, CUstream stream, CUdeviceptr pipelineParams, size_t pipelineParamsSize, const [OptixShaderBindingTable](#) *sbt, unsigned int width, unsigned int height, unsigned int depth)

Denoiser

- [OptixResult](#)(* [optixDenoiserCreate](#))(OptixDeviceContext context, const [OptixDenoiserOptions](#) *options, [OptixDenoiser](#) *returnHandle)
- [OptixResult](#)(* [optixDenoiserDestroy](#))(OptixDenoiser handle)
- [OptixResult](#)(* [optixDenoiserComputeMemoryResources](#))(const [OptixDenoiser](#) handle, unsigned int maximumInputWidth, unsigned int maximumInputHeight, [OptixDenoiserSizes](#) *returnSizes)
- [OptixResult](#)(* [optixDenoiserSetup](#))(OptixDenoiser denoiser, CUstream stream, unsigned int inputWidth, unsigned int inputHeight, CUdeviceptr state, size_t stateSizeInBytes, CUdeviceptr scratch, size_t scratchSizeInBytes)
- [OptixResult](#)(* [optixDenoiserInvoke](#))(OptixDenoiser denoiser, CUstream stream, const [OptixDenoiserParams](#) *params, CUdeviceptr denoiserState, size_t denoiserStateSizeInBytes, const [OptixImage2D](#) *inputLayers, unsigned int numInputLayers, unsigned int inputOffsetX, unsigned int inputOffsetY, const [OptixImage2D](#) *outputLayer, CUdeviceptr scratch, size_t scratchSizeInBytes)
- [OptixResult](#)(* [optixDenoiserSetModel](#))(OptixDenoiser handle, [OptixDenoiserModelKind](#) kind, void *data, size_t sizeInBytes)
- [OptixResult](#)(* [optixDenoiserComputeIntensity](#))(OptixDenoiser handle, CUstream stream, const [OptixImage2D](#) *inputImage, CUdeviceptr outputIntensity, CUdeviceptr scratch, size_t scratchSizeInBytes)

5.16.1 Detailed Description

The function table containing all API functions.

See [optixInit\(\)](#) and [optixInitWithHandle\(\)](#).

5.16.2 Member Data Documentation

5.16.2.1 [OptixResult](#)(* [OptixFunctionTable::optixAccelBuild](#))(OptixDeviceContext context, CUstream stream, const [OptixAccelBuildOptions](#) *accelOptions, const [OptixBuildInput](#) *buildInputs, unsigned int numBuildInputs, CUdeviceptr tempBuffer, size_t tempBufferSizeInBytes, CUdeviceptr outputBuffer, size_t outputBufferSizeInBytes, [OptixTraversableHandle](#) *outputHandle, const [OptixAccelEmitDesc](#) *emittedProperties, unsigned int numEmittedProperties)

See [optixAccelBuild\(\)](#).

5.16.2.2 `OptixResult( * OptixFunctionTable::optixAccelCheckRelocationCompatibility)(OptixDeviceContext context, const OptixAccelRelocationInfo *info, int *compatible)`

See [optixAccelCheckRelocationCompatibility\(\)](#).

5.16.2.3 `OptixResult( * OptixFunctionTable::optixAccelCompact)(OptixDeviceContext context, CUstream stream, OptixTraversableHandle inputHandle, CUdeviceptr outputBuffer, size_t outputBufferSizeInBytes, OptixTraversableHandle *outputHandle)`

See [optixAccelCompact\(\)](#).

5.16.2.4 `OptixResult( * OptixFunctionTable::optixAccelComputeMemoryUsage)(OptixDeviceContext context, const OptixAccelBuildOptions *accelOptions, const OptixBuildInput *buildInputs, unsigned int numBuildInputs, OptixAccelBufferSizes *bufferSizes)`

See [optixAccelComputeMemoryUsage\(\)](#).

5.16.2.5 `OptixResult( * OptixFunctionTable::optixAccelGetRelocationInfo)(OptixDeviceContext context, OptixTraversableHandle handle, OptixAccelRelocationInfo *info)`

See [optixAccelGetRelocationInfo\(\)](#).

5.16.2.6 `OptixResult(* OptixFunctionTable::optixAccelRelocate)(OptixDeviceContext context, CUstream stream, const OptixAccelRelocationInfo *info, CUdeviceptr instanceTraversableHandles, size_t numInstanceTraversableHandles, CUdeviceptr targetAccel, size_t targetAccelSizeInBytes, OptixTraversableHandle *targetHandle)`

See [optixAccelRelocate\(\)](#).

5.16.2.7 `OptixResult( * OptixFunctionTable::optixBuiltinISModuleGet)(OptixDeviceContext context, const OptixModuleCompileOptions *moduleCompileOptions, const OptixPipelineCompileOptions *pipelineCompileOptions, const OptixBuiltinISOptions *builtinISOptions, OptixModule *builtinModule)`

See [optixBuiltinISModuleGet\(\)](#).

5.16.2.8 `OptixResult( * OptixFunctionTable::optixConvertPointerToTraversableHandle)(OptixDeviceContext onDevice, CUdeviceptr pointer, OptixTraversableType traversableType, OptixTraversableHandle *traversableHandle)`

See [optixConvertPointerToTraversableHandle\(\)](#).

5.16.2.9 **OptixResult(* OptixFunctionTable::optixDenoiserComputeIntensity)(OptixDenoiser handle, CUstream stream, const OptixImage2D *inputImage, CUdeviceptr outputIntensity, CUdeviceptr scratch, size_t scratchSizeInBytes)**

See [optixDenoiserComputeIntensity\(\)](#).

5.16.2.10 **OptixResult(* OptixFunctionTable::optixDenoiserComputeMemoryResources)(const OptixDenoiser handle, unsigned int maximumInputWidth, unsigned int maximumInputHeight, OptixDenoiserSizes *returnSizes)**

See [optixDenoiserComputeMemoryResources\(\)](#).

5.16.2.11 **OptixResult(* OptixFunctionTable::optixDenoiserCreate)(OptixDeviceContext context, const OptixDenoiserOptions *options, OptixDenoiser *returnHandle)**

See [optixDenoiserCreate\(\)](#).

5.16.2.12 **OptixResult(* OptixFunctionTable::optixDenoiserDestroy)(OptixDenoiser handle)**

See [optixDenoiserDestroy\(\)](#).

5.16.2.13 **OptixResult(* OptixFunctionTable::optixDenoiserInvoke)(OptixDenoiser denoiser, CUstream stream, const OptixDenoiserParams *params, CUdeviceptr denoiserState, size_t denoiserStateSizeInBytes, const OptixImage2D *inputLayers, unsigned int numInputLayers, unsigned int inputOffsetX, unsigned int inputOffsetY, const OptixImage2D *outputLayer, CUdeviceptr scratch, size_t scratchSizeInBytes)**

See [optixDenoiserInvoke\(\)](#).

5.16.2.14 **OptixResult(* OptixFunctionTable::optixDenoiserSetModel)(OptixDenoiser handle, OptixDenoiserModelKind kind, void *data, size_t sizeInBytes)**

See [optixDenoiserSetModel\(\)](#).

5.16.2.15 **OptixResult(* OptixFunctionTable::optixDenoiserSetup)(OptixDenoiser denoiser, CUstream stream, unsigned int inputWidth, unsigned int inputHeight, CUdeviceptr state, size_t stateSizeInBytes, CUdeviceptr scratch, size_t scratchSizeInBytes)**

See [optixDenoiserSetup\(\)](#).

5.16.2.16 **OptixResult(* OptixFunctionTable::optixDeviceContextCreate)(CUcontext fromContext, const OptixDeviceContextOptions *options, OptixDeviceContext *context)**

See [optixDeviceContextCreate\(\)](#).

5.16.2.17 **OptixResult(* OptixFunctionTable::optixDeviceContextDestroy)(OptixDeviceContext context)**

See [optixDeviceContextDestroy\(\)](#).

5.16.2.18 **OptixResult(* OptixFunctionTable::optixDeviceContextGetCacheDatabaseSizes)(OptixDeviceContext context, size_t *lowWaterMark, size_t *highWaterMark)**

See [optixDeviceContextGetCacheDatabaseSizes\(\)](#).

5.16.2.19 **OptixResult(* OptixFunctionTable::optixDeviceContextGetCacheEnabled)(OptixDeviceContext context, int *enabled)**

See [optixDeviceContextGetCacheEnabled\(\)](#).

5.16.2.20 **OptixResult(* OptixFunctionTable::optixDeviceContextGetCacheLocation)(OptixDeviceContext context, char *location, size_t locationSize)**

See [optixDeviceContextGetCacheLocation\(\)](#).

5.16.2.21 **OptixResult(* OptixFunctionTable::optixDeviceContextGetProperty)(OptixDeviceContext context, OptixDeviceProperty property, void *value, size_t sizeInBytes)**

See [optixDeviceContextGetProperty\(\)](#).

5.16.2.22 **OptixResult(* OptixFunctionTable::optixDeviceContextSetCacheDatabaseSizes)(OptixDeviceContext context, size_t lowWaterMark, size_t highWaterMark)**

See [optixDeviceContextSetCacheDatabaseSizes\(\)](#).

5.16.2.23 **OptixResult(* OptixFunctionTable::optixDeviceContextSetCacheEnabled)(OptixDeviceContext context, int enabled)**

See [optixDeviceContextSetCacheEnabled\(\)](#).

5.16.2.24 **OptixResult(* OptixFunctionTable::optixDeviceContextSetCacheLocation)(OptixDeviceContext context, const char *location)**

See [optixDeviceContextSetCacheLocation\(\)](#).

5.16.2.25 **OptixResult(* OptixFunctionTable::optixDeviceContextSetLogCallback)(OptixDeviceContext context, OptixLogCallback callbackFunction, void *callbackData, unsigned int callbackLevel)**

See [optixDeviceContextSetLogCallback\(\)](#).

5.16.2.26 **const char*(* OptixFunctionTable::optixGetErrorName)(OptixResult result)**

See [optixGetErrorName\(\)](#).

5.16.2.27 **const char*(* OptixFunctionTable::optixGetErrorString)(OptixResult result)**

See [optixGetErrorString\(\)](#).

5.16.2.28 **OptixResult(* OptixFunctionTable::optixLaunch)(OptixPipeline pipeline, CUstream stream, CUdeviceptr pipelineParams, size_t pipelineParamsSize, const OptixShaderBindingTable *sbt, unsigned int width, unsigned int height, unsigned int depth)**

See [optixConvertPointerToTraversableHandle\(\)](#).

5.16.2.29 **OptixResult(* OptixFunctionTable::optixModuleCreateFromPTX)(OptixDeviceContext context, const OptixModuleCompileOptions *moduleCompileOptions, const OptixPipelineCompileOptions *pipelineCompileOptions, const char *PTX, size_t PTXsize, char *logString, size_t *logStringSize, OptixModule *module)**

See [optixModuleCreateFromPTX\(\)](#).

5.16.2.30 **OptixResult(* OptixFunctionTable::optixModuleDestroy)(OptixModule module)**

See [optixModuleDestroy\(\)](#).

5.16.2.31 **OptixResult(* OptixFunctionTable::optixPipelineCreate)(OptixDeviceContext context, const OptixPipelineCompileOptions *pipelineCompileOptions, const OptixPipelineLinkOptions *pipelineLinkOptions, const OptixProgramGroup *programGroups, unsigned int numProgramGroups, char *logString, size_t *logStringSize, OptixPipeline *pipeline)**

See [optixPipelineCreate\(\)](#).

5.16.2.32 **OptixResult(* OptixFunctionTable::optixPipelineDestroy)(OptixPipeline pipeline)**

See [optixPipelineDestroy\(\)](#).

5.16.2.33 `OptixResult( * OptixFunctionTable::optixPipelineSetStackSize)(OptixPipeline pipeline, unsigned int directCallableStackSizeFromTraversal, unsigned int directCallableStackSizeFromState, unsigned int continuationStackSize, unsigned int maxTraversableGraphDepth)`

See [optixPipelineSetStackSize\(\)](#).

5.16.2.34 `OptixResult( * OptixFunctionTable::optixProgramGroupCreate)(OptixDeviceContext context, const OptixProgramGroupDesc *programDescriptions, unsigned int numProgramGroups, const OptixProgramGroupOptions *options, char *logString, size_t *logStringSize, OptixProgramGroup *programGroups)`

See [optixProgramGroupCreate\(\)](#).

5.16.2.35 `OptixResult( * OptixFunctionTable::optixProgramGroupDestroy)(OptixProgramGroup programGroup)`

See [optixProgramGroupDestroy\(\)](#).

5.16.2.36 `OptixResult( * OptixFunctionTable::optixProgramGroupGetStackSize)(OptixProgramGroup programGroup, OptixStackSizes *stackSizes)`

See [optixProgramGroupGetStackSize\(\)](#).

5.16.2.37 `OptixResult( * OptixFunctionTable::optixSbtRecordPackHeader)(OptixProgramGroup programGroup, void *sbtRecordHeaderHostPointer)`

See [optixConvertPointerToTraversableHandle\(\)](#).

5.17 OptixImage2D Struct Reference

Public Attributes

- [CUdeviceptr](#) data
- unsigned int width
- unsigned int height
- unsigned int rowStrideInBytes
- unsigned int pixelStrideInBytes
- [OptixPixelFormat](#) format

5.17.1 Detailed Description

Image descriptor used by the denoiser.

See Also

[optixDenoiserInvoke\(\)](#), [optixDenoiserComputeIntensity\(\)](#)

5.17.2 Member Data Documentation

5.17.2.1 CUdeviceptr OptixImage2D::data

Pointer to the actual pixel data.

5.17.2.2 OptixPixelFormat OptixImage2D::format

Pixel format.

5.17.2.3 unsigned int OptixImage2D::height

Height of the image (in pixels)

5.17.2.4 unsigned int OptixImage2D::pixelStrideInBytes

Stride between subsequent pixels of the image (in bytes). For now, only 0 or the value that corresponds to a dense packing of pixels (no gaps) is supported.

5.17.2.5 unsigned int OptixImage2D::rowStrideInBytes

Stride between subsequent rows of the image (in bytes).

5.17.2.6 unsigned int OptixImage2D::width

Width of the image (in pixels)

5.18 OptixInstance Struct Reference

Public Attributes

- float [transform](#) [12]
- unsigned int [instanceId](#)
- unsigned int [sbtOffset](#)
- unsigned int [visibilityMask](#)
- unsigned int [flags](#)
- [OptixTraversableHandle](#) [traversableHandle](#)
- unsigned int [pad](#) [2]

5.18.1 Detailed Description

Instances.

See Also

[OptixBuildInputInstanceArray::instances](#)

5.18.2 Member Data Documentation

5.18.2.1 unsigned int OptixInstance::flags

Any combination of OptixInstanceFlags is allowed.

5.18.2.2 unsigned int OptixInstance::instanceId

Application supplied ID. The maximal ID can be queried using OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCE_ID.

5.18.2.3 unsigned int OptixInstance::pad[2]

round up to 80-byte, to ensure 16-byte alignment

5.18.2.4 unsigned int OptixInstance::sbtOffset

SBT record offset. Will only be used for instances of geometry acceleration structure (GAS) objects. Needs to be set to 0 for instances of instance acceleration structure (IAS) objects. The maximal SBT offset can be queried using OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCE_SBT_OFFSET.

5.18.2.5 float OptixInstance::transform[12]

affine object-to-world transformation as 3x4 matrix in row-major layout

5.18.2.6 OptixTraversableHandle OptixInstance::traversableHandle

Set with an OptixTraversableHandle.

5.18.2.7 unsigned int OptixInstance::visibilityMask

Visibility mask. If rayMask & instanceMask == 0 the instance is culled. The number of available bits can be queried using OPTIX_DEVICE_PROPERTY_LIMIT_NUM_BITS_INSTANCE_VISIBILITY_MASK.

5.19 OptixMatrixMotionTransform Struct Reference

Public Attributes

- [OptixTraversableHandle](#) child
- [OptixMotionOptions](#) motionOptions
- unsigned int pad [3]
- float transform [2][12]

5.19.1 Detailed Description

Represents a matrix motion transformation.

The device address of instances of this type must be a multiple of `OPTIX_TRANSFORM_BYTE_ALIGNMENT`.

This struct, as defined here, handles only N=2 motion keys due to the fixed array length of its transform member. The following example shows how to create instances for an arbitrary number N of motion keys:

```
float matrixData[N][12];
... // setup matrixData

size_t transformSizeInBytes = sizeof( OptixMatrixMotionTransform ) + ( N-2 ) * 12
    * sizeof( float );
OptixMatrixMotionTransform* matrixMoptionTransform = (
    OptixMatrixMotionTransform*) malloc( transformSizeInBytes );
memset( matrixMoptionTransform, 0, transformSizeInBytes );

... // setup other members of matrixMoptionTransform
matrixMoptionTransform->motionOptions.numKeys
memcpy( matrixMoptionTransform->transform, matrixData, N * 12 * sizeof( float ) );

... // copy matrixMoptionTransform to device memory
free( matrixMoptionTransform )
```

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

5.19.2 Member Data Documentation

5.19.2.1 OptixTraversableHandle OptixMatrixMotionTransform::child

The traversable that is transformed by this transformation.

5.19.2.2 OptixMotionOptions OptixMatrixMotionTransform::motionOptions

The motion options for this transformation.

5.19.2.3 unsigned int OptixMatrixMotionTransform::pad[3]

Padding to make the transformation 16 byte aligned.

5.19.2.4 float OptixMatrixMotionTransform::transform[2][12]

Affine object-to-world transformation as 3x4 matrix in row-major layout.

5.20 OptixModuleCompileOptions Struct Reference

Public Attributes

- int [maxRegisterCount](#)
- [OptixCompileOptimizationLevel](#) [optLevel](#)
- [OptixCompileDebugLevel](#) [debugLevel](#)

5.20.1 Detailed Description

Compilation options for module.

See Also

[optixModuleCreateFromPTX\(\)](#)

5.20.2 Member Data Documentation

5.20.2.1 [OptixCompileDebugLevel](#) [OptixModuleCompileOptions::debugLevel](#)

Generate debug information.

5.20.2.2 [int](#) [OptixModuleCompileOptions::maxRegisterCount](#)

Maximum number of registers allowed when compiling to SASS. Set to 0 for no explicit limit. May vary within a pipeline.

5.20.2.3 [OptixCompileOptimizationLevel](#) [OptixModuleCompileOptions::optLevel](#)

Optimization level. May vary within a pipeline.

5.21 OptixMotionOptions Struct Reference

Public Attributes

- unsigned short [numKeys](#)
- unsigned short [flags](#)
- float [timeBegin](#)
- float [timeEnd](#)

5.21.1 Detailed Description

Motion options.

See Also

[OptixAccelBuildOptions::motionOptions](#), [OptixMatrixMotionTransform::motionOptions](#),
[OptixSRTMotionTransform::motionOptions](#)

5.21.2 Member Data Documentation

5.21.2.1 unsigned short OptixMotionOptions::flags

Combinations of [OptixMotionFlags](#).

5.21.2.2 unsigned short OptixMotionOptions::numKeys

If numKeys > 1, motion is enabled. timeBegin, timeEnd and flags are all ignored when motion is disabled.

5.21.2.3 float OptixMotionOptions::timeBegin

Point in time where motion starts.

5.21.2.4 float OptixMotionOptions::timeEnd

Point in time where motion ends.

5.22 OptixPagingContext Struct Reference

Public Attributes

- unsigned int [maxVaSizeInPages](#)
- unsigned int * [usageBits](#)
- unsigned int * [residenceBits](#)
- unsigned long long * [pageTable](#)

5.22.1 Member Data Documentation

5.22.1.1 unsigned int OptixPagingContext::maxVaSizeInPages

5.22.1.2 unsigned long long* OptixPagingContext::pageTable

5.22.1.3 unsigned int* OptixPagingContext::residenceBits

5.22.1.4 unsigned int* OptixPagingContext::usageBits

5.23 OptixPagingOptions Struct Reference

Public Attributes

- unsigned int [maxVaSizeInPages](#)
- unsigned int [initialVaSizeInPages](#)

5.23.1 Member Data Documentation

5.23.1.1 unsigned int `OptixPagingOptions::initialVaSizeInPages`

5.23.1.2 unsigned int `OptixPagingOptions::maxVaSizeInPages`

5.24 OptixPagingSizes Struct Reference

Public Attributes

- unsigned int `pageTableSizeInBytes`
- unsigned int `usageBitsSizeInBytes`

5.24.1 Member Data Documentation

5.24.1.1 unsigned int `OptixPagingSizes::pageTableSizeInBytes`

5.24.1.2 unsigned int `OptixPagingSizes::usageBitsSizeInBytes`

5.25 OptixPipelineCompileOptions Struct Reference

Public Attributes

- int `usesMotionBlur`
- unsigned int `traversableGraphFlags`
- int `numPayloadValues`
- int `numAttributeValues`
- unsigned int `exceptionFlags`
- const char * `pipelineLaunchParamsVariableName`
- unsigned int `usesPrimitiveTypeFlags`

5.25.1 Detailed Description

Compilation options for all modules of a pipeline.

Similar to `OptixModuleCompileOptions`, but these options here need to be equal for all modules of a pipeline.

See Also

`optixModuleCreateFromPTX()`, `optixPipelineCreate()`

5.25.2 Member Data Documentation

5.25.2.1 unsigned int `OptixPipelineCompileOptions::exceptionFlags`

A bitmask of `OptixExceptionFlags` indicating which exceptions are enabled.

5.25.2.2 int OptixPipelineCompileOptions::numAttributeValues

How much storage, in 32b words, to make available for the attributes. The minimum number is 2. Values below that will automatically be changed to 2. [2..8].

5.25.2.3 int OptixPipelineCompileOptions::numPayloadValues

How much storage, in 32b words, to make available for the payload, [0..8].

5.25.2.4 const char* OptixPipelineCompileOptions::pipelineLaunchParamsVariableName

The name of the pipeline parameter variable. If 0, no pipeline parameter will be available. This will be ignored if the launch param variable was optimized out or was not found in the modules linked to the pipeline.

5.25.2.5 unsigned int OptixPipelineCompileOptions::traversableGraphFlags

Traversable graph bitfield. See OptixTraversableGraphFlags.

5.25.2.6 int OptixPipelineCompileOptions::usesMotionBlur

Boolean value indicating whether motion blur could be used.

5.25.2.7 unsigned int OptixPipelineCompileOptions::usesPrimitiveTypeFlags

Bit field enabling primitive types. See OptixPrimitiveTypeFlags. Setting to zero corresponds to enabling OPTIX_PRIMITIVE_TYPE_FLAGS_CUSTOM and OPTIX_PRIMITIVE_TYPE_FLAGS_TRIANGLE.

5.26 OptixPipelineLinkOptions Struct Reference**Public Attributes**

- unsigned int [maxTraceDepth](#)
- [OptixCompileDebugLevel](#) [debugLevel](#)

5.26.1 Detailed Description

Link options for a pipeline.

See Also

[optixPipelineCreate\(\)](#)

5.26.2 Member Data Documentation**5.26.2.1 OptixCompileDebugLevel OptixPipelineLinkOptions::debugLevel**

Generate debug information.

5.26.2.2 unsigned int OptixPipelineLinkOptions::maxTraceDepth

Maximum trace recursion depth. 0 means a ray generation program can be launched, but can't trace any rays. The maximum allowed value is 31.

5.27 OptixProgramGroupCallables Struct Reference

Public Attributes

- [OptixModule moduleDC](#)
- `const char * entryFunctionNameDC`
- [OptixModule moduleCC](#)
- `const char * entryFunctionNameCC`

5.27.1 Detailed Description

Program group representing callables.

Module and entry function name need to be valid for at least one of the two callables.

See Also

[#OptixProgramGroupDesc::callables](#)

5.27.2 Member Data Documentation

5.27.2.1 `const char* OptixProgramGroupCallables::entryFunctionNameCC`

Entry function name of the continuation callable (CC) program.

5.27.2.2 `const char* OptixProgramGroupCallables::entryFunctionNameDC`

Entry function name of the direct callable (DC) program.

5.27.2.3 `OptixModule OptixProgramGroupCallables::moduleCC`

Module holding the continuation callable (CC) program.

5.27.2.4 `OptixModule OptixProgramGroupCallables::moduleDC`

Module holding the direct callable (DC) program.

5.28 OptixProgramGroupDesc Struct Reference

Public Attributes

- [OptixProgramGroupKind kind](#)
- `unsigned int flags`

```

• union {
    OptixProgramGroupSingleModule raygen
    OptixProgramGroupSingleModule miss
    OptixProgramGroupSingleModule exception
    OptixProgramGroupCallables callables
    OptixProgramGroupHitgroup hitgroup
};

```

5.28.1 Detailed Description

Descriptor for program groups.

5.28.2 Member Data Documentation

5.28.2.1 union { ... }

5.28.2.2 OptixProgramGroupCallables OptixProgramGroupDesc::callables

See Also

[OPTIX_PROGRAM_GROUP_KIND_CALLABLES](#)

5.28.2.3 OptixProgramGroupSingleModule OptixProgramGroupDesc::exception

See Also

[OPTIX_PROGRAM_GROUP_KIND_EXCEPTION](#)

5.28.2.4 unsigned int OptixProgramGroupDesc::flags

See [OptixProgramGroupFlags](#).

5.28.2.5 OptixProgramGroupHitgroup OptixProgramGroupDesc::hitgroup

See Also

[OPTIX_PROGRAM_GROUP_KIND_HITGROUP](#)

5.28.2.6 OptixProgramGroupKind OptixProgramGroupDesc::kind

The kind of program group.

5.28.2.7 OptixProgramGroupSingleModule OptixProgramGroupDesc::miss

See Also

[OPTIX_PROGRAM_GROUP_KIND_MISS](#)

5.28.2.8 OptixProgramGroupSingleModule OptixProgramGroupDesc::raygen

See Also

[OPTIX_PROGRAM_GROUP_KIND_RAYGEN](#)

5.29 OptixProgramGroupHitgroup Struct Reference

Public Attributes

- [OptixModule moduleCH](#)
- `const char * entryFunctionNameCH`
- [OptixModule moduleAH](#)
- `const char * entryFunctionNameAH`
- [OptixModule moduleIS](#)
- `const char * entryFunctionNameIS`

5.29.1 Detailed Description

Program group representing the hitgroup.

For each of the three program types, module and entry function name might both be nullptr.

See Also

[OptixProgramGroupDesc::hitgroup](#)

5.29.2 Member Data Documentation

5.29.2.1 `const char*` [OptixProgramGroupHitgroup::entryFunctionNameAH](#)

Entry function name of the any hit (AH) program.

5.29.2.2 `const char*` [OptixProgramGroupHitgroup::entryFunctionNameCH](#)

Entry function name of the closest hit (CH) program.

5.29.2.3 `const char*` [OptixProgramGroupHitgroup::entryFunctionNameIS](#)

Entry function name of the intersection (IS) program.

5.29.2.4 [OptixModule](#) [OptixProgramGroupHitgroup::moduleAH](#)

Module holding the any hit (AH) program.

5.29.2.5 [OptixModule](#) [OptixProgramGroupHitgroup::moduleCH](#)

Module holding the closest hit (CH) program.

5.29.2.6 OptixModule OptixProgramGroupHitgroup::moduleIs

Module holding the intersection (Is) program.

5.30 OptixProgramGroupOptions Struct Reference

Public Attributes

- int [placeholder](#)

5.30.1 Detailed Description

Program group options.

See Also

[optixProgramGroupCreate\(\)](#)

5.30.2 Member Data Documentation

5.30.2.1 int OptixProgramGroupOptions::placeholder

Currently no options, so include a placeholder.

5.31 OptixProgramGroupSingleModule Struct Reference

Public Attributes

- [OptixModule](#) module
- const char * [entryFunctionName](#)

5.31.1 Detailed Description

Program group representing a single module.

Used for raygen, miss, and exception programs. In case of raygen and exception programs, module and entry function name need to be valid. For miss programs, module and entry function name might both be nullptr.

See Also

[OptixProgramGroupDesc::raygen](#), [OptixProgramGroupDesc::miss](#),
[OptixProgramGroupDesc::exception](#)

5.31.2 Member Data Documentation

5.31.2.1 `const char*` `OptixProgramGroupSingleModule::entryFunctionName`

Entry function name of the single program.

5.31.2.2 `OptixModule` `OptixProgramGroupSingleModule::module`

Module holding single program.

5.32 OptixShaderBindingTable Struct Reference

Public Attributes

- `CUdeviceptr` `raygenRecord`
- `CUdeviceptr` `exceptionRecord`

- `CUdeviceptr` `missRecordBase`
- `unsigned int` `missRecordStrideInBytes`
- `unsigned int` `missRecordCount`

- `CUdeviceptr` `hitgroupRecordBase`
- `unsigned int` `hitgroupRecordStrideInBytes`
- `unsigned int` `hitgroupRecordCount`

- `CUdeviceptr` `callablesRecordBase`
- `unsigned int` `callablesRecordStrideInBytes`
- `unsigned int` `callablesRecordCount`

5.32.1 Detailed Description

Describes the shader binding table (SBT)

See Also

`optixLaunch()`

5.32.2 Member Data Documentation

5.32.2.1 `CUdeviceptr` `OptixShaderBindingTable::callablesRecordBase`

Arrays of SBT records for callable programs. If the base address is not null, the stride and count must not be zero. If the base address is null, then the count needs to zero. The base address and the stride must be a multiple of `OPTIX_SBT_RECORD_ALIGNMENT`.

5.32.2.2 unsigned int OptixShaderBindingTable::callablesRecordCount

Arrays of SBT records for callable programs. If the base address is not null, the stride and count must not be zero. If the base address is null, then the count needs to zero. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.3 unsigned int OptixShaderBindingTable::callablesRecordStrideInBytes

Arrays of SBT records for callable programs. If the base address is not null, the stride and count must not be zero. If the base address is null, then the count needs to zero. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.4 CUdeviceptr OptixShaderBindingTable::exceptionRecord

Device address of the SBT record of the exception program. The address must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.5 CUdeviceptr OptixShaderBindingTable::hitgroupRecordBase

Arrays of SBT records for hit groups. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.6 unsigned int OptixShaderBindingTable::hitgroupRecordCount

Arrays of SBT records for hit groups. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.7 unsigned int OptixShaderBindingTable::hitgroupRecordStrideInBytes

Arrays of SBT records for hit groups. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.8 CUdeviceptr OptixShaderBindingTable::missRecordBase

Arrays of SBT records for miss programs. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.9 unsigned int OptixShaderBindingTable::missRecordCount

Arrays of SBT records for miss programs. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.10 unsigned int OptixShaderBindingTable::missRecordStrideInBytes

Arrays of SBT records for miss programs. The base address and the stride must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.32.2.11 CUdeviceptr OptixShaderBindingTable::raygenRecord

Device address of the SBT record of the ray gen program to start launch at. The address must be a multiple of OPTIX_SBT_RECORD_ALIGNMENT.

5.33 OptixSRTData Struct Reference

Public Attributes

Parameters describing the SRT transformation

- float `sx`
- float `a`
- float `b`
- float `pvx`
- float `sy`
- float `c`
- float `pvy`
- float `sz`
- float `pvz`
- float `qx`
- float `qy`
- float `qz`
- float `qw`
- float `tx`
- float `ty`
- float `tz`

5.33.1 Detailed Description

Represents an SRT transformation.

An SRT transformation can represent a smooth rotation with fewer motion keys than a matrix transformation. Each motion key is constructed from elements taken from a matrix *S*, a quaternion *R*, and a translation *T*.

$$[\text{sx} \quad \text{a} \quad \text{b} \quad \text{pvx}]$$

The scaling matrix $S = [\text{0} \text{ sy } \text{c} \text{ pvy}]$ defines an affine transformation that can include scale, shear, and a $[\text{0} \text{ 0} \text{ sz } \text{pvz}]$

translation. The translation allows to define the pivot point for the subsequent rotation.

The quaternion $R = [\text{qx}, \text{qy}, \text{qz}, \text{qw}]$ describes a rotation with angular component $\text{qw} = \cos(\text{theta}/2)$ and other components $[\text{qx}, \text{qy}, \text{qz}] = \sin(\text{theta}/2) * [\text{ax}, \text{ay}, \text{az}]$ where the axis $[\text{ax}, \text{ay}, \text{az}]$ is normalized.

$$[\text{1} \quad \text{0} \quad \text{0} \quad \text{tx}]$$

The translation $T = [\text{0} \text{ 1} \text{ 0} \text{ ty}]$ defines another translation that is applied after the rotation. Typically, this $[\text{0} \text{ 0} \text{ 1} \text{ tz}]$

translation includes the inverse translation from the matrix *S* to reverse its effect.

To obtain the effective transformation at time *t*, the elements of the components of *S*, *R*, and *T* will be interpolated linearly. The components are then multiplied to obtain the combined transformation $C = T * R * S$. The transformation *C* is the effective object-to-world transformations at time *t*, and C^{-1} is the effective world-to-object transformation at time *t*.

See Also

[OptixSRTMotionTransform::srtData](#), [optixConvertPointerToTraversableHandle\(\)](#)

5.33.2 Member Data Documentation

5.33.2.1 float [OptixSRTData::a](#)

5.33.2.2 float [OptixSRTData::b](#)

5.33.2.3 float [OptixSRTData::c](#)

5.33.2.4 float [OptixSRTData::pvx](#)

5.33.2.5 float [OptixSRTData::pvy](#)

5.33.2.6 float [OptixSRTData::pvz](#)

5.33.2.7 float [OptixSRTData::qw](#)

5.33.2.8 float [OptixSRTData::qx](#)

5.33.2.9 float [OptixSRTData::qy](#)

5.33.2.10 float [OptixSRTData::qz](#)

5.33.2.11 float [OptixSRTData::sx](#)

5.33.2.12 float [OptixSRTData::sy](#)

5.33.2.13 float [OptixSRTData::sz](#)

5.33.2.14 float [OptixSRTData::tx](#)

5.33.2.15 float [OptixSRTData::ty](#)

5.33.2.16 float [OptixSRTData::tz](#)

5.34 OptixSRTMotionTransform Struct Reference

Public Attributes

- [OptixTraversableHandle](#) child
- [OptixMotionOptions](#) motionOptions
- unsigned int pad [3]
- [OptixSRTData](#) srtData [2]

5.34.1 Detailed Description

Represents an SRT motion transformation.

The device address of instances of this type must be a multiple of `OPTIX_TRANSFORM_BYTE_ALIGNMENT`.

This struct, as defined here, handles only N=2 motion keys due to the fixed array length of its `srtData` member. The following example shows how to create instances for an arbitrary number N of motion keys:

```
OptixSRTData srtData[N];
... // setup srtData

size_t transformSizeInBytes = sizeof( OptixSRTMotionTransform ) + ( N-2 ) * sizeof(
    OptixSRTData );
OptixSRTMotionTransform* srtMotionTransform = (
    OptixSRTMotionTransform*) malloc( transformSizeInBytes );
memset( srtMotionTransform, 0, transformSizeInBytes );

... // setup other members of srtMotionTransform
srtMotionTransform->motionOptions.numKeys = N;
memcpy( srtMotionTransform->srtData, srtData, N * sizeof( OptixSRTData ) );

... // copy srtMotionTransform to device memory
free( srtMotionTransform )
```

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

5.34.2 Member Data Documentation

5.34.2.1 OptixTraversableHandle OptixSRTMotionTransform::child

The traversable transformed by this transformation.

5.34.2.2 OptixMotionOptions OptixSRTMotionTransform::motionOptions

The motion options for this transformation.

5.34.2.3 unsigned int OptixSRTMotionTransform::pad[3]

Padding to make the SRT data 16 byte aligned.

5.34.2.4 OptixSRTData OptixSRTMotionTransform::srtData[2]

The actual SRT data describing the transformation.

5.35 OptixStackSize Struct Reference

Public Attributes

- unsigned int [cssRG](#)
- unsigned int [cssMS](#)
- unsigned int [cssCH](#)
- unsigned int [cssAH](#)
- unsigned int [cssIS](#)
- unsigned int [cssCC](#)
- unsigned int [dssDC](#)

5.35.1 Detailed Description

Describes the stack size requirements of a program group.

See Also

[optixProgramGroupGetStackSize\(\)](#)

5.35.2 Member Data Documentation

5.35.2.1 unsigned int OptixStackSize::cssAH

Continuation stack size of AH programs in bytes.

5.35.2.2 unsigned int OptixStackSize::cssCC

Continuation stack size of CC programs in bytes.

5.35.2.3 unsigned int OptixStackSize::cssCH

Continuation stack size of CH programs in bytes.

5.35.2.4 unsigned int OptixStackSize::cssIS

Continuation stack size of IS programs in bytes.

5.35.2.5 unsigned int OptixStackSize::cssMS

Continuation stack size of MS programs in bytes.

5.35.2.6 unsigned int OptixStackSize::cssRG

Continuation stack size of RG programs in bytes.

5.35.2.7 unsigned int OptixStackSize::dssDC

Direct stack size of DC programs in bytes.

5.36 OptixStaticTransform Struct Reference

Public Attributes

- [OptixTraversableHandle](#) child
- unsigned int pad [2]
- float transform [12]
- float invTransform [12]

5.36.1 Detailed Description

Static transform.

The device address of instances of this type must be a multiple of OPTIX_TRANSFORM_BYTE_ALIGNMENT.

See Also

[optixConvertPointerToTraversableHandle\(\)](#)

5.36.2 Member Data Documentation

5.36.2.1 [OptixTraversableHandle](#) `OptixStaticTransform::child`

The traversable transformed by this transformation.

5.36.2.2 `float` `OptixStaticTransform::invTransform[12]`

Affine world-to-object transformation as 3x4 matrix in row-major layout Must be the inverse of the transform matrix.

5.36.2.3 `unsigned int` `OptixStaticTransform::pad[2]`

Padding to make the transformations 16 byte aligned.

5.36.2.4 `float` `OptixStaticTransform::transform[12]`

Affine object-to-world transformation as 3x4 matrix in row-major layout.

5.37 OptixUtilDenoiserImageTile Struct Reference

Public Attributes

- [OptixImage2D](#) input
- [OptixImage2D](#) output
- unsigned int inputOffsetX
- unsigned int inputOffsetY

5.37.1 Detailed Description

Tile definition.

see [optixUtilDenoiserSplitImage](#)

5.37.2 Member Data Documentation

5.37.2.1 OptixImage2D OptixUtilDenoiserImageTile::input

5.37.2.2 unsigned int OptixUtilDenoiserImageTile::inputOffsetX

5.37.2.3 unsigned int OptixUtilDenoiserImageTile::inputOffsetY

5.37.2.4 OptixImage2D OptixUtilDenoiserImageTile::output

5.38 PageMapping Struct Reference

Public Attributes

- unsigned int [id](#)
- unsigned long long [page](#)

5.38.1 Member Data Documentation

5.38.1.1 unsigned int PageMapping::id

5.38.1.2 unsigned long long PageMapping::page

6 File Documentation

6.1 optix.h File Reference

Macros

- `#define` [OPTIX_VERSION](#)

6.1.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation Includes the host api if compiling host code, includes the cuda api if compiling device code. For the math library routines include `optix_math.h`

6.1.2 Macro Definition Documentation

6.1.2.1 #define OPTIX_VERSION

Value:

```
60600 /* major = OPTIX_VERSION/10000,      *
      * minor = (OPTIX_VERSION%10000)/100,  *
      * micro = OPTIX_VERSION%100          */
```

6.2 optix_7_device.h File Reference

Functions

- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5)

- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5, unsigned int &p6)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5, unsigned int &p6, unsigned int &p7)
- static __forceinline__
__device__ void [optixSetPayload_0](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_1](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_2](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_3](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_4](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_5](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_6](#) (unsigned int p)
- static __forceinline__
__device__ void [optixSetPayload_7](#) (unsigned int p)
- static __forceinline__
__device__ unsigned int [optixGetPayload_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPayload_7](#) ()
- static __forceinline__
__device__ unsigned int [optixUndefinedValue](#) ()
- static __forceinline__
__device__ float3 [optixGetWorldRayOrigin](#) ()

- static __forceinline__
__device__ float3 [optixGetWorldRayDirection](#) ()
- static __forceinline__
__device__ float3 [optixGetObjectRayOrigin](#) ()
- static __forceinline__
__device__ float3 [optixGetObjectRayDirection](#) ()
- static __forceinline__
__device__ float [optixGetRayTmin](#) ()
- static __forceinline__
__device__ float [optixGetRayTmax](#) ()
- static __forceinline__
__device__ float [optixGetRayTime](#) ()
- static __forceinline__
__device__ unsigned int [optixGetRayFlags](#) ()
- static __forceinline__
__device__ unsigned int [optixGetRayVisibilityMask](#) ()
- static __forceinline__
__device__ void [optixGetTriangleVertexData](#) (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float3 data[3])
- static __forceinline__
__device__ void [optixGetLinearCurveVertexData](#) (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[2])
- static __forceinline__
__device__ void [optixGetQuadraticBSplineVertexData](#) (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[3])
- static __forceinline__
__device__ void [optixGetCubicBSplineVertexData](#) (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[4])
- static __forceinline__
__device__
[OptixTraversableHandle](#) [optixGetGASTraversableHandle](#) ()
- static __forceinline__
__device__ float [optixGetGASMotionTimeBegin](#) (OptixTraversableHandle gas)
- static __forceinline__
__device__ float [optixGetGASMotionTimeEnd](#) (OptixTraversableHandle gas)
- static __forceinline__
__device__ unsigned int [optixGetGASMotionStepCount](#) (OptixTraversableHandle gas)
- static __forceinline__
__device__ void [optixGetWorldToObjectTransformMatrix](#) (float m[12])
- static __forceinline__
__device__ void [optixGetObjectToWorldTransformMatrix](#) (float m[12])
- static __forceinline__
__device__ float3 [optixTransformPointFromWorldToObjectSpace](#) (float3 point)
- static __forceinline__
__device__ float3 [optixTransformVectorFromWorldToObjectSpace](#) (float3 vec)
- static __forceinline__
__device__ float3 [optixTransformNormalFromWorldToObjectSpace](#) (float3 normal)

- static __forceinline__
__device__ float3 optixTransformPointFromObjectToWorldSpace (float3 point)
- static __forceinline__
__device__ float3 optixTransformVectorFromObjectToWorldSpace (float3 vec)
- static __forceinline__
__device__ float3 optixTransformNormalFromObjectToWorldSpace (float3 normal)
- static __forceinline__
__device__ unsigned int optixGetTransformListSize ()
- static __forceinline__
__device__
OptixTraversableHandle optixGetTransformListHandle (unsigned int index)
- static __forceinline__
__device__ OptixTransformType optixGetTransformTypeFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const
OptixStaticTransform * optixGetStaticTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const
OptixSRTMotionTransform * optixGetSRTMotionTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const
OptixMatrixMotionTransform * optixGetMatrixMotionTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ unsigned int optixGetInstanceIdFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const float4 * optixGetInstanceTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ const float4 * optixGetInstanceInverseTransformFromHandle (OptixTraversableHandle handle)
- static __forceinline__
__device__ bool optixReportIntersection (float hitT, unsigned int hitKind)
- static __forceinline__
__device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0)
- static __forceinline__
__device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1)
- static __forceinline__
__device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2)
- static __forceinline__
__device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3)

- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5, unsigned int a6)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5, unsigned int a6, unsigned int a7)
- static __forceinline__
__device__ unsigned int [optixGetAttribute_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_7](#) ()
- static __forceinline__
__device__ void [optixTerminateRay](#) ()
- static __forceinline__
__device__ void [optixIgnoreIntersection](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPrimitiveIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetSbtGASIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetInstanceId](#) ()
- static __forceinline__
__device__ unsigned int [optixGetInstanceIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetHitKind](#) ()
- static __forceinline__
__device__ [OptixPrimitiveType](#) [optixGetPrimitiveType](#) (unsigned int hitKind)
- static __forceinline__
__device__ bool [optixIsFrontFaceHit](#) (unsigned int hitKind)

- static __forceinline__
__device__ bool [optixIsBackFaceHit](#) (unsigned int hitKind)
- static __forceinline__
__device__ [OptixPrimitiveType](#) [optixGetPrimitiveType](#) ()
- static __forceinline__
__device__ bool [optixIsFrontFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsBackFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleFrontFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleBackFaceHit](#) ()
- static __forceinline__
__device__ float2 [optixGetTriangleBarycentrics](#) ()
- static __forceinline__
__device__ float [optixGetCurveParameter](#) ()
- static __forceinline__
__device__ uint3 [optixGetLaunchIndex](#) ()
- static __forceinline__
__device__ uint3 [optixGetLaunchDimensions](#) ()
- static __forceinline__
__device__ CUdeviceptr [optixGetSbtDataPointer](#) ()
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3,
unsigned int exceptionDetail4)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3,
unsigned int exceptionDetail4, unsigned int exceptionDetail5)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3,
unsigned int exceptionDetail4, unsigned int exceptionDetail5, unsigned int exceptionDetail6)

- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3, unsigned int exceptionDetail4, unsigned int exceptionDetail5, unsigned int exceptionDetail6, unsigned int exceptionDetail7)
- static __forceinline__
__device__ int [optixGetExceptionCode](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_7](#) ()
- static __forceinline__
__device__
[OptixTraversableHandle](#) [optixGetExceptionInvalidTraversable](#) ()
- static __forceinline__
__device__ int [optixGetExceptionInvalidSbtOffset](#) ()
- static __forceinline__
__device__
[OptixInvalidRayExceptionDetails](#) [optixGetExceptionInvalidRay](#) ()
- static __forceinline__
__device__
[OptixParameterMismatchExceptionDetails](#) [optixGetExceptionParameterMismatch](#) ()
- static __forceinline__
__device__ char * [optixGetExceptionLineInfo](#) ()
- template<typename ReturnT , typename... ArgTypes>
static __forceinline__
__device__ ReturnT [optixDirectCall](#) (unsigned int sbtIndex, ArgTypes...args)
- template<typename ReturnT , typename... ArgTypes>
static __forceinline__
__device__ ReturnT [optixContinuationCall](#) (unsigned int sbtIndex, ArgTypes...args)

6.2.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation OptiX public API Reference - Device API declarations

6.3 optix_7_device_impl.h File Reference

Macros

- `#define OPTIX_DEFINE_optixSetPayload_BODY(which) asm volatile( "call _optix_set_payload_" #which ", (%0);" : : "r"( p ) : );`
- `#define OPTIX_DEFINE_optixGetPayload_BODY(which)`
- `#define OPTIX_DEFINE_optixGetAttribute_BODY(which)`
- `#define OPTIX_DEFINE_optixGetExceptionDetail_BODY(which)`

Functions

- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex)`
- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0)`
- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1)`
- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2)`
- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3)`
- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4)`
- `static __forceinline__`
`__device__ void optixTrace (OptixTraversableHandle handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, OptixVisibilityMask visibilityMask, unsigned int rayFlags,`

- unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5)
- static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5, unsigned int &p6)
 - static __forceinline__
__device__ void [optixTrace](#) ([OptixTraversableHandle](#) handle, float3 rayOrigin, float3 rayDirection, float tmin, float tmax, float rayTime, [OptixVisibilityMask](#) visibilityMask, unsigned int rayFlags, unsigned int SBTOffset, unsigned int SBTstride, unsigned int missSBTIndex, unsigned int &p0, unsigned int &p1, unsigned int &p2, unsigned int &p3, unsigned int &p4, unsigned int &p5, unsigned int &p6, unsigned int &p7)
 - static __forceinline__
__device__ void [optixSetPayload_0](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_1](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_2](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_3](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_4](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_5](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_6](#) (unsigned int p)
 - static __forceinline__
__device__ void [optixSetPayload_7](#) (unsigned int p)
 - static __forceinline__
__device__ unsigned int [optixGetPayload_0](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_1](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_2](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_3](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_4](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_5](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_6](#) ()
 - static __forceinline__
__device__ unsigned int [optixGetPayload_7](#) ()
 - static __forceinline__
__device__ unsigned int [optixUndefinedValue](#) ()

- static __forceinline__
__device__ float3 optixGetWorldRayOrigin ()
- static __forceinline__
__device__ float3 optixGetWorldRayDirection ()
- static __forceinline__
__device__ float3 optixGetObjectRayOrigin ()
- static __forceinline__
__device__ float3 optixGetObjectRayDirection ()
- static __forceinline__
__device__ float optixGetRayTmin ()
- static __forceinline__
__device__ float optixGetRayTmax ()
- static __forceinline__
__device__ float optixGetRayTime ()
- static __forceinline__
__device__ unsigned int optixGetRayFlags ()
- static __forceinline__
__device__ unsigned int optixGetRayVisibilityMask ()
- static __forceinline__
__device__ void optixGetTriangleVertexData (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float3 data[3])
- static __forceinline__
__device__ void optixGetLinearCurveVertexData (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[2])
- static __forceinline__
__device__ void optixGetQuadraticBSplineVertexData (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[3])
- static __forceinline__
__device__ void optixGetCubicBSplineVertexData (OptixTraversableHandle gas, unsigned int primIdx, unsigned int sbtGASIndex, float time, float4 data[4])
- static __forceinline__
__device__
OptixTraversableHandle optixGetGASTraversableHandle ()
- static __forceinline__
__device__ float optixGetGASMotionTimeBegin (OptixTraversableHandle handle)
- static __forceinline__
__device__ float optixGetGASMotionTimeEnd (OptixTraversableHandle handle)
- static __forceinline__
__device__ unsigned int optixGetGASMotionStepCount (OptixTraversableHandle handle)
- static __forceinline__
__device__ void optixGetWorldToObjectTransformMatrix (float m[12])
- static __forceinline__
__device__ void optixGetObjectToWorldTransformMatrix (float m[12])
- static __forceinline__
__device__ float3 optixTransformPointFromWorldToObjectSpace (float3 point)
- static __forceinline__
__device__ float3 optixTransformVectorFromWorldToObjectSpace (float3 vec)

- static __forceinline__
__device__ float3 [optixTransformNormalFromWorldToObjectSpace](#) (float3 normal)
- static __forceinline__
__device__ float3 [optixTransformPointFromObjectToWorldSpace](#) (float3 point)
- static __forceinline__
__device__ float3 [optixTransformVectorFromObjectToWorldSpace](#) (float3 vec)
- static __forceinline__
__device__ float3 [optixTransformNormalFromObjectToWorldSpace](#) (float3 normal)
- static __forceinline__
__device__ unsigned int [optixGetTransformListSize](#) ()
- static __forceinline__
__device__
[OptixTraversableHandle](#) [optixGetTransformListHandle](#) (unsigned int index)
- static __forceinline__
__device__ [OptixTransformType](#) [optixGetTransformTypeFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ const
[OptixStaticTransform](#) * [optixGetStaticTransformFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ const
[OptixSRTMotionTransform](#) * [optixGetSRTMotionTransformFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ const
[OptixMatrixMotionTransform](#) * [optixGetMatrixMotionTransformFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ unsigned int [optixGetInstanceIdFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ const float4 * [optixGetInstanceTransformFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ const float4 * [optixGetInstanceInverseTransformFromHandle](#) ([OptixTraversableHandle](#) handle)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3)

- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5, unsigned int a6)
- static __forceinline__
__device__ bool [optixReportIntersection](#) (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1, unsigned int a2, unsigned int a3, unsigned int a4, unsigned int a5, unsigned int a6, unsigned int a7)
- static __forceinline__
__device__ unsigned int [optixGetAttribute_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetAttribute_7](#) ()
- static __forceinline__
__device__ void [optixTerminateRay](#) ()
- static __forceinline__
__device__ void [optixIgnoreIntersection](#) ()
- static __forceinline__
__device__ unsigned int [optixGetPrimitiveIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetSbtGASIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetInstanceId](#) ()
- static __forceinline__
__device__ unsigned int [optixGetInstanceIndex](#) ()
- static __forceinline__
__device__ unsigned int [optixGetHitKind](#) ()
- static __forceinline__
__device__ [OptixPrimitiveType](#) [optixGetPrimitiveType](#) (unsigned int hitKind)
- static __forceinline__
__device__ bool [optixIsBackFaceHit](#) (unsigned int hitKind)

- static __forceinline__
__device__ bool [optixIsFrontFaceHit](#) (unsigned int hitKind)
- static __forceinline__
__device__ [OptixPrimitiveType](#) [optixGetPrimitiveType](#) ()
- static __forceinline__
__device__ bool [optixIsBackFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsFrontFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleFrontFaceHit](#) ()
- static __forceinline__
__device__ bool [optixIsTriangleBackFaceHit](#) ()
- static __forceinline__
__device__ float [optixGetCurveParameter](#) ()
- static __forceinline__
__device__ float2 [optixGetTriangleBarycentrics](#) ()
- static __forceinline__
__device__ uint3 [optixGetLaunchIndex](#) ()
- static __forceinline__
__device__ uint3 [optixGetLaunchDimensions](#) ()
- static __forceinline__
__device__ CUdeviceptr [optixGetSbtDataPointer](#) ()
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3,
unsigned int exceptionDetail4)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3,
unsigned int exceptionDetail4, unsigned int exceptionDetail5)
- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0,
unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3,
unsigned int exceptionDetail4, unsigned int exceptionDetail5, unsigned int exceptionDetail6)

- static __forceinline__
__device__ void [optixThrowException](#) (int exceptionCode, unsigned int exceptionDetail0, unsigned int exceptionDetail1, unsigned int exceptionDetail2, unsigned int exceptionDetail3, unsigned int exceptionDetail4, unsigned int exceptionDetail5, unsigned int exceptionDetail6, unsigned int exceptionDetail7)
- static __forceinline__
__device__ int [optixGetExceptionCode](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_0](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_1](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_2](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_3](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_4](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_5](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_6](#) ()
- static __forceinline__
__device__ unsigned int [optixGetExceptionDetail_7](#) ()
- static __forceinline__
__device__
[OptixTraversableHandle](#) [optixGetExceptionInvalidTraversable](#) ()
- static __forceinline__
__device__ int [optixGetExceptionInvalidSbtOffset](#) ()
- static __forceinline__
__device__
[OptixInvalidRayExceptionDetails](#) [optixGetExceptionInvalidRay](#) ()
- static __forceinline__
__device__
[OptixParameterMismatchExceptionDetails](#) [optixGetExceptionParameterMismatch](#) ()
- static __forceinline__
__device__ char * [optixGetExceptionLineInfo](#) ()
- template<typename ReturnT , typename... ArgTypes>
static __forceinline__
__device__ ReturnT [optixDirectCall](#) (unsigned int sbtIndex, ArgTypes...args)
- template<typename ReturnT , typename... ArgTypes>
static __forceinline__
__device__ ReturnT [optixContinuationCall](#) (unsigned int sbtIndex, ArgTypes...args)

6.3.1 Detailed Description

OptiX public API.

Author

NVIDIA Corporation OptiX public API Reference - Device side implementation

6.3.2 Macro Definition Documentation

6.3.2.1 `#define OPTIX_DEFINE_optixGetAttribute_BODY(     which )`

Value:

```
unsigned int ret;
    \
asm( "call (%0), _optix_get_attribute_" #which ", ();" : "=r"( ret ) : );
    \
return ret;
```

6.3.2.2 `#define OPTIX_DEFINE_optixGetExceptionDetail_BODY(     which )`

Value:

```
unsigned int ret;
    \
asm( "call (%0), _optix_get_exception_detail_" #which ", ();" : "=r"( ret ) : );
    \
return ret;
```

6.3.2.3 `#define OPTIX_DEFINE_optixGetPayload_BODY(     which )`

Value:

```
unsigned int result;
    \
asm volatile( "call (%0), _optix_get_payload_" #which ", ();" : "=r"( result ) : );
    \
return result;
```

6.3.2.4 `#define OPTIX_DEFINE_optixSetPayload_BODY(     which ) asm volatile( "call _optix_set_payload_" #which ", (%0);" : : "r"( p ) : );`

6.3.3 Function Documentation

6.3.3.1 `template<typename ReturnT , typename... ArgTypes> static __forceinline__ __device__ ReturnT optixContinuationCall (`

```

    unsigned int sbtIndex,
    ArgTypes... args ) [static]

```

```

6.3.3.2 template<typename ReturnT , typename... ArgTypes> static __forceinline__
__device__ ReturnT optixDirectCall (
    unsigned int sbtIndex,
    ArgTypes... args ) [static]

```

```

6.3.3.3 static __forceinline__ __device__ unsigned int optixGetAttribute_0 ( ) [static]

```

```

6.3.3.4 static __forceinline__ __device__ unsigned int optixGetAttribute_1 ( ) [static]

```

```

6.3.3.5 static __forceinline__ __device__ unsigned int optixGetAttribute_2 ( ) [static]

```

```

6.3.3.6 static __forceinline__ __device__ unsigned int optixGetAttribute_3 ( ) [static]

```

```

6.3.3.7 static __forceinline__ __device__ unsigned int optixGetAttribute_4 ( ) [static]

```

```

6.3.3.8 static __forceinline__ __device__ unsigned int optixGetAttribute_5 ( ) [static]

```

```

6.3.3.9 static __forceinline__ __device__ unsigned int optixGetAttribute_6 ( ) [static]

```

```

6.3.3.10 static __forceinline__ __device__ unsigned int optixGetAttribute_7 ( ) [static]

```

```

6.3.3.11 static __forceinline__ __device__ void optixGetCubicBSplineVertexData (
    OptixTraversableHandle gas,
    unsigned int primIdx,
    unsigned int sbtGASIndex,
    float time,

```

float4 data[4]) [static]

6.3.3.12 static __forceinline__ __device__ float optixGetCurveParameter () [static]

6.3.3.13 static __forceinline__ __device__ int optixGetExceptionCode () [static]

6.3.3.14 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_0 () [static]

6.3.3.15 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_1 () [static]

6.3.3.16 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_2 () [static]

6.3.3.17 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_3 () [static]

6.3.3.18 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_4 () [static]

6.3.3.19 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_5 () [static]

6.3.3.20 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_6 () [static]

6.3.3.21 static __forceinline__ __device__ unsigned int optixGetExceptionDetail_7 () [static]

**6.3.3.22 static __forceinline__ __device__ OptixInvalidRayExceptionDetails
optixGetExceptionInvalidRay () [static]**

6.3.3.23 static __forceinline__ __device__ int optixGetExceptionInvalidSbtOffset () [static]

**6.3.3.24 static __forceinline__ __device__ OptixTraversableHandle
optixGetExceptionInvalidTraversable () [static]**

6.3.3.25 static __forceinline__ __device__ char* optixGetExceptionLineInfo () [static]

**6.3.3.26 static __forceinline__ __device__ OptixParameterMismatchExceptionDetails
optixGetExceptionParameterMismatch () [static]**

**6.3.3.27 static __forceinline__ __device__ unsigned int optixGetGASMotionStepCount (
OptixTraversableHandle *handle*) [static]**

6.3.3.28 static __forceinline__ __device__ float optixGetGASMotionTimeBegin (

OptixTraversableHandle *handle*) [static]

6.3.3.29 static __forceinline__ __device__ float optixGetGASMotionTimeEnd (
OptixTraversableHandle *handle*) [static]

6.3.3.30 static __forceinline__ __device__ OptixTraversableHandle
optixGetGASTraversableHandle () [static]

6.3.3.31 static __forceinline__ __device__ unsigned int optixGetHitKind () [static]

6.3.3.32 static __forceinline__ __device__ unsigned int optixGetInstanceId () [static]

6.3.3.33 static __forceinline__ __device__ unsigned int optixGetInstanceIdFromHandle (
OptixTraversableHandle *handle*) [static]

6.3.3.34 static __forceinline__ __device__ unsigned int optixGetInstanceIndex () [static]

6.3.3.35 static __forceinline__ __device__ const float4* optixGetInstanceInverseTransform-
FromHandle (
OptixTraversableHandle *handle*) [static]

6.3.3.36 static __forceinline__ __device__ const float4* optixGetInstanceTransformFromHandle
(
OptixTraversableHandle *handle*) [static]

6.3.3.37 static __forceinline__ __device__ uint3 optixGetLaunchDimensions () [static]

6.3.3.38 static __forceinline__ __device__ uint3 optixGetLaunchIndex () [static]

6.3.3.39 static __forceinline__ __device__ void optixGetLinearCurveVertexData (
OptixTraversableHandle *gas*,
unsigned int *primIdx*,
unsigned int *sbtGASIndex*,
float *time*,
float4 *data*[2]) [static]

6.3.3.40 static __forceinline__ __device__ const OptixMatrixMotionTransform*
optixGetMatrixMotionTransformFromHandle (
OptixTraversableHandle *handle*) [static]

6.3.3.41 static __forceinline__ __device__ float3 optixGetObjectRayDirection () [static]

6.3.3.42 static __forceinline__ __device__ float3 optixGetObjectRayOrigin () [static]

6.3.3.43 static __forceinline__ __device__ void optixGetObjectToWorldTransformMatrix (

float *m[12]*) [static]

6.3.3.44 static __forceinline__ __device__ unsigned int optixGetPayload_0 () [static]

6.3.3.45 static __forceinline__ __device__ unsigned int optixGetPayload_1 () [static]

6.3.3.46 static __forceinline__ __device__ unsigned int optixGetPayload_2 () [static]

6.3.3.47 static __forceinline__ __device__ unsigned int optixGetPayload_3 () [static]

6.3.3.48 static __forceinline__ __device__ unsigned int optixGetPayload_4 () [static]

6.3.3.49 static __forceinline__ __device__ unsigned int optixGetPayload_5 () [static]

6.3.3.50 static __forceinline__ __device__ unsigned int optixGetPayload_6 () [static]

6.3.3.51 static __forceinline__ __device__ unsigned int optixGetPayload_7 () [static]

6.3.3.52 static __forceinline__ __device__ unsigned int optixGetPrimitiveIndex () [static]

**6.3.3.53 static __forceinline__ __device__ OptixPrimitiveType optixGetPrimitiveType (
 unsigned int *hitKind*) [static]**

**6.3.3.54 static __forceinline__ __device__ OptixPrimitiveType optixGetPrimitiveType ()
 [static]**

**6.3.3.55 static __forceinline__ __device__ void optixGetQuadraticBSplineVertexData (
 OptixTraversableHandle *gas*,
 unsigned int *primIdx*,
 unsigned int *sbtGASIndex*,
 float *time*,**

float4 *data*[3]) [static]

6.3.3.56 static __forceinline__ __device__ unsigned int optixGetRayFlags () [static]

6.3.3.57 static __forceinline__ __device__ float optixGetRayTime () [static]

6.3.3.58 static __forceinline__ __device__ float optixGetRayTmax () [static]

6.3.3.59 static __forceinline__ __device__ float optixGetRayTmin () [static]

**6.3.3.60 static __forceinline__ __device__ unsigned int optixGetRayVisibilityMask ()
[static]**

6.3.3.61 static __forceinline__ __device__ CUdeviceptr optixGetSbtDataPointer () [static]

6.3.3.62 static __forceinline__ __device__ unsigned int optixGetSbtGASIndex () [static]

**6.3.3.63 static __forceinline__ __device__ const OptixSRTMotionTransform*
optixGetSRTMotionTransformFromHandle (
OptixTraversableHandle *handle*) [static]**

**6.3.3.64 static __forceinline__ __device__ const OptixStaticTransform*
optixGetStaticTransformFromHandle (
OptixTraversableHandle *handle*) [static]**

**6.3.3.65 static __forceinline__ __device__ OptixTraversableHandle
optixGetTransformListHandle (
unsigned int *index*) [static]**

**6.3.3.66 static __forceinline__ __device__ unsigned int optixGetTransformListSize ()
[static]**

**6.3.3.67 static __forceinline__ __device__ OptixTransformType optixGetTransformType-
FromHandle (
OptixTraversableHandle *handle*) [static]**

6.3.3.68 static __forceinline__ __device__ float2 optixGetTriangleBarycentrics () [static]

**6.3.3.69 static __forceinline__ __device__ void optixGetTriangleVertexData (
OptixTraversableHandle *gas*,
unsigned int *primIdx*,
unsigned int *sbtGASIndex*,
float *time*,**

float3 data[3]) [static]

6.3.3.70 static __forceinline__ __device__ float3 optixGetWorldRayDirection () [static]

6.3.3.71 static __forceinline__ __device__ float3 optixGetWorldRayOrigin () [static]

6.3.3.72 static __forceinline__ __device__ void optixGetWorldToObjectTransformMatrix (float m[12]) [static]

6.3.3.73 static __forceinline__ __device__ void optixIgnoreIntersection () [static]

6.3.3.74 static __forceinline__ __device__ bool optixIsBackFaceHit (unsigned int hitKind) [static]

6.3.3.75 static __forceinline__ __device__ bool optixIsBackFaceHit () [static]

6.3.3.76 static __forceinline__ __device__ bool optixIsFrontFaceHit (unsigned int hitKind) [static]

6.3.3.77 static __forceinline__ __device__ bool optixIsFrontFaceHit () [static]

6.3.3.78 static __forceinline__ __device__ bool optixIsTriangleBackFaceHit () [static]

6.3.3.79 static __forceinline__ __device__ bool optixIsTriangleFrontFaceHit () [static]

6.3.3.80 static __forceinline__ __device__ bool optixIsTriangleHit () [static]

6.3.3.81 static __forceinline__ __device__ bool optixReportIntersection (float hitT, unsigned int hitKind) [static]

6.3.3.82 static __forceinline__ __device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0) [static]

6.3.3.83 static __forceinline__ __device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1) [static]

6.3.3.84 static __forceinline__ __device__ bool optixReportIntersection (float hitT, unsigned int hitKind, unsigned int a0, unsigned int a1,

unsigned int *a2*) [static]

6.3.3.85 static __forceinline__ __device__ bool optixReportIntersection (
 float *hitT*,
 unsigned int *hitKind*,
 unsigned int *a0*,
 unsigned int *a1*,
 unsigned int *a2*,
 unsigned int *a3*) [static]

6.3.3.86 static __forceinline__ __device__ bool optixReportIntersection (
 float *hitT*,
 unsigned int *hitKind*,
 unsigned int *a0*,
 unsigned int *a1*,
 unsigned int *a2*,
 unsigned int *a3*,
 unsigned int *a4*) [static]

6.3.3.87 static __forceinline__ __device__ bool optixReportIntersection (
 float *hitT*,
 unsigned int *hitKind*,
 unsigned int *a0*,
 unsigned int *a1*,
 unsigned int *a2*,
 unsigned int *a3*,
 unsigned int *a4*,
 unsigned int *a5*) [static]

6.3.3.88 static __forceinline__ __device__ bool optixReportIntersection (
 float *hitT*,
 unsigned int *hitKind*,
 unsigned int *a0*,
 unsigned int *a1*,
 unsigned int *a2*,
 unsigned int *a3*,
 unsigned int *a4*,
 unsigned int *a5*,
 unsigned int *a6*) [static]

6.3.3.89 static __forceinline__ __device__ bool optixReportIntersection (
 float *hitT*,
 unsigned int *hitKind*,

```

    unsigned int a0,
    unsigned int a1,
    unsigned int a2,
    unsigned int a3,
    unsigned int a4,
    unsigned int a5,
    unsigned int a6,
    unsigned int a7 ) [static]

```

```

6.3.3.90 static __forceinline__ __device__ void optixSetPayload_0 (
    unsigned int p ) [static]

```

```

6.3.3.91 static __forceinline__ __device__ void optixSetPayload_1 (
    unsigned int p ) [static]

```

```

6.3.3.92 static __forceinline__ __device__ void optixSetPayload_2 (
    unsigned int p ) [static]

```

```

6.3.3.93 static __forceinline__ __device__ void optixSetPayload_3 (
    unsigned int p ) [static]

```

```

6.3.3.94 static __forceinline__ __device__ void optixSetPayload_4 (
    unsigned int p ) [static]

```

```

6.3.3.95 static __forceinline__ __device__ void optixSetPayload_5 (
    unsigned int p ) [static]

```

```

6.3.3.96 static __forceinline__ __device__ void optixSetPayload_6 (
    unsigned int p ) [static]

```

```

6.3.3.97 static __forceinline__ __device__ void optixSetPayload_7 (
    unsigned int p ) [static]

```

```

6.3.3.98 static __forceinline__ __device__ void optixTerminateRay ( ) [static]

```

```

6.3.3.99 static __forceinline__ __device__ void optixThrowException (
    int exceptionCode ) [static]

```

```

6.3.3.100 static __forceinline__ __device__ void optixThrowException (
    int exceptionCode,
    unsigned int exceptionDetail0 ) [static]

```

```

6.3.3.101 static __forceinline__ __device__ void optixThrowException (
    int exceptionCode,
    unsigned int exceptionDetail0,

```

unsigned int *exceptionDetail1*) [static]

6.3.3.102 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*,
unsigned int *exceptionDetail2*) [static]

6.3.3.103 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*,
unsigned int *exceptionDetail2*,
unsigned int *exceptionDetail3*) [static]

6.3.3.104 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*,
unsigned int *exceptionDetail2*,
unsigned int *exceptionDetail3*,
unsigned int *exceptionDetail4*) [static]

6.3.3.105 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*,
unsigned int *exceptionDetail2*,
unsigned int *exceptionDetail3*,
unsigned int *exceptionDetail4*,
unsigned int *exceptionDetail5*) [static]

6.3.3.106 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,
unsigned int *exceptionDetail0*,
unsigned int *exceptionDetail1*,
unsigned int *exceptionDetail2*,
unsigned int *exceptionDetail3*,
unsigned int *exceptionDetail4*,
unsigned int *exceptionDetail5*,
unsigned int *exceptionDetail6*) [static]

6.3.3.107 static __forceinline__ __device__ void optixThrowException (
int *exceptionCode*,

```

    unsigned int exceptionDetail0,
    unsigned int exceptionDetail1,
    unsigned int exceptionDetail2,
    unsigned int exceptionDetail3,
    unsigned int exceptionDetail4,
    unsigned int exceptionDetail5,
    unsigned int exceptionDetail6,
    unsigned int exceptionDetail7 ) [static]

```

6.3.3.108 static __forceinline__ __device__ void optixTrace (

```

    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex ) [static]

```

6.3.3.109 static __forceinline__ __device__ void optixTrace (

```

    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0 ) [static]

```

6.3.3.110 static __forceinline__ __device__ void optixTrace (

```

    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,

```

```

OptixVisibilityMask visibilityMask,
unsigned int rayFlags,
unsigned int SBTOffset,
unsigned int SBTstride,
unsigned int missSBTIndex,
unsigned int & p0,
unsigned int & p1 ) [static]

```

```

6.3.3.111 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2 ) [static]

```

```

6.3.3.112 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3 ) [static]

```

```

6.3.3.113 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,

```

```

float3 rayOrigin,
float3 rayDirection,
float tmin,
float tmax,
float rayTime,
OptixVisibilityMask visibilityMask,
unsigned int rayFlags,
unsigned int SBTOffset,
unsigned int SBTstride,
unsigned int missSBTIndex,
unsigned int & p0,
unsigned int & p1,
unsigned int & p2,
unsigned int & p3,
unsigned int & p4 ) [static]

```

6.3.3.114 static __forceinline__ __device__ void optixTrace (

```

    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3,
    unsigned int & p4,
    unsigned int & p5 ) [static]

```

6.3.3.115 static __forceinline__ __device__ void optixTrace (

```

    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,

```

```

OptixVisibilityMask visibilityMask,
unsigned int rayFlags,
unsigned int SBTOffset,
unsigned int SBTstride,
unsigned int missSBTIndex,
unsigned int & p0,
unsigned int & p1,
unsigned int & p2,
unsigned int & p3,
unsigned int & p4,
unsigned int & p5,
unsigned int & p6 ) [static]

```

```

6.3.3.116 static __forceinline__ __device__ void optixTrace (
    OptixTraversableHandle handle,
    float3 rayOrigin,
    float3 rayDirection,
    float tmin,
    float tmax,
    float rayTime,
    OptixVisibilityMask visibilityMask,
    unsigned int rayFlags,
    unsigned int SBTOffset,
    unsigned int SBTstride,
    unsigned int missSBTIndex,
    unsigned int & p0,
    unsigned int & p1,
    unsigned int & p2,
    unsigned int & p3,
    unsigned int & p4,
    unsigned int & p5,
    unsigned int & p6,
    unsigned int & p7 ) [static]

```

```

6.3.3.117 static __forceinline__ __device__ float3 optixTransformNormalFromObject-
    ToWorldSpace (
        float3 normal ) [static]

```

```

6.3.3.118 static __forceinline__ __device__ float3 optixTransformNormalFromWorldToOb-
    jectSpace (

```

float3 *normal*) [static]

6.3.3.119 static __forceinline__ __device__ float3 optixTransformPointFromObjectToWorldSpace (float3 *point*) [static]

6.3.3.120 static __forceinline__ __device__ float3 optixTransformPointFromWorldToObjectSpace (float3 *point*) [static]

6.3.3.121 static __forceinline__ __device__ float3 optixTransformVectorFromObjectToWorldSpace (float3 *vec*) [static]

6.3.3.122 static __forceinline__ __device__ float3 optixTransformVectorFromWorldToObjectSpace (float3 *vec*) [static]

6.3.3.123 static __forceinline__ __device__ unsigned int optixUndefinedValue () [static]

6.4 optix_7_device_impl_exception.h File Reference

Namespaces

- [optix_impl](#)

Constant Groups

- [optix_impl](#)

Functions

- static __forceinline__ __device__ void [optix_impl::optixDumpStaticTransformFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__ __device__ void [optix_impl::optixDumpMotionMatrixTransformFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__ __device__ void [optix_impl::optixDumpSrtMatrixTransformFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__ __device__ void [optix_impl::optixDumpInstanceFromHandle](#) (OptixTraversableHandle handle)
- static __forceinline__ __device__ void [optix_impl::optixDumpTransform](#) (OptixTraversableHandle handle)
- static __forceinline__ __device__ void [optix_impl::optixDumpTransformList](#) ()

- static `__forceinline__`
`__device__ void optix_impl::optixDumpExceptionDetails ()`

6.4.1 Detailed Description

OptiX public API.

Author

NVIDIA Corporation OptiX public API Reference - Device side implementation for exception helper function.

6.5 optix_7_device_impl_transformations.h File Reference

Namespaces

- `optix_impl`

Constant Groups

- `optix_impl`

Functions

- static `__forceinline__`
`__device__ float4 optix_impl::optixAddFloat4 (const float4 &a, const float4 &b)`
- static `__forceinline__`
`__device__ float4 optix_impl::optixMulFloat4 (const float4 &a, float b)`
- static `__forceinline__`
`__device__ uint4 optix_impl::optixLdg (unsigned long long addr)`
- template<class T >
`static __forceinline__ __device__ T optix_impl::optixLoadReadOnlyAlign16 (const T *ptr)`
- static `__forceinline__`
`__device__ float4 optix_impl::optixMultiplyRowMatrix (const float4 vec, const float4 m0, const float4 m1, const float4 m2)`
- static `__forceinline__`
`__device__ void optix_impl::optixGetMatrixFromSrt (float4 &m0, float4 &m1, float4 &m2, const OptixSRTData &srt)`
- static `__forceinline__`
`__device__ void optix_impl::optixInvertMatrix (float4 &m0, float4 &m1, float4 &m2)`
- static `__forceinline__`
`__device__ void optix_impl::optixLoadInterpolatedMatrixKey (float4 &m0, float4 &m1, float4 &m2, const float4 *matrix, const float t1)`
- static `__forceinline__`
`__device__ void optix_impl::optixLoadInterpolatedSrtKey (float4 &srt0, float4 &srt1, float4 &srt2, float4 &srt3, const float4 *srt, const float t1)`

- static __forceinline__
__device__ void [optix_impl::optixResolveMotionKey](#) (float &localt, int &key, const [OptixMotionOptions](#) &options, const float globalt)
- static __forceinline__
__device__ void [optix_impl::optixGetInterpolatedTransformation](#) (float4 &trf0, float4 &trf1, float4 &trf2, const [OptixMatrixMotionTransform](#) *transformData, const float time)
- static __forceinline__
__device__ void [optix_impl::optixGetInterpolatedTransformation](#) (float4 &trf0, float4 &trf1, float4 &trf2, const [OptixSRTMotionTransform](#) *transformData, const float time)
- static __forceinline__
__device__ void [optix_impl::optixGetInterpolatedTransformationFromHandle](#) (float4 &trf0, float4 &trf1, float4 &trf2, const [OptixTraversableHandle](#) handle, const float time, const bool objectToWorld)
- static __forceinline__
__device__ void [optix_impl::optixGetWorldToObjectTransformMatrix](#) (float4 &m0, float4 &m1, float4 &m2)
- static __forceinline__
__device__ void [optix_impl::optixGetObjectToWorldTransformMatrix](#) (float4 &m0, float4 &m1, float4 &m2)
- static __forceinline__
__device__ float3 [optix_impl::optixTransformPoint](#) (const float4 &m0, const float4 &m1, const float4 &m2, const float3 &p)
- static __forceinline__
__device__ float3 [optix_impl::optixTransformVector](#) (const float4 &m0, const float4 &m1, const float4 &m2, const float3 &v)
- static __forceinline__
__device__ float3 [optix_impl::optixTransformNormal](#) (const float4 &m0, const float4 &m1, const float4 &m2, const float3 &n)

6.5.1 Detailed Description

OptiX public API.

Author

NVIDIA Corporation OptiX public API Reference - Device side implementation for transformation helper functions.

6.6 optix_7_host.h File Reference

Functions

- const char * [optixGetErrorName](#) ([OptixResult](#) result)
- const char * [optixGetErrorString](#) ([OptixResult](#) result)
- [OptixResult](#) [optixDeviceContextCreate](#) (CUcontext fromContext, const [OptixDeviceContextOptions](#) *options, [OptixDeviceContext](#) *context)
- [OptixResult](#) [optixDeviceContextDestroy](#) ([OptixDeviceContext](#) context)

- [OptixResult optixDeviceContextGetProperty](#) ([OptixDeviceContext](#) context, [OptixDeviceProperty](#) property, void *value, size_t sizeInBytes)
- [OptixResult optixDeviceContextSetLogCallback](#) ([OptixDeviceContext](#) context, [OptixLogCallback](#) callbackFunction, void *callbackData, unsigned int callbackLevel)
- [OptixResult optixDeviceContextSetCacheEnabled](#) ([OptixDeviceContext](#) context, int enabled)
- [OptixResult optixDeviceContextSetCacheLocation](#) ([OptixDeviceContext](#) context, const char *location)
- [OptixResult optixDeviceContextSetCacheDatabaseSizes](#) ([OptixDeviceContext](#) context, size_t lowWaterMark, size_t highWaterMark)
- [OptixResult optixDeviceContextGetCacheEnabled](#) ([OptixDeviceContext](#) context, int *enabled)
- [OptixResult optixDeviceContextGetCacheLocation](#) ([OptixDeviceContext](#) context, char *location, size_t locationSize)
- [OptixResult optixDeviceContextGetCacheDatabaseSizes](#) ([OptixDeviceContext](#) context, size_t *lowWaterMark, size_t *highWaterMark)
- [OptixResult optixPipelineCreate](#) ([OptixDeviceContext](#) context, const [OptixPipelineCompileOptions](#) *pipelineCompileOptions, const [OptixPipelineLinkOptions](#) *pipelineLinkOptions, const [OptixProgramGroup](#) *programGroups, unsigned int numProgramGroups, char *logString, size_t *logStringSize, [OptixPipeline](#) *pipeline)
- [OptixResult optixPipelineDestroy](#) ([OptixPipeline](#) pipeline)
- [OptixResult optixPipelineSetStackSize](#) ([OptixPipeline](#) pipeline, unsigned int directCallableStackSizeFromTraversal, unsigned int directCallableStackSizeFromState, unsigned int continuationStackSize, unsigned int maxTraversableGraphDepth)
- [OptixResult optixModuleCreateFromPTX](#) ([OptixDeviceContext](#) context, const [OptixModuleCompileOptions](#) *moduleCompileOptions, const [OptixPipelineCompileOptions](#) *pipelineCompileOptions, const char *PTX, size_t PTXsize, char *logString, size_t *logStringSize, [OptixModule](#) *module)
- [OptixResult optixModuleDestroy](#) ([OptixModule](#) module)
- [OptixResult optixBuiltinISModuleGet](#) ([OptixDeviceContext](#) context, const [OptixModuleCompileOptions](#) *moduleCompileOptions, const [OptixPipelineCompileOptions](#) *pipelineCompileOptions, const [OptixBuiltinISOptions](#) *builtinISOptions, [OptixModule](#) *builtinModule)
- [OptixResult optixProgramGroupGetStackSize](#) ([OptixProgramGroup](#) programGroup, [OptixStackSizes](#) *stackSizes)
- [OptixResult optixProgramGroupCreate](#) ([OptixDeviceContext](#) context, const [OptixProgramGroupDesc](#) *programDescriptions, unsigned int numProgramGroups, const [OptixProgramGroupOptions](#) *options, char *logString, size_t *logStringSize, [OptixProgramGroup](#) *programGroups)
- [OptixResult optixProgramGroupDestroy](#) ([OptixProgramGroup](#) programGroup)
- [OptixResult optixLaunch](#) ([OptixPipeline](#) pipeline, CUstream stream, CUdeviceptr pipelineParams, size_t pipelineParamsSize, const [OptixShaderBindingTable](#) *sbt, unsigned int width, unsigned int height, unsigned int depth)
- [OptixResult optixSbtRecordPackHeader](#) ([OptixProgramGroup](#) programGroup, void *sbtRecordHeaderHostPointer)
- [OptixResult optixAccelComputeMemoryUsage](#) ([OptixDeviceContext](#) context, const [OptixAccelBuildOptions](#) *accelOptions, const [OptixBuildInput](#) *buildInputs, unsigned int numBuildInputs, [OptixAccelBufferSizes](#) *bufferSizes)

- `OptixResult optixAccelBuild` (`OptixDeviceContext` context, `CUstream` stream, const `OptixAccelBuildOptions` *accelOptions, const `OptixBuildInput` *buildInputs, unsigned int numBuildInputs, `CUdeviceptr` tempBuffer, size_t tempBufferSizeInBytes, `CUdeviceptr` outputBuffer, size_t outputBufferSizeInBytes, `OptixTraversableHandle` *outputHandle, const `OptixAccelEmitDesc` *emittedProperties, unsigned int numEmittedProperties)
- `OptixResult optixAccelGetRelocationInfo` (`OptixDeviceContext` context, `OptixTraversableHandle` handle, `OptixAccelRelocationInfo` *info)
- `OptixResult optixAccelCheckRelocationCompatibility` (`OptixDeviceContext` context, const `OptixAccelRelocationInfo` *info, int *compatible)
- `OptixResult optixAccelRelocate` (`OptixDeviceContext` context, `CUstream` stream, const `OptixAccelRelocationInfo` *info, `CUdeviceptr` instanceTraversableHandles, size_t numInstanceTraversableHandles, `CUdeviceptr` targetAccel, size_t targetAccelSizeInBytes, `OptixTraversableHandle` *targetHandle)
- `OptixResult optixAccelCompact` (`OptixDeviceContext` context, `CUstream` stream, `OptixTraversableHandle` inputHandle, `CUdeviceptr` outputBuffer, size_t outputBufferSizeInBytes, `OptixTraversableHandle` *outputHandle)
- `OptixResult optixConvertPointerToTraversableHandle` (`OptixDeviceContext` onDevice, `CUdeviceptr` pointer, `OptixTraversableType` traversableType, `OptixTraversableHandle` *traversableHandle)
- `OptixResult optixDenoiserCreate` (`OptixDeviceContext` context, const `OptixDenoiserOptions` *options, `OptixDenoiser` *denoiser)
- `OptixResult optixDenoiserSetModel` (`OptixDenoiser` denoiser, `OptixDenoiserModelKind` kind, void *data, size_t sizeInBytes)
- `OptixResult optixDenoiserDestroy` (`OptixDenoiser` denoiser)
- `OptixResult optixDenoiserComputeMemoryResources` (const `OptixDenoiser` denoiser, unsigned int outputWidth, unsigned int outputHeight, `OptixDenoiserSizes` *returnSizes)
- `OptixResult optixDenoiserSetup` (`OptixDenoiser` denoiser, `CUstream` stream, unsigned int inputWidth, unsigned int inputHeight, `CUdeviceptr` denoiserState, size_t denoiserStateSizeInBytes, `CUdeviceptr` scratch, size_t scratchSizeInBytes)
- `OptixResult optixDenoiserInvoke` (`OptixDenoiser` denoiser, `CUstream` stream, const `OptixDenoiserParams` *params, `CUdeviceptr` denoiserState, size_t denoiserStateSizeInBytes, const `OptixImage2D` *inputLayers, unsigned int numInputLayers, unsigned int inputOffsetX, unsigned int inputOffsetY, const `OptixImage2D` *outputLayer, `CUdeviceptr` scratch, size_t scratchSizeInBytes)
- `OptixResult optixDenoiserComputeIntensity` (`OptixDenoiser` denoiser, `CUstream` stream, const `OptixImage2D` *inputImage, `CUdeviceptr` outputIntensity, `CUdeviceptr` scratch, size_t scratchSizeInBytes)

6.6.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation OptiX host include file – includes the host api if compiling host code. For the math library routines include `optix_math.h`

6.7 optix_7_types.h File Reference

Classes

- struct [OptixDeviceContextOptions](#)
- struct [OptixBuildInputTriangleArray](#)
- struct [OptixBuildInputCurveArray](#)
- struct [OptixAabb](#)
- struct [OptixBuildInputCustomPrimitiveArray](#)
- struct [OptixBuildInputInstanceArray](#)
- struct [OptixBuildInput](#)
- struct [OptixInstance](#)
- struct [OptixMotionOptions](#)
- struct [OptixAccelBuildOptions](#)
- struct [OptixAccelBufferSizes](#)
- struct [OptixAccelEmitDesc](#)
- struct [OptixAccelRelocationInfo](#)
- struct [OptixStaticTransform](#)
- struct [OptixMatrixMotionTransform](#)
- struct [OptixSRTData](#)
- struct [OptixSRTMotionTransform](#)
- struct [OptixImage2D](#)
- struct [OptixDenoiserOptions](#)
- struct [OptixDenoiserParams](#)
- struct [OptixDenoiserSizes](#)
- struct [OptixModuleCompileOptions](#)
- struct [OptixProgramGroupSingleModule](#)
- struct [OptixProgramGroupHitgroup](#)
- struct [OptixProgramGroupCallables](#)
- struct [OptixProgramGroupDesc](#)
- struct [OptixProgramGroupOptions](#)
- struct [OptixPipelineCompileOptions](#)
- struct [OptixPipelineLinkOptions](#)
- struct [OptixShaderBindingTable](#)
- struct [OptixStackSizes](#)
- struct [OptixBuiltinISOOptions](#)

Macros

- [#define OPTIX_SBT_RECORD_HEADER_SIZE \(\(size_t\)32 \)](#)
- [#define OPTIX_SBT_RECORD_ALIGNMENT 16ull](#)
- [#define OPTIX_ACCEL_BUFFER_BYTE_ALIGNMENT 128ull](#)
- [#define OPTIX_INSTANCE_BYTE_ALIGNMENT 16ull](#)
- [#define OPTIX_AABB_BUFFER_BYTE_ALIGNMENT 8ull](#)
- [#define OPTIX_GEOMETRY_TRANSFORM_BYTE_ALIGNMENT 16ull](#)
- [#define OPTIX_TRANSFORM_BYTE_ALIGNMENT 64ull](#)
- [#define OPTIX_COMPILE_DEFAULT_MAX_REGISTER_COUNT 0](#)

Typedefs

- typedef unsigned long long [CUdeviceptr](#)
- typedef struct
 [OptixDeviceContext_t](#) * [OptixDeviceContext](#)
- typedef struct [OptixModule_t](#) * [OptixModule](#)
- typedef struct
 [OptixProgramGroup_t](#) * [OptixProgramGroup](#)
- typedef struct [OptixPipeline_t](#) * [OptixPipeline](#)
- typedef struct [OptixDenoiser_t](#) * [OptixDenoiser](#)
- typedef unsigned long long [OptixTraversableHandle](#)
- typedef unsigned int [OptixVisibilityMask](#)
- typedef enum [OptixResult](#) [OptixResult](#)
- typedef enum [OptixDeviceProperty](#) [OptixDeviceProperty](#)
- typedef void(* [OptixLogCallback](#))(unsigned int level, const char *tag, const char *message, void *cbdata)
- typedef struct
 [OptixDeviceContextOptions](#) [OptixDeviceContextOptions](#)
- typedef enum [OptixGeometryFlags](#) [OptixGeometryFlags](#)
- typedef enum [OptixHitKind](#) [OptixHitKind](#)
- typedef enum [OptixIndicesFormat](#) [OptixIndicesFormat](#)
- typedef enum [OptixVertexFormat](#) [OptixVertexFormat](#)
- typedef enum [OptixTransformFormat](#) [OptixTransformFormat](#)
- typedef struct
 [OptixBuildInputTriangleArray](#) [OptixBuildInputTriangleArray](#)
- typedef enum [OptixPrimitiveType](#) [OptixPrimitiveType](#)
- typedef enum
 [OptixPrimitiveTypeFlags](#) [OptixPrimitiveTypeFlags](#)
- typedef struct
 [OptixBuildInputCurveArray](#) [OptixBuildInputCurveArray](#)
- typedef struct [OptixAabb](#) [OptixAabb](#)
- typedef struct
 [OptixBuildInputCustomPrimitiveArray](#) [OptixBuildInputCustomPrimitiveArray](#)
- typedef struct
 [OptixBuildInputInstanceArray](#) [OptixBuildInputInstanceArray](#)
- typedef enum [OptixBuildInputType](#) [OptixBuildInputType](#)
- typedef struct [OptixBuildInput](#) [OptixBuildInput](#)
- typedef enum [OptixInstanceFlags](#) [OptixInstanceFlags](#)
- typedef struct [OptixInstance](#) [OptixInstance](#)
- typedef enum [OptixBuildFlags](#) [OptixBuildFlags](#)
- typedef enum [OptixBuildOperation](#) [OptixBuildOperation](#)
- typedef enum [OptixMotionFlags](#) [OptixMotionFlags](#)
- typedef struct [OptixMotionOptions](#) [OptixMotionOptions](#)
- typedef struct
 [OptixAccelBuildOptions](#) [OptixAccelBuildOptions](#)
- typedef struct
 [OptixAccelBufferSizes](#) [OptixAccelBufferSizes](#)

- typedef enum [OptixAccelPropertyType](#) [OptixAccelPropertyType](#)
- typedef struct [OptixAccelEmitDesc](#) [OptixAccelEmitDesc](#)
- typedef struct [OptixAccelRelocationInfo](#) [OptixAccelRelocationInfo](#)
- typedef struct [OptixStaticTransform](#) [OptixStaticTransform](#)
- typedef struct [OptixMatrixMotionTransform](#) [OptixMatrixMotionTransform](#)
- typedef struct [OptixSRTData](#) [OptixSRTData](#)
- typedef struct [OptixSRTMotionTransform](#) [OptixSRTMotionTransform](#)
- typedef enum [OptixTraversableType](#) [OptixTraversableType](#)
- typedef enum [OptixPixelFormat](#) [OptixPixelFormat](#)
- typedef struct [OptixImage2D](#) [OptixImage2D](#)
- typedef enum [OptixDenoiserInputKind](#) [OptixDenoiserInputKind](#)
- typedef enum [OptixDenoiserModelKind](#) [OptixDenoiserModelKind](#)
- typedef struct [OptixDenoiserOptions](#) [OptixDenoiserOptions](#)
- typedef struct [OptixDenoiserParams](#) [OptixDenoiserParams](#)
- typedef struct [OptixDenoiserSizes](#) [OptixDenoiserSizes](#)
- typedef enum [OptixRayFlags](#) [OptixRayFlags](#)
- typedef enum [OptixTransformType](#) [OptixTransformType](#)
- typedef enum [OptixTraversableGraphFlags](#) [OptixTraversableGraphFlags](#)
- typedef enum [OptixCompileOptimizationLevel](#) [OptixCompileOptimizationLevel](#)
- typedef enum [OptixCompileDebugLevel](#) [OptixCompileDebugLevel](#)
- typedef struct [OptixModuleCompileOptions](#) [OptixModuleCompileOptions](#)
- typedef enum [OptixProgramGroupKind](#) [OptixProgramGroupKind](#)
- typedef enum [OptixProgramGroupFlags](#) [OptixProgramGroupFlags](#)
- typedef struct [OptixProgramGroupSingleModule](#) [OptixProgramGroupSingleModule](#)
- typedef struct [OptixProgramGroupHitgroup](#) [OptixProgramGroupHitgroup](#)
- typedef struct [OptixProgramGroupCallables](#) [OptixProgramGroupCallables](#)
- typedef struct [OptixProgramGroupDesc](#) [OptixProgramGroupDesc](#)
- typedef struct [OptixProgramGroupOptions](#) [OptixProgramGroupOptions](#)
- typedef enum [OptixExceptionCodes](#) [OptixExceptionCodes](#)
- typedef enum [OptixExceptionFlags](#) [OptixExceptionFlags](#)
- typedef struct [OptixPipelineCompileOptions](#) [OptixPipelineCompileOptions](#)
- typedef struct [OptixPipelineLinkOptions](#) [OptixPipelineLinkOptions](#)
- typedef struct [OptixShaderBindingTable](#) [OptixShaderBindingTable](#)

- typedef struct OptixStackSizes OptixStackSizes
- typedef enum
OptixQueryFunctionTableOptions OptixQueryFunctionTableOptions
- typedef OptixResult(OptixQueryFunctionTable_t)(int abild, unsigned int numOptions,
OptixQueryFunctionTableOptions *, const void **, void *functionTable, size_t sizeOfTable)
- typedef struct
OptixBuiltinISOOptions OptixBuiltinISOOptions

Enumerations

- enum OptixResult {
OPTIX_SUCCESS = 0,
OPTIX_ERROR_INVALID_VALUE = 7001,
OPTIX_ERROR_HOST_OUT_OF_MEMORY = 7002,
OPTIX_ERROR_INVALID_OPERATION = 7003,
OPTIX_ERROR_FILE_IO_ERROR = 7004,
OPTIX_ERROR_INVALID_FILE_FORMAT = 7005,
OPTIX_ERROR_DISK_CACHE_INVALID_PATH = 7010,
OPTIX_ERROR_DISK_CACHE_PERMISSION_ERROR = 7011,
OPTIX_ERROR_DISK_CACHE_DATABASE_ERROR = 7012,
OPTIX_ERROR_DISK_CACHE_INVALID_DATA = 7013,
OPTIX_ERROR_LAUNCH_FAILURE = 7050,
OPTIX_ERROR_INVALID_DEVICE_CONTEXT = 7051,
OPTIX_ERROR_CUDA_NOT_INITIALIZED = 7052,
OPTIX_ERROR_INVALID_PTX = 7200,
OPTIX_ERROR_INVALID_LAUNCH_PARAMETER = 7201,
OPTIX_ERROR_INVALID_PAYLOAD_ACCESS = 7202,
OPTIX_ERROR_INVALID_ATTRIBUTE_ACCESS = 7203,
OPTIX_ERROR_INVALID_FUNCTION_USE = 7204,
OPTIX_ERROR_INVALID_FUNCTION_ARGUMENTS = 7205,
OPTIX_ERROR_PIPELINE_OUT_OF_CONSTANT_MEMORY = 7250,
OPTIX_ERROR_PIPELINE_LINK_ERROR = 7251,
OPTIX_ERROR_INTERNAL_COMPILER_ERROR = 7299,
OPTIX_ERROR_DENOISER_MODEL_NOT_SET = 7300,
OPTIX_ERROR_DENOISER_NOT_INITIALIZED = 7301,
OPTIX_ERROR_ACCEL_NOT_COMPATIBLE = 7400,
OPTIX_ERROR_NOT_SUPPORTED = 7800,
OPTIX_ERROR_UNSUPPORTED_ABI_VERSION = 7801,
OPTIX_ERROR_FUNCTION_TABLE_SIZE_MISMATCH = 7802,
OPTIX_ERROR_INVALID_ENTRY_FUNCTION_OPTIONS = 7803,
OPTIX_ERROR_LIBRARY_NOT_FOUND = 7804,
OPTIX_ERROR_ENTRY_SYMBOL_NOT_FOUND = 7805,
OPTIX_ERROR_CUDA_ERROR = 7900,
OPTIX_ERROR_INTERNAL_ERROR = 7990,
OPTIX_ERROR_UNKNOWN = 7999 }
- enum OptixDeviceProperty {
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_TRACE_DEPTH = 0x2001,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_TRAVERSABLE_GRAPH_DEPTH = 0x2002,

```

OPTIX_DEVICE_PROPERTY_LIMIT_MAX_PRIMITIVES_PER_GAS = 0x2003,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCES_PER_IAS = 0x2004,
OPTIX_DEVICE_PROPERTY_RTCORE_VERSION = 0x2005,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCE_ID = 0x2006,
OPTIX_DEVICE_PROPERTY_LIMIT_NUM_BITS_INSTANCE_VISIBILITY_MASK = 0x2007,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_RECORDS_PER_GAS = 0x2008,
OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_OFFSET = 0x2009 }

• enum OptixGeometryFlags {
    OPTIX_GEOMETRY_FLAG_NONE = 0,
    OPTIX_GEOMETRY_FLAG_DISABLE_ANYHIT = 1u << 0,
    OPTIX_GEOMETRY_FLAG_REQUIRE_SINGLE_ANYHIT_CALL = 1u << 1 }

• enum OptixHitKind {
    OPTIX_HIT_KIND_TRIANGLE_FRONT_FACE = 0xFE,
    OPTIX_HIT_KIND_TRIANGLE_BACK_FACE = 0xFF }

• enum OptixIndicesFormat {
    OPTIX_INDICES_FORMAT_NONE = 0,
    OPTIX_INDICES_FORMAT_UNSIGNED_SHORT3 = 0x2102,
    OPTIX_INDICES_FORMAT_UNSIGNED_INT3 = 0x2103 }

• enum OptixVertexFormat {
    OPTIX_VERTEX_FORMAT_NONE = 0,
    OPTIX_VERTEX_FORMAT_FLOAT3 = 0x2121,
    OPTIX_VERTEX_FORMAT_FLOAT2 = 0x2122,
    OPTIX_VERTEX_FORMAT_HALF3 = 0x2123,
    OPTIX_VERTEX_FORMAT_HALF2 = 0x2124,
    OPTIX_VERTEX_FORMAT_SNORM16_3 = 0x2125,
    OPTIX_VERTEX_FORMAT_SNORM16_2 = 0x2126 }

• enum OptixTransformFormat {
    OPTIX_TRANSFORM_FORMAT_NONE = 0,
    OPTIX_TRANSFORM_FORMAT_MATRIX_FLOAT12 = 0x21E1 }

• enum OptixPrimitiveType {
    OPTIX_PRIMITIVE_TYPE_CUSTOM = 0x2500,
    OPTIX_PRIMITIVE_TYPE_ROUND_QUADRATIC_BSPLINE = 0x2501,
    OPTIX_PRIMITIVE_TYPE_ROUND_CUBIC_BSPLINE = 0x2502,
    OPTIX_PRIMITIVE_TYPE_ROUND_LINEAR = 0x2503,
    OPTIX_PRIMITIVE_TYPE_TRIANGLE = 0x2531 }

• enum OptixPrimitiveTypeFlags {
    OPTIX_PRIMITIVE_TYPE_FLAGS_CUSTOM = 1 << 0,
    OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_QUADRATIC_BSPLINE = 1 << 1,
    OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_CUBIC_BSPLINE = 1 << 2,
    OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_LINEAR = 1 << 3,
    OPTIX_PRIMITIVE_TYPE_FLAGS_TRIANGLE = 1 << 31 }

• enum OptixBuildInputType {
    OPTIX_BUILD_INPUT_TYPE_TRIANGLES = 0x2141,
    OPTIX_BUILD_INPUT_TYPE_CUSTOM_PRIMITIVES = 0x2142,
    OPTIX_BUILD_INPUT_TYPE_INSTANCES = 0x2143,
    OPTIX_BUILD_INPUT_TYPE_INSTANCE_POINTERS = 0x2144,
    OPTIX_BUILD_INPUT_TYPE_CURVES = 0x2145 }

• enum OptixInstanceFlags {

```

```

OPTIX_INSTANCE_FLAG_NONE = 0,
OPTIX_INSTANCE_FLAG_DISABLE_TRIANGLE_FACE_CULLING = 1u << 0,
OPTIX_INSTANCE_FLAG_FLIP_TRIANGLE_FACING = 1u << 1,
OPTIX_INSTANCE_FLAG_DISABLE_ANYHIT = 1u << 2,
OPTIX_INSTANCE_FLAG_ENFORCE_ANYHIT = 1u << 3,
OPTIX_INSTANCE_FLAG_DISABLE_TRANSFORM = 1u << 6 }

• enum OptixBuildFlags {
    OPTIX_BUILD_FLAG_NONE = 0,
    OPTIX_BUILD_FLAG_ALLOW_UPDATE = 1u << 0,
    OPTIX_BUILD_FLAG_ALLOW_COMPACTION = 1u << 1,
    OPTIX_BUILD_FLAG_PREFER_FAST_TRACE = 1u << 2,
    OPTIX_BUILD_FLAG_PREFER_FAST_BUILD = 1u << 3,
    OPTIX_BUILD_FLAG_ALLOW_RANDOM_VERTEX_ACCESS = 1u << 4 }

• enum OptixBuildOperation {
    OPTIX_BUILD_OPERATION_BUILD = 0x2161,
    OPTIX_BUILD_OPERATION_UPDATE = 0x2162 }

• enum OptixMotionFlags {
    OPTIX_MOTION_FLAG_NONE = 0,
    OPTIX_MOTION_FLAG_START_VANISH = 1u << 0,
    OPTIX_MOTION_FLAG_END_VANISH = 1u << 1 }

• enum OptixAccelPropertyType {
    OPTIX_PROPERTY_TYPE_COMPACTED_SIZE = 0x2181,
    OPTIX_PROPERTY_TYPE_AABBS = 0x2182 }

• enum OptixTraversableType {
    OPTIX_TRAVERSABLE_TYPE_STATIC_TRANSFORM = 0x21C1,
    OPTIX_TRAVERSABLE_TYPE_MATRIX_MOTION_TRANSFORM = 0x21C2,
    OPTIX_TRAVERSABLE_TYPE_SRT_MOTION_TRANSFORM = 0x21C3 }

• enum OptixPixelFormat {
    OPTIX_PIXEL_FORMAT_HALF3 = 0x2201,
    OPTIX_PIXEL_FORMAT_HALF4 = 0x2202,
    OPTIX_PIXEL_FORMAT_FLOAT3 = 0x2203,
    OPTIX_PIXEL_FORMAT_FLOAT4 = 0x2204,
    OPTIX_PIXEL_FORMAT_UCHAR3 = 0x2205,
    OPTIX_PIXEL_FORMAT_UCHAR4 = 0x2206 }

• enum OptixDenoiserInputKind {
    OPTIX_DENOISER_INPUT_RGB = 0x2301,
    OPTIX_DENOISER_INPUT_RGB_ALBEDO = 0x2302,
    OPTIX_DENOISER_INPUT_RGB_ALBEDO_NORMAL = 0x2303 }

• enum OptixDenoiserModelKind {
    OPTIX_DENOISER_MODEL_KIND_USER = 0x2321,
    OPTIX_DENOISER_MODEL_KIND_LDR = 0x2322,
    OPTIX_DENOISER_MODEL_KIND_HDR = 0x2323 }

• enum OptixRayFlags {
    OPTIX_RAY_FLAG_NONE = 0u,
    OPTIX_RAY_FLAG_DISABLE_ANYHIT = 1u << 0,
    OPTIX_RAY_FLAG_ENFORCE_ANYHIT = 1u << 1,
    OPTIX_RAY_FLAG_TERMINATE_ON_FIRST_HIT = 1u << 2,
    OPTIX_RAY_FLAG_DISABLE_CLOSESTHIT = 1u << 3,

```

```

OPTIX_RAY_FLAG_CULL_BACK_FACING_TRIANGLES = 1u << 4,
OPTIX_RAY_FLAG_CULL_FRONT_FACING_TRIANGLES = 1u << 5,
OPTIX_RAY_FLAG_CULL_DISABLED_ANYHIT = 1u << 6,
OPTIX_RAY_FLAG_CULL_ENFORCED_ANYHIT = 1u << 7 }

• enum OptixTransformType {
    OPTIX_TRANSFORM_TYPE_NONE = 0,
    OPTIX_TRANSFORM_TYPE_STATIC_TRANSFORM = 1,
    OPTIX_TRANSFORM_TYPE_MATRIX_MOTION_TRANSFORM = 2,
    OPTIX_TRANSFORM_TYPE_SRT_MOTION_TRANSFORM = 3,
    OPTIX_TRANSFORM_TYPE_INSTANCE = 4 }

• enum OptixTraversableGraphFlags {
    OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_ANY = 0,
    OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_GAS = 1u << 0,
    OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_LEVEL_INSTANCING = 1u << 1 }

• enum OptixCompileOptimizationLevel {
    OPTIX_COMPILE_OPTIMIZATION_DEFAULT = 0,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_0 = 0x2340,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_1 = 0x2341,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_2 = 0x2342,
    OPTIX_COMPILE_OPTIMIZATION_LEVEL_3 = 0x2343 }

• enum OptixCompileDebugLevel {
    OPTIX_COMPILE_DEBUG_LEVEL_DEFAULT = 0,
    OPTIX_COMPILE_DEBUG_LEVEL_NONE = 0x2350,
    OPTIX_COMPILE_DEBUG_LEVEL_LINEINFO = 0x2351,
    OPTIX_COMPILE_DEBUG_LEVEL_FULL = 0x2352 }

• enum OptixProgramGroupKind {
    OPTIX_PROGRAM_GROUP_KIND_RAYGEN = 0x2421,
    OPTIX_PROGRAM_GROUP_KIND_MISS = 0x2422,
    OPTIX_PROGRAM_GROUP_KIND_EXCEPTION = 0x2423,
    OPTIX_PROGRAM_GROUP_KIND_HITGROUP = 0x2424,
    OPTIX_PROGRAM_GROUP_KIND_CALLABLES = 0x2425 }

• enum OptixProgramGroupFlags { OPTIX_PROGRAM_GROUP_FLAGS_NONE = 0 }

• enum OptixExceptionCodes {
    OPTIX_EXCEPTION_CODE_STACK_OVERFLOW = -1,
    OPTIX_EXCEPTION_CODE_TRACE_DEPTH_EXCEEDED = -2,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_DEPTH_EXCEEDED = -3,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_TRAVERSABLE = -5,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_MISS_SBT = -6,
    OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_HIT_SBT = -7,
    OPTIX_EXCEPTION_CODE_UNSUPPORTED_PRIMITIVE_TYPE = -8,
    OPTIX_EXCEPTION_CODE_INVALID_RAY = -9,
    OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH = -10,
    OPTIX_EXCEPTION_CODE_BUILTIN_IS_MISMATCH = -11,
    OPTIX_EXCEPTION_CODE_UNSUPPORTED_SINGLE_LEVEL_GAS = -15 }

• enum OptixExceptionFlags {
    OPTIX_EXCEPTION_FLAG_NONE = 0,
    OPTIX_EXCEPTION_FLAG_STACK_OVERFLOW = 1u << 0,
    OPTIX_EXCEPTION_FLAG_TRACE_DEPTH = 1u << 1,
    OPTIX_EXCEPTION_FLAG_USER = 1u << 2,

```

```
OPTIX_EXCEPTION_FLAG_DEBUG = 1u << 3 }
```

- enum `OptixQueryFunctionTableOptions` {
`OPTIX_QUERY_FUNCTION_TABLE_OPTION_DUMMY = 0` }

6.7.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation OptiX types include file – defines types and enums used by the API. For the math library routines include `optix_math.h`

6.8 optix_denoiser_tiling.h File Reference

Classes

- struct `OptixUtilDenoiserImageTile`

Functions

- unsigned int `optixUtilGetPixelStride` (const `OptixImage2D` &image)
- `OptixResult` `optixUtilDenoiserSplitImage` (const `OptixImage2D` &input, const `OptixImage2D` &output, unsigned int overlapWindowSizeInPixels, unsigned int tileWidth, unsigned int tileHeight, std::vector< `OptixUtilDenoiserImageTile` > &tiles)
- `OptixResult` `optixUtilDenoiserInvokeTiled` (`OptixDenoiser` &denoiser, CUstream stream, const `OptixDenoiserParams` *params, CUdeviceptr denoiserState, size_t denoiserStateSizeInBytes, const `OptixImage2D` *inputLayers, unsigned int numInputLayers, const `OptixImage2D` *outputLayer, CUdeviceptr scratch, size_t scratchSizeInBytes, unsigned int overlapWindowSizeInPixels, unsigned int tileWidth, unsigned int tileHeight)

6.8.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation

6.9 optix_device.h File Reference

Macros

- #define `__UNDEF_OPTIX_INCLUDE_INTERNAL_HEADERS_OPTIX_DEVICE_H__`

6.9.1 Detailed Description

OptiX public API.

Author

NVIDIA Corporation OptiX public API Reference - Host/Device side

6.9.2 Macro Definition Documentation

6.9.2.1 #define __UNDEF_OPTIX_INCLUDE_INTERNAL_HEADERS_OPTIX_DEVICE_H__

6.10 optix_function_table.h File Reference

Classes

- struct [OptixFunctionTable](#)

Macros

- #define [OPTIX_ABI_VERSION](#) 36

Typedefs

- typedef struct [OptixFunctionTable](#) [OptixFunctionTable](#)

6.10.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation

6.10.2 Macro Definition Documentation

6.10.2.1 #define OPTIX_ABI_VERSION 36

The OptiX ABI version.

6.11 optix_function_table_definition.h File Reference

Variables

- [OptixFunctionTable](#) [g_optixFunctionTable](#)

6.11.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation

6.12 optix_host.h File Reference

Macros

- `#define __UNDEF_OPTIX_INCLUDE_INTERNAL_HEADERS_OPTIX_HOST_H__`

6.12.1 Detailed Description

OptiX public API.

Author

NVIDIA Corporation OptiX public API Reference - Host side

6.12.2 Macro Definition Documentation

6.12.2.1 `#define __UNDEF_OPTIX_INCLUDE_INTERNAL_HEADERS_OPTIX_HOST_H__`

6.13 optix_stack_size.h File Reference

Functions

- `OptixResult optixUtilAccumulateStackSizes (OptixProgramGroup programGroup, OptixStackSizes *stackSizes)`
- `OptixResult optixUtilComputeStackSizes (const OptixStackSizes *stackSizes, unsigned int maxTraceDepth, unsigned int maxCCDepth, unsigned int maxDCDepth, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)`
- `OptixResult optixUtilComputeStackSizesDCSplit (const OptixStackSizes *stackSizes, unsigned int dssDCFromTraversal, unsigned int dssDCFromState, unsigned int maxTraceDepth, unsigned int maxCCDepth, unsigned int maxDCDepthFromTraversal, unsigned int maxDCDepthFromState, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)`
- `OptixResult optixUtilComputeStackSizesCssCCTree (const OptixStackSizes *stackSizes, unsigned int cssCCTree, unsigned int maxTraceDepth, unsigned int maxDCDepth, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)`

- `OptixResult optixUtilComputeStackSizesSimplePathTracer` (`OptixProgramGroup` programGroupRG, `OptixProgramGroup` programGroupMS1, const `OptixProgramGroup` *programGroupCH1, unsigned int programGroupCH1Count, `OptixProgramGroup` programGroupMS2, const `OptixProgramGroup` *programGroupCH2, unsigned int programGroupCH2Count, unsigned int *directCallableStackSizeFromTraversal, unsigned int *directCallableStackSizeFromState, unsigned int *continuationStackSize)

6.13.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation

6.14 optix_stubs.h File Reference

Macros

- `#define WIN32_LEAN_AND_MEAN` 1

Functions

- static void * `optixLoadWindowsDllFromName` (const char *optixDllName)
- static void * `optixLoadWindowsDll` ()
- `OptixResult optixInitWithHandle` (void **handlePtr)
- `OptixResult optixInit` (void)

Variables

- `OptixFunctionTable g_optixFunctionTable`

6.14.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation

6.14.2 Macro Definition Documentation

6.14.2.1 `#define WIN32_LEAN_AND_MEAN 1`

6.14.3 Function Documentation

6.14.3.1 `static void* optixLoadWindowsDll( ) [static]`

6.14.3.2 `static void* optixLoadWindowsDllFromName (const char * optixDllName ) [static]`

6.15 optix_types.h File Reference

Macros

- `#define __OPTIX_INCLUDE_INTERNAL_HEADERS__`
- `#define __UNDEF_OPTIX_INCLUDE_INTERNAL_HEADERS_OPTIX_TYPES_H__`

6.15.1 Detailed Description

OptiX public API header.

Author

NVIDIA Corporation

6.15.2 Macro Definition Documentation

6.15.2.1 `#define __OPTIX_INCLUDE_INTERNAL_HEADERS__`

6.15.2.2 `#define __UNDEF_OPTIX_INCLUDE_INTERNAL_HEADERS_OPTIX_TYPES_H__`

6.16 optixPaging.cu File Reference

Functions

- `__host__ void optixPagingPullRequests (OptixPagingContext *context, unsigned int *devRequestedPages, unsigned int numRequestedPages, unsigned int *devStalePages, unsigned int numStalePages, unsigned int *devEvictablePages, unsigned int numEvictablePages, unsigned int *devNumPagesReturned)`
- `__host__ void optixPagingPushMappings (OptixPagingContext *context, PageMapping *devFilledPages, int filledPageCount, unsigned int *devInvalidatedPages, int invalidatedPageCount)`
- `__host__ void optixPagingCreate (OptixPagingOptions *options, OptixPagingContext **context)`

- `__host__ void optixPagingDestroy (OptixPagingContext *context)`
- `__host__ void optixPagingCalculateSizes (unsigned int vaSizeInPages, OptixPagingSizes &sizes)`
- `__host__ void optixPagingSetup (OptixPagingContext *context, const OptixPagingSizes &sizes, int numWorkers)`

6.16.1 Function Documentation

6.16.1.1 `__host__ void optixPagingCalculateSizes (`
 `unsigned int vaSizeInPages,`
 `OptixPagingSizes & sizes )`

6.16.1.2 `__host__ void optixPagingCreate (`
 `OptixPagingOptions * options,`
 `OptixPagingContext ** context )`

6.16.1.3 `__host__ void optixPagingDestroy (`
 `OptixPagingContext * context )`

6.16.1.4 `__host__ void optixPagingPullRequests (`
 `OptixPagingContext * context,`
 `unsigned int * devRequestedPages,`
 `unsigned int numRequestedPages,`
 `unsigned int * devStalePages,`
 `unsigned int numStalePages,`
 `unsigned int * devEvictablePages,`
 `unsigned int numEvictablePages,`
 `unsigned int * devNumPagesReturned )`

6.16.1.5 `__host__ void optixPagingPushMappings (`
 `OptixPagingContext * context,`
 `PageMapping * devFilledPages,`
 `int filledPageCount,`
 `unsigned int * devInvalidatedPages,`
 `int invalidatedPageCount )`

6.16.1.6 `__host__ void optixPagingSetup (`
 `OptixPagingContext * context,`
 `const OptixPagingSizes & sizes,`
 `int numWorkers )`

6.17 optixPaging.h File Reference

Classes

- struct [PageMapping](#)

- struct [OptixPagingSizes](#)
- struct [OptixPagingOptions](#)
- struct [OptixPagingContext](#)

Macros

- `#define OPTIX_PAGING_CHECK_CUDA_ERROR(err) optixPagingCheckCudaError( err )`

Functions

- `bool optixPagingCheckCudaError (cudaError_t err)`
- `template<typename T > __host__ __device__ T minimum (T lhs, T rhs)`
- `template<typename T > __host__ __device__ T maximum (T lhs, T rhs)`
- `__host__ void optixPagingCreate (OptixPagingOptions *options, OptixPagingContext **context)`
- `__host__ void optixPagingDestroy (OptixPagingContext *context)`
- `__host__ void optixPagingCalculateSizes (unsigned int vaSizeInPages, OptixPagingSizes &sizes)`
- `__host__ void optixPagingSetup (OptixPagingContext *context, const OptixPagingSizes &sizes, int numWorkers)`
- `__host__ void optixPagingPullRequests (OptixPagingContext *context, unsigned int *devRequestedPages, unsigned int numRequestedPages, unsigned int *devStalePages, unsigned int numStalePages, unsigned int *devEvictablePages, unsigned int numEvictablePages, unsigned int *devNumPagesReturned)`
- `__host__ void optixPagingPushMappings (OptixPagingContext *context, PageMapping *devFilledPages, int filledPageCount, unsigned int *devInvalidatedPages, int invalidatedPageCount)`

Variables

- `const int MAX_WORKER_THREADS = 32`

6.17.1 Macro Definition Documentation

6.17.1.1 `#define OPTIX_PAGING_CHECK_CUDA_ERROR( err ) optixPagingCheckCudaError( err )`

6.17.2 Function Documentation

6.17.2.1 `template<typename T > __host__ __device__ T maximum ( T lhs, T rhs )`

6.17.2.2 `template<typename T > __host__ __device__ T minimum ( T lhs,`

T rhs)

6.17.2.3 __host__ void optixPagingCalculateSizes (
 unsigned int vaSizeInPages,
 OptixPagingSizes & sizes)

6.17.2.4 bool optixPagingCheckCudaError (
 cudaError_t err) [inline]

6.17.2.5 __host__ void optixPagingCreate (
 OptixPagingOptions * options,
 OptixPagingContext ** context)

6.17.2.6 __host__ void optixPagingDestroy (
 OptixPagingContext * context)

6.17.2.7 __host__ void optixPagingPullRequests (
 OptixPagingContext * context,
 unsigned int * devRequestedPages,
 unsigned int numRequestedPages,
 unsigned int * devStalePages,
 unsigned int numStalePages,
 unsigned int * devEvictablePages,
 unsigned int numEvictablePages,
 unsigned int * devNumPagesReturned)

6.17.2.8 __host__ void optixPagingPushMappings (
 OptixPagingContext * context,
 PageMapping * devFilledPages,
 int filledPageCount,
 unsigned int * devInvalidatedPages,
 int invalidatedPageCount)

6.17.2.9 __host__ void optixPagingSetup (
 OptixPagingContext * context,
 const OptixPagingSizes & sizes,

int numWorkers)

6.17.3 Variable Documentation

6.17.3.1 const int MAX_WORKER_THREADS = 32

6.18 optixPagingImpl.cpp File Reference

Functions

- `__device__ __forceinline__`
unsigned int `countSetBitsAndCalcIndex` (const unsigned int *laneld*, const unsigned int *pageBits*, unsigned int **pageCount*)
- `__device__ __forceinline__ void` `addPagesToList` (unsigned int *startingIndex*, unsigned int *pageBits*, unsigned int *pageBitOffset*, unsigned int *maxCount*, unsigned int **outputArray*)
- `__global__ void` `devicePullRequests` (unsigned int **usageBits*, unsigned int **residenceBits*, unsigned int *maxVaSizeInPages*, unsigned int **devRequestedPages*, unsigned int *numRequestedPages*, unsigned int **numRequestedPagesReturned*, unsigned int **devStalePages*, unsigned int *numStalePages*, unsigned int **numStalePagesReturned*, unsigned int **devEvictablePages*, unsigned int *numEvictablePages*, unsigned int **numEvictablePagesReturned*)
- `__global__ void` `deviceFillPages` (unsigned long long **pageTable*, unsigned int **residenceBits*, `PageMapping` **devFilledPages*, int *filledPageCount*)
- `__global__ void` `deviceInvalidatePages` (unsigned int **residenceBits*, unsigned int **devInvalidatedPages*, int *invalidatedPageCount*)

6.18.1 Function Documentation

6.18.1.1 `__device__ __forceinline__ void addPagesToList (`
 unsigned int *startingIndex*,
 unsigned int *pageBits*,
 unsigned int *pageBitOffset*,
 unsigned int *maxCount*,
 unsigned int * *outputArray*)

6.18.1.2 `__device__ __forceinline__ unsigned int countSetBitsAndCalcIndex (`
 const unsigned int *laneld*,
 const unsigned int *pageBits*,
 unsigned int * *pageCount*)

6.18.1.3 `__global__ void deviceFillPages (`
 unsigned long long * *pageTable*,
 unsigned int * *residenceBits*,
 `PageMapping` * *devFilledPages*,

int *filledPageCount*)

6.18.1.4 __global__ void deviceInvalidatePages (
 unsigned int * *residenceBits*,
 unsigned int * *devInvalidatedPages*,
 int *invalidatedPageCount*)

6.18.1.5 __global__ void devicePullRequests (
 unsigned int * *usageBits*,
 unsigned int * *residenceBits*,
 unsigned int *maxVaSizeInPages*,
 unsigned int * *devRequestedPages*,
 unsigned int *numRequestedPages*,
 unsigned int * *numRequestedPagesReturned*,
 unsigned int * *devStalePages*,
 unsigned int *numStalePages*,
 unsigned int * *numStalePagesReturned*,
 unsigned int * *devEvictablePages*,
 unsigned int *numEvictablePages*,
 unsigned int * *numEvictablePagesReturned*)